

Solutions to End of Section Chapter 1 Problems

Section 1.2:

- 1.2.1 (a) Let us label the 4 math professors as A, B, C , and D . Using ordered pairs to represent which professor was asked by students 1 and 2, a possible sample space to use (consisting of 16 outcomes) is

$$S = \{AA, AB, AC, AD, BA, BB, BC, BD, CA, CB, CC, CD, DA, DB, DC, DD\}.$$

- (b) Let $E =$ “both students ask the same professor”. Since $E = \{AA, BB, CC, DD\}$, it follows that

$$P(E) = \frac{4}{16} = \frac{1}{4}.$$

Therefore, the odds in favour of E are given by

$$\frac{P(E)}{1 - P(E)} = \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3}, \text{ or equivalently } 1 : 3.$$

- 1.2.2 (a) We can simply use

$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}.$$

- (b) Since the coin is fair, all 8 outcomes in S are equally likely. If $A =$ “getting exactly 2 tails”, then

$$A = \{HTT, THT, TTH\}.$$

It follows that $P(A) = \frac{3}{8}$. Therefore, the odds against A are given by

$$\frac{1 - P(A)}{P(A)} = \frac{\frac{5}{8}}{\frac{3}{8}} = \frac{5}{3}, \text{ or equivalently } 5 : 3.$$

- 1.2.3 A sample space of all possible pairs is given by

$$S = \{(1, 2), (1, 3), (1, 4), (1, 5), (2, 3), (2, 4), (2, 5), (3, 4), (3, 5), (4, 5), \\ (2, 1), (3, 1), (4, 1), (5, 1), (3, 2), (4, 2), (5, 2), (4, 3), (5, 3), (5, 4)\}.$$

If $A =$ “numbers chosen differ by 1”, then

$$A = \{(1, 2), (2, 3), (3, 4), (4, 5), (2, 1), (3, 2), (4, 3), (5, 4)\}.$$

Therefore, $P(A) = \frac{|A|}{|S|} = \frac{8}{20} = \frac{2}{5}$.

- 1.2.4 (a) Let $abcd$ represent the outcome that letter a is in W 's envelope, letter b is in X 's envelope, letter c is in Y 's envelope, and letter d is in Z 's envelope. With this notation, the set of 24 possible outcomes for this experiment is given as follows:

$$S = \{WXYZ, XWYZ, YWXZ, ZWXY, WXZY, XWZY, YWZX, ZWYX, \\ WYXZ, XYWZ, YXWZ, ZXWY, WYZX, XYZW, YXZW, ZXYW, \\ WZXY, XZWY, YZWX, ZYXW, WZYX, XZYW, YZXW, ZYWX\}.$$

- (b) Using the sample space in part (a), we have:

$$A = \{WXYZ, WXZY, WYXZ, WYZX, WZXY, WZYX\}, \\ B = \{XWZY, XYZW, XZWY, YWZX, YZWX, YZXW, ZWXY, ZYXW, ZYWX\}, \\ C = \{WXZY, WYXZ, WZYX, ZXYW, YXWZ, XWYZ\}, \\ D = \emptyset.$$

(c) $P(A) = \frac{6}{24} = \frac{1}{4}, \quad P(B) = \frac{9}{24} = \frac{3}{8}, \quad P(C) = \frac{6}{24} = \frac{1}{4}, \quad P(D) = P(\emptyset) = 0.$

- 1.2.5 (a) Let ijk represent the outcome that ball 1 is in box i , ball 2 is in box j , and ball 3 is in box k . With this notation, the set of 27 possible outcomes for this experiment is given as follows:

$$S = \{111, 222, 333, 112, 121, 211, 113, 131, 311, \\ 221, 212, 122, 223, 232, 322, 331, 313, 133, \\ 332, 323, 233, 123, 132, 213, 231, 312, 321\}.$$

Using the above sample space, we therefore have:

$$A = \{222, 333, 223, 232, 322, 332, 323, 233\} \implies P(A) = \frac{8}{27}, \\ B = \{333\} \implies P(B) = \frac{1}{27}, \\ C = \{123, 132, 213, 231, 312, 321\} \implies P(C) = \frac{6}{27} = \frac{2}{9}.$$

- (b) Moving from 3 boxes to n boxes where $n \geq 3$, we readily deduce that the sample space would increase from $3^3 = 27$ outcomes to n^3 outcomes. As a result, the probabilities of events A , B , and C would generalize to become

$$P(A) = \frac{(n-1)^3}{n^3}, \quad P(B) = \frac{(n-2)^3}{n^3}, \quad \text{and} \quad P(C) = \frac{n(n-1)(n-2)}{n^3}.$$

Note that when $n = 3$, the above probabilities agree with those in part (a).

- (c) Moving from 3 balls to k balls under the assumption of n boxes such that $n \geq k$, the outcomes in the corresponding sample space would now look like $i_1 i_2 \cdots i_k$, intended to denote that ball 1 is in box i_1 , ball 2 is in box $i_2, \dots$, and ball k is in box i_k . As a result, the probabilities of events A, B , and C would generalize to become

$$P(A) = \frac{(n-1)^k}{n^k}, \quad P(B) = \frac{(n-2)^k}{n^k}, \quad \text{and} \quad P(C) = \frac{n(n-1) \cdots (n-k+1)}{n^k}.$$

Note that when $k = 3$, the above probabilities agree with those in part (b).

- 1.2.6 (a) Out of the 1000 people who took the inexpensive test, 18 persons with the disease tested positive. Therefore, we would estimate the desired fraction of the population to be $\frac{18}{1000} = 0.018$.
- (b) Out of the 1000 people who took the inexpensive test, $18 + 2 = 20$ persons had the disease. Therefore, we would estimate the desired fraction of the population to be $\frac{20}{1000} = 0.02$.
- (c) Out of the $60 + 18 = 78$ people who had a positive test, 18 of them actually had the disease. Therefore, we would estimate the desired probability to be $\frac{18}{78} = \frac{3}{13}$.

- 1.2.7 (a) We would use $S = \{(x, y) : x = 0, 1, \dots, 9 \text{ and } y = 0, 1, \dots, 9\}$.

- (b) The following table displays the probabilities for each point (x, y) in S :

		y									
		0	1	2	3	4	5	6	7	8	9
x	0	0.096	0	0	0	0	0	0.004	0	0	0
	1	0	0.1	0	0	0	0	0	0	0	0
	2	0	0	0.1	0	0	0	0	0	0	0
	3	0	0	0	0.1	0	0	0	0	0	0
	4	0	0	0	0	0.098	0	0	0.002	0	0
	5	0	0	0	0	0	0.095	0	0	0	0.005
	6	0.004	0	0	0	0	0	0.096	0	0	0
	7	0	0	0	0	0.002	0	0	0.098	0	0
	8	0	0	0	0	0	0	0	0	0.1	0
	9	0	0	0	0	0	0.005	0	0	0	0.095

The fraction of numbers which are correctly identified by the machine is given by

$$\sum_{x=0}^9 P((x, x)) = 2 \cdot 0.096 + 2 \cdot 0.095 + 2 \cdot 0.098 + 4 \cdot 0.1 = 0.978.$$

Section 1.3:

- 1.3.1 (a) Suppose we list the outcomes in S in a particular order (i.e., choice of student A , choice of student B , and choice of student C), so that the outcome $(1, 2, 3)$ denotes A chose section 1, B chose section 2, and C chose section 3. Therefore, S looks like

$$S = \{(1, 1, 1), (1, 1, 2), (1, 1, 3), \dots, (4, 4, 4)\}.$$

Since each student can choose a section in 4 ways regardless of the choice of the other two students, S has $4 \times 4 \times 4 = 64$ outcomes by the multiplication rule.

- (i) Let $E_1 =$ “all 3 students end up in the same section”. To satisfy the desired condition, we note that the first student, say student A , can choose a section in 4 ways and the others then only have 1 section they can go in. Therefore, we readily obtain

$$P(E_1) = \frac{4 \times 1 \times 1}{64} = \frac{1}{16}.$$

- (ii) Let $E_2 =$ “all 3 students end up in different sections”. To satisfy the desired condition, we note that the first student, say student A , has 4 ways to choose a section, the next student, say student B , has 3 ways to choose a section, and the last student has 2 sections to pick from. Therefore, we readily obtain

$$P(E_2) = \frac{4 \times 3 \times 2}{64} = \frac{3}{8}.$$

- (iii) Let $E_3 =$ “no one picks section 1”. To satisfy the desired condition, we note that each student has 3 ways to choose a section (since we are excluding section 1 as a choice). Therefore, we readily obtain

$$P(E_3) = \frac{3 \times 3 \times 3}{64} = \frac{27}{64}.$$

- (b) In this generalized version of the problem, we remark that S now consists of n^s possible outcomes, having the form

$$S = \{(i_1, i_2, \dots, i_s) : i_j = 1, 2, \dots, n \text{ for each } j = 1, 2, \dots, s\}.$$

- (i) $P(E_1) = \frac{n \times 1 \times 1 \times \dots \times 1}{n^s} = \frac{1}{n^{s-1}}.$
(ii) $P(E_2) = \frac{n \times (n-1) \times (n-2) \times \dots \times (n-s+1)}{n^s} = \frac{n^{(s)}}{n^s}.$
(iii) $P(E_3) = \frac{(n-1) \times (n-1) \times (n-1) \times \dots \times (n-1)}{n^s} = \frac{(n-1)^s}{n^s}.$

Note that when $n = 4$ and $s = 3$, the above probabilities agree with those in part (a).

- 1.3.2 (a) There are 26 ways to choose each of the 3 letters in a postal code, so in all, the letters can be chosen in $26 \times 26 \times 26 = 26^3$ ways. If all 3 letters are to be the same, then there are 26 ways to choose the first letter and only 1 way to choose the remaining 2 letters. If A denotes the event that all 3 letters are the same, it follows that

$$P(A) = \frac{26 \times 1 \times 1}{26^3} = \frac{1}{26^2} = \frac{1}{676}.$$

- (b) Turning our attention now to the digits, we note that there are $10 \times 10 \times 10 = 10^3$ ways to choose the 3 digits in a postal code. The number of ways to choose all even digits is $5 \times 5 \times 5 = 5^3$. Similarly, the number of ways to choose all odd digits is $5 \times 5 \times 5 = 5^3$. If B denotes the event that the digits are all even or all odd, it follows by the addition rule that

$$P(B) = \frac{5^3 + 5^3}{10^3} = \frac{250}{1000} = \frac{1}{4}.$$

- 1.3.3 Each element of a binary sequence can be one of two possible values (i.e., either 0 or 1). Therefore, the total number of possible binary sequences of length 10 would be given by 2^{10} by the multiplication rule. Let us consider a binary sequence that has exactly 5 zeros such as (1, 0, 1, 1, 0, 0, 1, 0, 0, 1). In this sequence, note that a 0 occupies positions 2, 5, 6, 8, and 9 of the sequence. Therefore, to determine the total number of binary sequences with exactly 5 zeros, we simply need to determine all combinations of size 5 chosen from the set of positions $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. This number is given by $\binom{10}{5}$. Since we are assuming a uniform probability model, we immediately obtain

$$P(\text{sequence has exactly 5 zeros}) = \frac{\binom{10}{5}}{2^{10}} = \frac{252}{1024} = \frac{63}{256}.$$

- 1.3.4 (a) The number of possible 4-digit numbers which can be formed is $7^{(4)}$. The issue of whether a number is even or odd is determined by the last digit of the number. To ensure that the number is even, the last digit must be either a 2, 4, or 6 (i.e., 3 possible choices). Once this digit is chosen to be even, the remaining 3 digits of the number have no other restrictions. Therefore, it follows by the multiplication rule that

$$P(\text{number is even}) = \frac{6 \times 5 \times 4 \times 3}{7^{(4)}} = \frac{3}{7}.$$

- (b) To ensure that the number is over 3000, there are 5 ways to choose the first digit (i.e., 3, 4, 5, 6, or 7), 6 ways to choose the second digit, 5 ways to choose the third digit, and 4 ways to choose the fourth digit. Therefore, by the multiplication rule, we obtain

$$P(\text{number is over 3000}) = \frac{5 \times 6 \times 5 \times 4}{7^{(4)}} = \frac{5}{7}.$$

(c) To ensure that the number is even and over 3000, we consider two cases:

- (i) If the first digit is an even number, then there are 2 ways to choose the first digit (i.e., 4 or 6). With the first digit being either a 4 or 6, there are now 2 remaining even digits to choose from for the last digit of the number. With these digits now selected, the middle two digits can be selected in $5 \times 4 = 20$ ways. Therefore, the total number of ways is given by $2 \times 20 \times 2 = 80$ ways.
- (ii) If the first digit is an odd number, then there are 3 ways to choose the first digit (i.e., 3, 5, or 7). With the first digit being odd, there are 3 even digits to choose from for the last digit of the number. As in case (i), the middle two digits can be selected in $5 \times 4 = 20$ ways. Therefore, the total number of ways is given by $3 \times 20 \times 2 = 120$ ways.

Therefore, it follows by the addition rule that

$$P(\text{an even number over 3000}) = \frac{80 + 120}{7^{(4)}} = \frac{200}{840} = \frac{10}{42}.$$

1.3.5 First of all, the 8 cars can be chosen (without replacement) in $\binom{160}{8}$ ways. In a sample of size 8, it is possible to have x with faulty emission controls and $8 - x$ with non-faulty ones in $\binom{35}{x} \times \binom{125}{8-x}$ ways. Note that $x \in \{0, 1, 2, \dots, 8\}$. Therefore, applying the addition rule, it follows that

$$P(\text{at least 3 faulty emission controls}) = \frac{\sum_{x=3}^8 \binom{35}{x} \binom{125}{8-x}}{\binom{160}{8}}.$$

1.3.6 (a) Suppose we list the outcomes in S in the following manner: $(i_1, i_2, \dots, i_r)$ will represent individual 1 selecting floor i_1 , individual 2 selecting floor $i_2, \dots$, and individual r selecting floor i_r . As a result, S consists of n^r possible outcomes, having the form

$$S = \{(i_1, i_2, \dots, i_r) : i_j = 1, 2, \dots, n \text{ for each } j = 1, 2, \dots, r\}.$$

Therefore, eliminating floor 1 as a possible option, it immediately follows that

$$P(\text{no individual gets off at floor 1}) = \frac{(n-1) \times (n-1) \times (n-1) \times \dots \times (n-1)}{n^r} = \frac{(n-1)^r}{n^r}.$$

(b) We note that when a floor gets selected by an individual, that floor is removed from consideration for all subsequent selections. Therefore, it immediately follows that

$$P(\text{individuals all get off at different floors}) = \frac{n \times (n-1) \times (n-2) \times \dots \times (n-r+1)}{n^r} = \frac{n^{(r)}}{n^r}.$$

1.3.7 Let $A =$ “2 passengers get off at one stop and two passengers get off at another stop”. Let us label the 5 stops as 1, 2, 3, 4, and 5. Since each of the 4 passengers can select any of these 5 stops, we can use a sample space for this experiment consisting of 5^4 possible outcomes, having the form

$$S = \{(i_1, i_2, i_3, i_4) : i_j = 1, 2, 3, 4, 5 \text{ for each } j = 1, 2, 3, 4\}.$$

The number of possible pairs of stops to choose from is given by $\binom{5}{2}$. This set of 10 pairs would like

$$\{(1, 2), (1, 3), (1, 4), (1, 5), (2, 3), (2, 4), (2, 5), (3, 4), (3, 5), (4, 5)\}.$$

For a given pair (i, j) from the above set, note that:

- passengers 1 and 2 can be assigned to stop i and passengers 3 and 4 to stop j ,
- passengers 1 and 3 can be assigned to stop i and passengers 2 and 4 to stop j ,
- passengers 1 and 4 can be assigned to stop i and passengers 2 and 3 to stop j ,
- passengers 2 and 3 can be assigned to stop i and passengers 1 and 4 to stop j ,
- passengers 2 and 4 can be assigned to stop i and passengers 1 and 3 to stop j ,
- passengers 3 and 4 can be assigned to stop i and passengers 1 and 2 to stop j .

In other words, there are $\binom{4}{2} = 6$ assignments possible. Therefore, applying the multiplication rule, we ultimately obtain

$$P(A) = \frac{\binom{5}{2} \times \binom{4}{2}}{5^4} = \frac{60}{625} = \frac{12}{125}.$$

1.3.8 (a) If the draws are made without replacement, the sample space S consists of $10^{(3)}$ ordered arrangements of size 3 taken from the set $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Therefore, the following outcomes involving, say the digits 1, 4, and 8, would be included in S :

$$(1, 4, 8), (1, 8, 4), (4, 1, 8), (4, 8, 1), (8, 1, 4), (8, 4, 1).$$

Of these $3! = 6$ permutations of the digits 1, 4, and 8, note that only one of them is in increasing order (i.e., $(1, 4, 8)$). As a result of this observation, it suffices to simply count the number of subsets of size 3 chosen from a group of 10 distinct objects. In other words, we have that

$$P(\text{digits drawn in increasing order}) = \frac{\binom{10}{3}}{10^{(3)}} = \frac{120}{720} = \frac{1}{6}.$$

- (b) If the draws are now made with replacement, the sample space S consists of 10^3 ordered arrangements of size 3 taken from the set $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Therefore, the following outcomes involving, say the digits 1, 4, and 8, would be included in S :

(1, 1, 1), (1, 1, 4), (1, 1, 8), (1, 4, 1), (1, 4, 4), (1, 4, 8), (1, 8, 1), (1, 8, 4), (1, 8, 8),
 (4, 1, 1), (4, 1, 4), (4, 1, 8), (4, 4, 1), (4, 4, 4), (4, 4, 8), (4, 8, 1), (4, 8, 4), (4, 8, 8),
 (8, 1, 1), (8, 1, 4), (8, 1, 8), (8, 4, 1), (8, 4, 4), (8, 4, 8), (8, 8, 1), (8, 8, 4), (8, 8, 8).

Of these $3^3 = 27$ permutations of the digits 1, 4, and 8, note that once again only one of them is in increasing order (i.e., (1, 4, 8)). As a result, it follows that

$$P(\text{digits drawn in increasing order}) = \frac{\binom{10}{3}}{10^3} = \frac{120}{1000} = \frac{3}{25}.$$

- 1.3.9 (a) If the ticket numbered 7337 is drawn, prizes would be paid on the 6 tickets numbered 3377, 3737, 3773, 7337, 7373, and 7733. Therefore, $P(\text{prize is won}) = \frac{6}{10000} = 0.0006$. On the other hand, if the ticket numbered 7235 is drawn, prizes would be paid on the $4! = 24$ arrangements of the ticket digits 7, 2, 3, and 5. In this case,

$$P(\text{prize is won}) = \frac{24}{10000} = 0.0024.$$

Based on these winning probabilities, you would advise someone to purchase a ticket having 4 distinct digits.

- (b) Assuming all tickets are sold for \$1 each, the operator initially collects \$10,000. Of the 10,000 tickets sold, we remark that there are:
- (i) $10^{(4)} = 5040$ tickets having all distinct digits (e.g., 7235) $\implies 4! = 24$ prizes of \$500 will be awarded if such a ticket is drawn for a grand payout of $24 \cdot 500 = \$12,000$.
 - (ii) $\binom{10}{2} \times 6 = 270$ tickets having 2 digits of one kind and 2 digits of second kind (e.g., 7337) $\implies 6$ prizes of \$500 will be awarded if such a ticket is drawn for a grand payout of $6 \cdot 500 = \$3,000$.
 - (iii) $10^{(2)} \times 4 = 360$ tickets having 3 digits of one kind and 1 digit of second kind (e.g., 3533) $\implies 4$ prizes of \$500 will be awarded if such a ticket is drawn for a grand payout of $4 \cdot 500 = \$2,000$.
 - (iv) $\binom{10}{1} = 10$ tickets having all the same digits (e.g., 2222) $\implies 1$ prize of \$500 will be awarded if such a ticket is drawn for a grand payout of \$500.

Therefore, there are $10000 - 5040 - 270 - 360 - 10 = 4320$ tickets having 2 digits of one kind, 1 digit of a second kind, and 1 digit of a third kind (e.g., 2342). Under this case, 12 prizes of \$500 will be awarded for such a ticket (*Why?*), resulting in a grand payout of $12 \cdot 500 = \$6,000$. Looking at the results, there is only one scenario where the raffle operator will lose money (i.e., case (i)), and this happens with probability $\frac{10^{(4)}}{10000} = 0.504$.

1.3.10 Let A_x denote the event that x matches are obtained, $x \in \{0, 1, 2, 3, 4, 5, 6\}$. Of the 49 numbers available to be chosen, note that 6 of them correspond to numbers found on your purchased ticket and 43 of them to numbers not found on your purchased ticket. Therefore, in a randomly selected draw of size 6 (chosen without replacement), it is possible to obtain x matches and $6 - x$ non-matches in $\binom{6}{x} \times \binom{43}{6-x}$ ways. Since the 6 numbers can be chosen in $\binom{49}{6}$ total ways, it follows that

$$P(A_x) = \frac{\binom{6}{x} \binom{43}{6-x}}{\binom{49}{6}}, \quad x = 0, 1, 2, 3, 4, 5, 6.$$

Plugging in the individual values of x leads to the following table of probabilities:

x	0	1	2	3	4	5	6
$P(A_x)$	$\frac{435461}{998844}$	$\frac{68757}{166474}$	$\frac{44075}{332948}$	$\frac{8815}{499422}$	$\frac{645}{665896}$	$\frac{43}{2330636}$	$\frac{1}{13983816}$

1.3.11 Since each person has 365 choices for the day of their birthday, it follows that the sample space S for this experiment would consist of 365^r possible outcomes. Since each person is to have a different birthday, it immediately follows that

$$P(\text{no two persons have the same birthday}) = \frac{365^{(r)}}{365^r}.$$

With the aid of computer software, we can calculate the following numerical probabilities:

- When $r = 20$: $P(\text{no two persons have the same birthday}) = 0.588562$,
- When $r = 40$: $P(\text{no two persons have the same birthday}) = 0.108768$,
- When $r = 60$: $P(\text{no two persons have the same birthday}) = 0.00587734$.

Solutions to End of Section Chapter 2 Problems

Section 2.1:

2.1.1 A sample space to use for this experiment is

$$S = \{HHHH, HHHT, HHTH, HTHH, HHTT, HTHT, HTTH, HTTT, THHH, THHT, THTH, TTHH, THTT, TTHT, TTTH, TTTT\}.$$

Based on the definitions of A and B , it follows that

$$A = \{HHTT, HTHT, HTTH, THHT, THTH, TTHH\}$$

and

$$B = \{HHHT, HHTH, HHTT, HTHT, HTTH, HTTT, THHT, THTH, THTT, TTHT, TTTH, TTTT\}.$$

Furthermore, the following events can be expressed as

$$A \cup B = \{TTHH, HHHT, HHTH, HHTT, HTHT, HTTH, HTTT, THHT, THTH, THTT, TTHT, TTTH, TTTT\},$$

$$A \cap B = \{HHTT, HTHT, HTTH, THHT, THTH\},$$

$$\bar{A} = \{HHHH, HHHT, HHTH, HTHH, THHH, TTTH, TTHT, THTT, HTTT, TTTT\},$$

and

$$\bar{B} = \{HHHH, HTHH, THHH, TTHH\}.$$

Using the above events, note that

$$\overline{A \cup B} = \{HHHH, HTHH, THHH\} \text{ and } \bar{A} \cap \bar{B} = \{HHHH, HTHH, THHH\}.$$

Clearly, $\overline{A \cup B} = \bar{A} \cap \bar{B}$. In a similar fashion, we have that

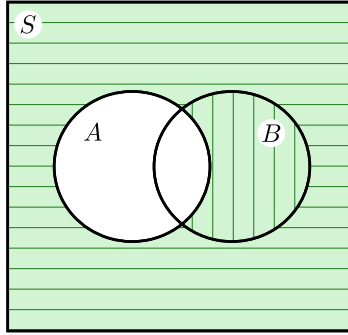
$$\overline{A \cap B} = \{HHHH, HHHT, HHTH, HTHH, THHH, TTTH, TTHT, THTT, HTTT, TTTT, TTHH\}$$

and

$$\bar{A} \cup \bar{B} = \{HHHH, HHHT, HHTH, HTHH, THHH, TTTH, TTHT, THTT, HTTT, TTTT, TTHH\},$$

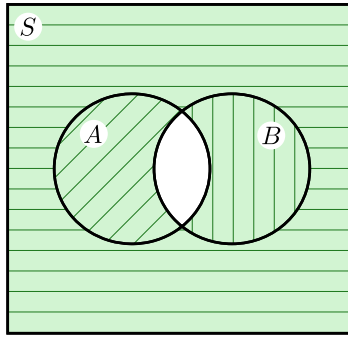
implying that $\overline{A \cap B} = \bar{A} \cup \bar{B}$.

2.1.2 (a) We use the following Venn diagram to prove the result:



Note $\bar{A} \cap B$ is indicated with vertical lines and $\bar{A} \cap \bar{B}$ with horizontal lines. Therefore, $(\bar{A} \cap B) \cup (\bar{A} \cap \bar{B})$ is the region with both vertical and horizontal lines. Note that this agrees with the shaded region representing $\bar{A}$.

(b) We use the following Venn diagram to prove the result:



Note $\bar{A} \cap B$ is indicated with vertical lines, $\bar{A} \cap \bar{B}$ with horizontal lines, and $A \cap \bar{B}$ with diagonal lines. Note that this agrees with the shaded region representing $\bar{A} \cap \bar{B}$.

(c) A sample space to use for this experiment is

$$S = \{(i, j) : i = 1, 2, 3, 4, 5, 6 \text{ and } j = 1, 2, 3, 4, 5, 6\}.$$

Based on the definitions of A and B , we have that

$$\begin{aligned} A = \{ & (1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 1), (2, 3), (2, 5), \\ & (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), (4, 1), (4, 3), (4, 5), \\ & (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), (6, 1), (6, 3), (6, 5) \} \end{aligned}$$

and

$$B = \{(3, 6), (4, 5), (4, 6), (5, 4), (5, 5), (5, 6), (6, 3), (6, 4), (6, 5), (6, 6)\}.$$

Furthermore, the following events can be expressed as

$$A \cap B = \{(3, 6), (4, 5), (5, 4), (5, 5), (5, 6), (6, 3), (6, 5)\},$$

$$\bar{A} = \{(2, 2), (2, 4), (2, 6), (4, 2), (4, 4), (4, 6), (6, 2), (6, 4), (6, 6)\},$$

and

$$\begin{aligned} \bar{B} = & \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 1), (2, 2), (2, 3), \\ & (2, 4), (2, 5), (2, 6), (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (4, 1), \\ & (4, 2), (4, 3), (4, 4), (5, 1), (5, 2), (5, 3), (6, 1), (6, 2)\}. \end{aligned}$$

Therefore, we can now find

$$\begin{aligned} \overline{A \cap B} = & \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 1), (2, 2), \\ & (2, 3), (2, 4), (2, 5), (2, 6), (3, 1), (3, 2), (3, 3), (3, 4), \\ & (3, 5), (4, 1), (4, 2), (4, 3), (4, 4), (4, 6), (5, 1), (5, 2), \\ & (5, 3), (6, 1), (6, 2), (6, 4), (6, 6)\}, \end{aligned}$$

$$\bar{A} \cap B = \{(4, 6), (6, 4), (6, 6)\},$$

$$\bar{A} \cap \bar{B} = \{(2, 2), (2, 4), (2, 6), (4, 2), (4, 4), (6, 2)\},$$

and

$$\begin{aligned} A \cap \bar{B} = & \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 1), (2, 3), (2, 5), (3, 1), \\ & (3, 2), (3, 3), (3, 4), (3, 5), (4, 1), (4, 3), (5, 1), (5, 2), (5, 3), (6, 1)\}. \end{aligned}$$

Based on the above representations, note that $\overline{A \cap B} = (\bar{A} \cap B) \cup (\bar{A} \cap \bar{B}) \cup (A \cap \bar{B})$.

2.1.3 (a) Let us consider each identity in turn.

- (i) If a is a point such that $a \in A \cup (\bigcap_{i=1}^{\infty} B_i)$, then there are two possibilities. Either $a \in A$, in which case $a \in A \cup B_i \forall i$, and so therefore it would also belong to $\bigcap_{i=1}^{\infty} (A \cup B_i)$. Alternatively, $a \in \bigcap_{i=1}^{\infty} B_i$, implying that $a \in B_i \forall i$. Therefore, $a \in A \cup B_i \forall i$, and so therefore it would also belong to $\bigcap_{i=1}^{\infty} (A \cup B_i)$. This proves that $A \cup (\bigcap_{i=1}^{\infty} B_i) \subseteq \bigcap_{i=1}^{\infty} (A \cup B_i)$.

Conversely, if $a \in \bigcap_{i=1}^{\infty} (A \cup B_i)$, then $a \in A \cup B_i \forall i$. If $a \in A$, then it would certainly belong to $A \cup (\bigcap_{i=1}^{\infty} B_i)$. Otherwise, $a \in B_i \forall i$, in which case $a \in \bigcap_{i=1}^{\infty} B_i$, and so therefore it would also belong to $A \cup (\bigcap_{i=1}^{\infty} B_i)$. This proves that $\bigcap_{i=1}^{\infty} (A \cup B_i) \subseteq A \cup (\bigcap_{i=1}^{\infty} B_i)$. Hence, the identity is proven.

(ii) If a is a point such that $a \in A \cap (\bigcup_{i=1}^{\infty} B_i)$, then $a \in A$ and $a \in \bigcup_{i=1}^{\infty} B_i$. Since $a \in \bigcup_{i=1}^{\infty} B_i$, there exists a positive integer j such that $a \in B_j$. Therefore, it follows that $a \in A \cap B_j$, which in turn implies that $a \in \bigcup_{i=1}^{\infty} (A \cap B_i)$. This proves that $A \cap (\bigcup_{i=1}^{\infty} B_i) \subseteq \bigcup_{i=1}^{\infty} (A \cap B_i)$.

Conversely, if $a \in \bigcup_{i=1}^{\infty} (A \cap B_i)$, then there exists a positive integer j such that $a \in A \cap B_j$. This implies that $a \in A$ and $a \in B_j$. Since $a \in B_j$, we must also have that $a \in \bigcup_{i=1}^{\infty} B_i$. Therefore, it follows that $a \in A \cap (\bigcup_{i=1}^{\infty} B_i)$. This proves that $\bigcup_{i=1}^{\infty} (A \cap B_i) \subseteq A \cap (\bigcup_{i=1}^{\infty} B_i)$. Hence, the identity is proven.

(b) Using the results of parts (i) and (ii), we know that

$$A \cup \left(\bigcap_{i=1}^{\infty} C_i \right) = \bigcap_{i=1}^{\infty} (A \cup C_i) \quad (1)$$

and

$$A \cap \left(\bigcup_{i=1}^{\infty} C_i \right) = \bigcup_{i=1}^{\infty} (A \cap C_i). \quad (2)$$

For $n \in \{1, 2, 3, \dots\}$, let us now define $\{C_i\}_{i=1}^{\infty}$ as follows:

$$C_i = \begin{cases} B_i & \text{if } i = 1, 2, \dots, n, \\ B_n & \text{if } i = n + 1, n + 2, n + 3, \dots \end{cases}$$

With this definition, the left-hand and right-hand sides of (1) equate to

$$A \cup \left(\bigcap_{i=1}^{\infty} C_i \right) = A \cup \left(\bigcap_{i=1}^n B_i \right)$$

and

$$\bigcap_{i=1}^{\infty} (A \cup C_i) = \bigcap_{i=1}^n (A \cup B_i),$$

as required.

Alternatively, let us instead define $\{C_i\}_{i=1}^{\infty}$ as follows:

$$C_i = \begin{cases} B_i & \text{if } i = 1, 2, \dots, n, \\ \emptyset & \text{if } i = n + 1, n + 2, n + 3, \dots \end{cases}$$

Therefore, the left-hand and right-hand sides of (2) equate to

$$A \cap \left(\bigcup_{i=1}^{\infty} C_i \right) = A \cap \left(\bigcup_{i=1}^n B_i \right)$$

and

$$\bigcup_{i=1}^{\infty} (A \cap C_i) = \bigcup_{i=1}^n (A \cap B_i),$$

as required.

Section 2.2:

- 2.2.1 We define the events F = “person is female” and R = “person is politically to the right”. We are given that $P(F) = 0.55$, $P(R) = 0.55$, and $P(\bar{F}\bar{R}) = 0.15$. We wish to determine $P(FR)$. Since $P(F \cup R) = P(F) + P(R) - P(FR)$, it follows that

$$\begin{aligned} P(FR) &= P(F) + P(R) - P(F \cup R) \\ &= P(F) + P(R) - (1 - P(\bar{F} \cup \bar{R})) \\ &= P(F) + P(R) - 1 + P(\bar{F}\bar{R}) \text{ by De Morgan's Law} \\ &= 0.55 + 0.55 - 1 + 0.15 \\ &= 0.25. \end{aligned}$$

2.2.2 (a) $P(\bar{A}) = 1 - P(A) = 1 - 0.25 = 0.75$.

(b) $P(\bar{B}) = 1 - P(B) = 1 - 0.4 = 0.6$.

(c) Since A and B are mutually exclusive, $P(A \cup B) = P(A) + P(B) = 0.25 + 0.4 = 0.65$.

(d) Since A and B are mutually exclusive, $P(A \cap B) = P(\emptyset) = 0$.

(e) Using De Morgan's Law, $P(\bar{A} \cup \bar{B}) = P(\overline{A \cap B}) = 1 - P(A \cap B) = 1 - 0 = 1$.

(f) Using De Morgan's Law, $P(\bar{A} \cap \bar{B}) = P(\overline{A \cup B}) = 1 - P(A \cup B) = 1 - 0.65 = 0.35$.

(g) $P(\overline{A \cap B}) = 1 - P(A \cap B) = 1 - 0 = 1$.

- 2.2.3 Note that

$$P(A \cup B) = 1 - P(\overline{A \cup B}) = 1 - P(\bar{A} \bar{B}) \text{ by De Morgan's Law.}$$

- 2.2.4 Since $A_1 \subseteq A_2 \subseteq A_3 \subseteq \dots$, we can construct a sequence $\{B_i\}_{i=1}^{\infty}$ of mutually exclusive events given by

$$\begin{aligned} B_1 &= A_1, \\ B_2 &= \bar{A}_1 A_2, \\ B_3 &= \bar{A}_2 A_3, \\ &\vdots \\ B_i &= \bar{A}_{i-1} A_i, \quad i = 2, 3, 4, \dots \end{aligned}$$

With this construction, we have that $\cup_{i=1}^{\infty} A_i = \cup_{i=1}^{\infty} B_i$. To prove this, let $a \in \cup_{i=1}^{\infty} A_i$. If $a \in A_1$, then $a \in B_1$ and so $a \in \cup_{i=1}^{\infty} B_i$. Otherwise, there exists a *smallest* $k \in \{2, 3, 4, \dots\}$ such that $a \in A_k$, implying that $a \in \bar{A}_j \forall j = 1, 2, \dots, k-1$. Thus, $a \in \bar{A}_{k-1} A_k = B_k$ so that $a \in \cup_{i=1}^{\infty} B_i$. This proves that $\cup_{i=1}^{\infty} A_i \subseteq \cup_{i=1}^{\infty} B_i$. Conversely, suppose that $a \in \cup_{i=1}^{\infty} B_i$, which implies that $a \in B_\ell$ for some $\ell \in \{1, 2, 3, \dots\}$. If $\ell = 1$,

then clearly $a \in A_1$. If $\ell > 1$, then $a \in \bar{A}_{\ell-1}A_\ell$ and so $a \in A_\ell$. In either case, $a \in A_\ell$ so that $a \in \cup_{i=1}^\infty A_i$. This proves that $\cup_{i=1}^\infty B_i \subseteq \cup_{i=1}^\infty A_i$. Therefore, $\cup_{i=1}^\infty A_i = \cup_{i=1}^\infty B_i$.

We now consider

$$\begin{aligned}
P(\cup_{i=1}^\infty A_i) &= P(\cup_{i=1}^\infty B_i) \\
&= \sum_{i=1}^\infty P(B_i) \text{ by the rule of countable additivity} \\
&= \lim_{n \rightarrow \infty} \sum_{i=1}^n P(B_i) \\
&= \lim_{n \rightarrow \infty} \left(P(A_1) + \sum_{i=2}^n P(\bar{A}_{i-1}A_i) \right). \tag{3}
\end{aligned}$$

For $i \geq 2$, note that $A_i = A_{i-1}A_i \cup \bar{A}_{i-1}A_i$. Since $A_{i-1}A_i$ and $\bar{A}_{i-1}A_i$ are mutually exclusive, we have that

$$\begin{aligned}
P(A_i) &= P(A_{i-1}A_i) + P(\bar{A}_{i-1}A_i) \\
P(A_i) &= P(A_{i-1}) + P(\bar{A}_{i-1}A_i) \text{ since } A_{i-1} \subseteq A_i \\
P(\bar{A}_{i-1}A_i) &= P(A_i) - P(A_{i-1}). \tag{4}
\end{aligned}$$

Substituting (4) into (3) yields

$$\begin{aligned}
P(\cup_{i=1}^\infty A_i) &= \lim_{n \rightarrow \infty} \left(P(A_1) + \sum_{i=2}^n (P(A_i) - P(A_{i-1})) \right) \\
&= \lim_{n \rightarrow \infty} \left(P(A_1) + \underbrace{(P(A_2) - P(A_1)) + (P(A_3) - P(A_2)) + \cdots + (P(A_n) - P(A_{n-1}))}_{\text{telescoping series}} \right) \\
&= \lim_{n \rightarrow \infty} P(A_n).
\end{aligned}$$

One equivalent definition of continuity of a real-valued function of a real variable, $f : \mathbb{R} \mapsto \mathbb{R}$, is “sequential continuity”. Specifically, f is said to be continuous at $x \in \mathbb{R}$ if for any sequence x_n such that $\lim_{n \rightarrow \infty} x_n = x$, $\lim_{n \rightarrow \infty} f(x_n) = f(x)$ (i.e., “continuity” means that *the limit of the function is the function of the limit*). Notice that the “continuity” property of P is similar to this for increasing (also decreasing) sequences of sets, since in this case for any n , $\cup_{i=1}^n A_i = A_n$, and so we may think of

$$A = \cup_{i=1}^\infty A_i = \lim_{n \rightarrow \infty} \cup_{i=1}^n A_i = \lim_{n \rightarrow \infty} A_n.$$

2.2.5 (a) We begin by showing that $\cup_{i=1}^{\infty} A_i = \cup_{i=1}^{\infty} B_i$, where we define

$$B_1 = A_1 \quad \text{and} \quad B_i = A_i \cap (\cap_{j=1}^{i-1} \bar{A}_j), \quad i = 2, 3, 4, \dots$$

Let $a \in \cup_{i=1}^{\infty} A_i$. If $a \in A_1$, then $a \in B_1$ and so $a \in \cup_{i=1}^{\infty} B_i$. Otherwise, there exists a *smallest* $m \in \{2, 3, 4, \dots\}$ such that $a \in A_m$, implying that $a \notin A_j \forall j = 1, 2, \dots, m-1$. In other words, $a \in \bar{A}_j \forall j = 1, 2, \dots, m-1$, and so $a \in \cap_{j=1}^{m-1} \bar{A}_j$. As a result, $a \in A_m \cap (\cap_{j=1}^{m-1} \bar{A}_j) = B_m$, and so $a \in \cup_{i=1}^{\infty} B_i$. This proves that $\cup_{i=1}^{\infty} A_i \subseteq \cup_{i=1}^{\infty} B_i$. Conversely, suppose that $a \in \cup_{i=1}^{\infty} B_i$, which implies that $a \in B_n$ for some $n \in \{1, 2, 3, \dots\}$. If $n = 1$, then clearly $a \in A_1$. If $n > 1$, then $a \in A_n \cap (\cap_{j=1}^{n-1} \bar{A}_j)$ and so $a \in A_n$. In either case, $a \in A_n$ so that $a \in \cup_{i=1}^{\infty} A_i$. This proves that $\cup_{i=1}^{\infty} B_i \subseteq \cup_{i=1}^{\infty} A_i$. Therefore, $\cup_{i=1}^{\infty} A_i = \cup_{i=1}^{\infty} B_i$.

Next, consider positive integers k and ℓ such that $k < \ell$. If $k = 1$, note that

$$B_1 \cap B_\ell = A_1 \cap A_\ell \cap (\cap_{j=1}^{\ell-1} \bar{A}_j) = \underbrace{A_1 \cap \bar{A}_1}_{=\emptyset} \cap A_\ell \cap (\cap_{j=2}^{\ell-1} \bar{A}_j) = \emptyset,$$

and if $k > 1$, then

$$\begin{aligned} B_k \cap B_\ell &= A_k \cap (\cap_{j=1}^{k-1} \bar{A}_j) \cap A_\ell \cap (\cap_{j=1}^{\ell-1} \bar{A}_j) \\ &= A_k \cap A_\ell \cap (\cap_{j=1}^{\ell-1} \bar{A}_j) \text{ since } \cap_{j=1}^{\ell-1} \bar{A}_j \subseteq \cap_{j=1}^{k-1} \bar{A}_j \\ &= \underbrace{A_k \cap \bar{A}_k}_{=\emptyset} \cap A_\ell \cap (\cap_{\substack{j=1 \\ j \neq k}}^{\ell-1} \bar{A}_j) \\ &= \emptyset. \end{aligned}$$

By definition, $\{B_i\}_{i=1}^{\infty}$ is a sequence of mutually exclusive events. Therefore, it follows that

$$\begin{aligned} P(\cup_{i=1}^{\infty} A_i) &= P(\cup_{i=1}^{\infty} B_i) \\ &= \sum_{i=1}^{\infty} P(B_i) \text{ by the rule of countable additivity} \\ &= P(A_1) + \sum_{i=2}^{\infty} P(A_i \cap (\cap_{j=1}^{i-1} \bar{A}_j)) \\ &\leq P(A_1) + \sum_{i=2}^{\infty} P(A_i) \text{ since } A_i \cap (\cap_{j=1}^{i-1} \bar{A}_j) \subseteq A_i \\ &= \sum_{i=1}^{\infty} P(A_i). \end{aligned}$$

(b) First of all, we show that

$$\overline{\bigcap_{i=1}^{\infty} A_i} = \bigcup_{i=1}^{\infty} \bar{A}_i. \quad (5)$$

To do so, suppose that $a \in \bigcup_{i=1}^{\infty} \bar{A}_i$. This implies that there exists $n \in \{1, 2, 3, \dots\}$ such that $a \in \bar{A}_n$, or equivalently, $a \notin A_n$. Since $a \notin A_n$, we immediately have that $a \notin \bigcap_{i=1}^{\infty} A_i$, or equivalently, $a \in \overline{\bigcap_{i=1}^{\infty} A_i}$. This proves that $\bigcup_{i=1}^{\infty} \bar{A}_i \subseteq \overline{\bigcap_{i=1}^{\infty} A_i}$. Since the converse inclusion is established by simply reversing the above argument, (5) subsequently follows.

Using (5) and the result of part (a), we have that

$$P\left(\overline{\bigcap_{i=1}^{\infty} A_i}\right) = P\left(\bigcup_{i=1}^{\infty} \bar{A}_i\right) \leq \sum_{i=1}^{\infty} P(\bar{A}_i),$$

or equivalently,

$$1 - P\left(\bigcap_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} P(\bar{A}_i).$$

Therefore, it immediately follows that

$$P\left(\bigcap_{i=1}^{\infty} A_i\right) \geq 1 - \sum_{i=1}^{\infty} P(\bar{A}_i).$$

2.2.6 (a) Note that

$$\begin{aligned} P(A \cup B \cup C) &= P(A) + P(B) + P(C) - P(AB) - P(AC) - P(BC) + P(ABC) \\ &= 0.2 + 0.5 + 0.3 - 0.1 - (P(AC) + P(BC) - P(ABC)) \\ &= 0.9 - (P(AC) + P(BC) - P(AC \cap BC)) \\ &= 0.9 - P(AC \cup BC). \end{aligned}$$

Therefore, it follows that $P(A \cup B \cup C) = 0.9$ is the largest possible value, and this occurs when $P(AC \cup BC) = 0$.

- (b) Assuming each outcome in the sample space has a strictly positive probability, then if $P(AC \cup BC) = 0$, it must be that $AC = \emptyset$ and $BC = \emptyset$, thereby implying that A and C are mutually exclusive (and B and C are mutually exclusive as well).

2.2.7 Let us define the following events:

$$\begin{aligned} A &= \text{“student has math average} > 80\% \text{”}, \\ B &= \text{“student has STAT 230 mark} > 80\% \text{”}, \text{ and} \\ C &= \text{“student has overall average} > 80\% \text{”}. \end{aligned}$$

In terms of these events, we are given the following information:

$$\begin{aligned} P(A) &= 0.22, & P(B) &= 0.24, & P(C) &= 0.2, \\ P(AB) &= 0.14, & P(BC) &= 0.13, & P(ABC) &= 0.1, \\ & & P(\bar{A} \bar{B} \bar{C}) &= 0.67. \end{aligned}$$

We wish to determine $P(A\bar{B}C)$. To do so, we first see that

$$\begin{aligned}
 P(A \cup B \cup C) &= P(A) + P(B) + P(C) - P(AB) - P(AC) - P(BC) + P(ABC) \\
 1 - P(\overline{A \cup B \cup C}) &= 0.22 + 0.24 + 0.2 - 0.14 - P(AC) - 0.13 + 0.1 \\
 1 - P(\bar{A}\bar{B}\bar{C}) &= 0.49 - P(AC) \text{ by De Morgan's Law} \\
 1 - 0.67 &= 0.49 - P(AC) \\
 P(AC) &= 0.16.
 \end{aligned}$$

Since $AC = ABC \cup A\bar{B}C$ where ABC and $A\bar{B}C$ are mutually exclusive, it follows that

$$P(AC) = P(ABC) + P(A\bar{B}C) \implies P(A\bar{B}C) = P(AC) - P(ABC) = 0.16 - 0.1 = 0.06.$$

Section 2.3:

2.3.1 We have that

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(AB) \\ &= P(A) + P(B) - P(A)P(B) \text{ since } A \text{ and } B \text{ are independent} \\ &= 0.3 + 0.2 - (0.3)(0.2) \\ &= 0.44. \end{aligned}$$

2.3.2 In order to show that A is independent of $\bar{B} \cup C$, we consider

$$\begin{aligned} &P(A \cap (\bar{B} \cup C)) \\ &= P(A\bar{B} \cup AC) \\ &= P(A\bar{B}) + P(AC) - P(A\bar{B}C) \\ &= P(A\bar{B}) + P(AC) - (P(AC) - P(ABC)) \\ &= P(A)P(\bar{B}) + P(A)P(BC) \text{ since } A, B, \text{ and } C \text{ are independent} \\ &= P(A)(P(\bar{B}) + P(BC)) \\ &= P(A)(P(\bar{B}) + P(C) - P(\bar{B}C)) \\ &= P(A)P(\bar{B} \cup C). \end{aligned}$$

By definition, A and $\bar{B} \cup C$ are independent events.

2.3.3 First of all, we note that

$$P(E \cap F) = P((A \cup B) \cap (A \cap B)) = P(A \cap B).$$

On the other hand, since E and F are independent events, we also have that

$$P(E \cap F) = P(E)P(F) = P(A \cup B)P(A \cap B).$$

Comparing the above two equations, it must be that

$$P(A \cap B) = P(A \cup B)P(A \cap B),$$

or equivalently,

$$P(AB)(1 - P(A \cup B)) = 0. \tag{6}$$

However, note that

$$1 - P(A \cup B) = P(\overline{A \cup B}) = P(\bar{A} \bar{B}) \text{ by De Morgan's Law.}$$

Substituting the above equation into (6) leads to

$$P(AB)P(\bar{A} \bar{B}) = 0,$$

which immediately implies that either $P(AB) = 0$ or $P(\bar{A} \bar{B}) = 0$.

- 2.3.4 (a) Since the selection process is with replacement, the three digit selections are independent of one another. Therefore, we obtain

$$P(\text{all 3 digits are identical}) = \sum_{i=0}^9 P(\text{all 3 digits are } i\text{'s}) = \sum_{i=0}^9 (0.1)^3 = 10(0.1)^3 = 0.01.$$

- (b) Since there are 5 digits which exceed 4, $P(\text{all 3 digits exceed 4}) = (0.5)^3 = 0.125$.

(c) $P(\text{all 3 digits are different}) = \frac{10^{(3)}}{10^3} = \frac{720}{1000} = \frac{18}{25} = 0.72$.

- (d) Since there are 5 digits which are odd and 5 digits which are even,

$$\begin{aligned} P(\text{all 3 digits have the same parity}) &= P(\text{all 3 digits are odd}) + P(\text{all 3 digits are even}) \\ &= (0.5)^3 + (0.5)^3 \\ &= 0.25. \end{aligned}$$

- (e) Looking at the complementary event, it follows that

$$P(\text{at least 2 of the digits are } > 0) = 1 - P(\text{no digits are } > 0) - P(\text{exactly 1 digit is } > 0).$$

Clearly, $P(\text{no digits are } > 0) = (0.1)^3 = 0.001$. Moreover, we have that

$$P(\text{exactly 1 digit is } > 0) = \sum_{i=1}^3 P(\text{digit } > 0 \text{ occurs in the } i^{\text{th}} \text{ position}) = 3(0.9)(0.1)^2 = 0.027.$$

Therefore, we obtain $P(\text{at least 2 of the digits are } > 0) = 1 - 0.001 - 0.027 = 0.972$.

- 2.3.5 (a) Let A be the event that the number 1 does not occur. Since events determined by different throws of the die are independent of each another, it follows that

$$P(A) = (1 - 0.3)^5 = (0.7)^5 = 0.16807.$$

- (b) If B denotes the event that the number 2 does not occur, then

$$P(B) = (1 - 0.1)^5 = (0.9)^5 = 0.59049.$$

- (c) We wish to find $P(A \cap B) = (1 - 0.3 - 0.1)^5 = (0.6)^5 = 0.07776$.

- (d) We wish to find $P(\bar{A} \cap \bar{B})$. To do so, we observe that

$$\begin{aligned} P(\bar{A} \cap \bar{B}) &= P(\overline{A \cup B}) \text{ by De Morgan's Law} \\ &= 1 - P(A \cup B) \\ &= 1 - (P(A) + P(B) - P(A \cap B)) \\ &= 1 - (0.7)^5 - (0.9)^5 + (0.6)^5 \\ &= 0.3192. \end{aligned}$$

- 2.3.6 (a) For $i = 1, 2, 3, 4, 5$, let A_i = “customer i pays by credit card”. Since customer choices (i.e., to pay by credit card or with cash) are independent of each other, it follows that A_1, A_2, A_3, A_4 , and A_5 are independent events. As a result, the probability that the first three customers pay by credit card would be

$$P(A_1 A_2 A_3 \bar{A}_4 \bar{A}_5) = P(A_1)P(A_2)P(A_3)P(\bar{A}_4)P(\bar{A}_5) = (0.7)^3(0.3)^2.$$

However, this is not the only way three out of the five customers could pay by credit card. There would be $\binom{5}{3} = 10$ ways that this could happen, and so

$$P(3 \text{ out of } 5 \text{ customers pay by credit card}) = 10(0.7)^3(0.3)^2 = 0.3087.$$

- (b) If we wish the 5th customer to be the 3rd one to pay by credit card, then this implies that among the first four customers, exactly two pay by credit card and two pay with cash. This observation leads to

$$\begin{aligned} & P(5^{\text{th}} \text{ customer is } 3^{\text{rd}} \text{ one to pay by credit card}) \\ &= P(A_1 A_2 \bar{A}_3 \bar{A}_4 A_5) + P(A_1 \bar{A}_2 A_3 \bar{A}_4 A_5) + P(A_1 \bar{A}_2 \bar{A}_3 A_4 A_5) \\ &\quad + P(\bar{A}_1 A_2 A_3 \bar{A}_4 A_5) + P(\bar{A}_1 A_2 \bar{A}_3 A_4 A_5) + P(\bar{A}_1 \bar{A}_2 A_3 A_4 A_5) \\ &= 6(0.7)^3(0.3)^2 \\ &= 0.18522. \end{aligned}$$

- 2.3.7 (a) For a given value of i , let A_i denote the event that team A wins the i^{th} game of the series (so that $\bar{A}_i$ denotes the event that team B wins the i^{th} game of the series). These win probabilities vary depending on which team's field the game is played. We assume that the first two games are played on team A 's field, the next three games on team B 's field, and the last two games on team A 's field. Since the results of the games are independent of each other, we immediately obtain

$$\begin{aligned} P(\text{team } A \text{ wins the series in 4 games}) &= P(A_1 A_2 A_3 A_4) \\ &= P(A_1)P(A_2)P(A_3)P(A_4) \\ &= (0.7)^2(0.5)^2 \\ &= 0.1225. \end{aligned}$$

- (b) If team B is to win the series in 5 games, then team A must win exactly one of the first 4 games of the series. This leads to

$$\begin{aligned} & P(\text{team } B \text{ wins the series in 5 games}) \\ &= P(A_1 \bar{A}_2 \bar{A}_3 \bar{A}_4 \bar{A}_5) + P(\bar{A}_1 A_2 \bar{A}_3 \bar{A}_4 \bar{A}_5) + P(\bar{A}_1 \bar{A}_2 A_3 \bar{A}_4 \bar{A}_5) + P(\bar{A}_1 \bar{A}_2 \bar{A}_3 A_4 \bar{A}_5) \\ &= P(A_1)P(\bar{A}_2)P(\bar{A}_3)P(\bar{A}_4)P(\bar{A}_5) + P(\bar{A}_1)P(A_2)P(\bar{A}_3)P(\bar{A}_4)P(\bar{A}_5) \\ &\quad + P(\bar{A}_1)P(\bar{A}_2)P(A_3)P(\bar{A}_4)P(\bar{A}_5) + P(\bar{A}_1)P(\bar{A}_2)P(\bar{A}_3)P(A_4)P(\bar{A}_5) \\ &= (0.7)(0.3)(0.5)^3 + (0.3)(0.7)(0.5)^3 + (0.3)^2(0.5)^3 + (0.3)^2(0.5)^3 \\ &= 2(0.3)(0.7)(0.5)^3 + 2(0.3)^2(0.5)^3 \\ &= 0.075. \end{aligned}$$

(c) We observe that

$$\begin{aligned}
& P(\text{series does not go to 6 games}) \\
&= P(\text{team } A \text{ wins the series in 4 games}) + P(\text{team } B \text{ wins the series in 4 games}) \\
&\quad + P(\text{team } A \text{ wins the series in 5 games}) + P(\text{team } B \text{ wins the series in 5 games}).
\end{aligned}$$

Following the same approach as in parts (a) and (b), we have that

$$P(\text{team } B \text{ wins the series in 4 games}) = P(\bar{A}_1 \bar{A}_2 \bar{A}_3 \bar{A}_4) = (0.3)^2 (0.5)^2 = 0.0225$$

and

$$\begin{aligned}
& P(\text{team } A \text{ wins the series in 5 games}) \\
&= P(\bar{A}_1 A_2 A_3 A_4 A_5) + P(A_1 \bar{A}_2 A_3 A_4 A_5) + P(A_1 A_2 \bar{A}_3 A_4 A_5) + P(A_1 A_2 A_3 \bar{A}_4 A_5) \\
&= (0.3)(0.7)(0.5)^3 + (0.7)(0.3)(0.5)^3 + (0.7)^2(0.5)^3 + (0.7)^2(0.5)^3 \\
&= 2(0.3)(0.7)(0.5)^3 + 2(0.7)^2(0.5)^3 \\
&= 0.175.
\end{aligned}$$

Therefore, we get

$$P(\text{series does not go to 6 games}) = 0.1225 + 0.0225 + 0.175 + 0.075 = 0.395.$$

2.3.8 Based on the given information, we deduce the following:

$$P(A) = \frac{F}{F+M}, \quad P(B) = \frac{f+m}{F+M}, \quad \text{and} \quad P(AB) = \frac{f}{F+M}.$$

A and B are independent events if and only if

$$\begin{aligned}
P(AB) &= P(A)P(B) \\
\frac{f}{F+M} &= \left(\frac{F}{F+M} \right) \left(\frac{f+m}{F+M} \right) \\
f(F+M) &= F(f+m) \\
fF + fM &= fF + mF \\
\frac{f}{F} &= \frac{m}{M}.
\end{aligned}$$

Section 2.4:

- 2.4.1 Since $A = (A \cap B) \cup (A \cap \bar{B})$ where $A \cap B$ and $A \cap \bar{B}$ are mutually exclusive, it follows that

$$\begin{aligned}
 P(A|\bar{B}) &= \frac{P(A \cap \bar{B})}{P(\bar{B})} \\
 &= \frac{P(A) - P(A \cap B)}{1 - P(B)} \\
 &= \frac{0.3 - P(B)P(A|B)}{1 - 0.4} \\
 &= \frac{0.3 - (0.4)(0.5)}{0.6} \\
 &= \frac{1}{6}.
 \end{aligned}$$

- 2.4.2 We define the events H = “day has a temperature $> 22^\circ\text{C}$ ” and R = “day has rain”. We are given that $P(H) = 0.2$ (so that $P(\bar{H}) = 0.8$), $P(\bar{R}|H) = 0.4$, and $P(\bar{R}|\bar{H}) = 0.7$. We simply wish to calculate

$$P(\bar{H} \cap R) = P(\bar{H})P(R|\bar{H}) = P(\bar{H})(1 - P(\bar{R}|\bar{H})) = (0.8)(0.3) = 0.24.$$

- 2.4.3 We define the events B = “bicycle is taken to work” and L = “arrive late to work”. We are given that $P(B) = 0.7$ (so that $P(\bar{B}) = 0.3$), $P(L|B) = 0.1$, and $P(L|\bar{B}) = 0.2$. We wish to calculate

$$\begin{aligned}
 P(\bar{B}|L) &= \frac{P(\bar{B})P(L|\bar{B})}{P(B)P(L|B) + P(\bar{B})P(L|\bar{B})} \text{ by Bayes' Rule} \\
 &= \frac{(0.3)(0.2)}{(0.7)(0.1) + (0.3)(0.2)} \\
 &= \frac{0.06}{0.13} \\
 &= \frac{6}{13}.
 \end{aligned}$$

- 2.4.4 For $i = 1, 2, 3, 4, 5, 6$, let A_i = “heads appears on the i^{th} coin toss”. Also, we define the event F = “coin picked is a fair one”. Based on the given information, we deduce that $P(F) = 0.75$. If we let B be the event of getting 5 heads in 6 tosses of a coin, it follows that (since coin tosses are independent regardless of which coin is being tossed)

$$\begin{aligned}
 P(B|F) &= P(A_1A_2A_3A_4A_5\bar{A}_6|F) + P(A_1A_2A_3A_4\bar{A}_5A_6|F) + P(A_1A_2A_3\bar{A}_4A_5A_6|F) \\
 &\quad + P(A_1A_2\bar{A}_3A_4A_5A_6|F) + P(A_1\bar{A}_2A_3A_4A_5A_6|F) + P(\bar{A}_1A_2A_3A_4A_5A_6|F) \\
 &= 6(0.5)^6.
 \end{aligned}$$

In a similar fashion, we would immediately obtain that $P(B|\bar{F}) = 6(0.2)(0.8)^5$. Therefore, applying Bayes' Rule, we get

$$\begin{aligned}
 P(F|B) &= \frac{P(F)P(B|F)}{P(F)P(B|F) + P(\bar{F})P(B|\bar{F})} \\
 &= \frac{0.75 \cdot 6(0.5)^6}{0.75 \cdot 6(0.5)^6 + 0.25 \cdot 6(0.2)(0.8)^5} \\
 &= \frac{\frac{3}{256}}{\frac{3}{256} + \frac{256}{15625}} \\
 &= \frac{46875}{112411} \\
 &\approx 0.416997.
 \end{aligned}$$

2.4.5 We define the events D = “car has defective headlights” and F = “car has a faulty muffler”. We are given that $P(D \cap F) = 0.1$ and $P(D \cap \bar{F}) = 0.15$. Note that $D = (D \cap F) \cup (D \cap \bar{F})$ where $D \cap F$ and $D \cap \bar{F}$ are mutually exclusive, and so

$$P(F|D) = \frac{P(D \cap F)}{P(D)} = \frac{P(D \cap F)}{P(D \cap F) + P(D \cap \bar{F})} = \frac{0.1}{0.1 + 0.15} = \frac{2}{5}.$$

2.4.6 We define the events F = “student speaks French” and W = “student is a woman”. We are given that $P(F) = 0.15$, $P(W) = 0.45$, and $P(F|W) = 0.2$. Using the product rule for two events, we obtain $P(F \cap W) = P(W)P(F|W) = (0.45)(0.2) = 0.09$. As a result, we have that

$$\begin{aligned}
 P(\bar{F} \cap \bar{W}) &= P(\overline{F \cup W}) \text{ by De Morgan's Law} \\
 &= 1 - P(F \cup W) \\
 &= 1 - (P(F) + P(W) - P(F \cap W)) \\
 &= 1 - 0.15 - 0.45 + 0.09 \\
 &= 0.49.
 \end{aligned}$$

Let us next define the events

A = “at least one French speaking student is on the committee”

and

B = “at least one woman student is on the committee”.

We wish to determine $P(A \cap B)$ which is given by

$$P(A \cap B) = 1 - P(\overline{A \cap B}) = 1 - P(\bar{A} \cup \bar{B}) = 1 - (P(\bar{A}) + P(\bar{B}) - P(\bar{A} \cap \bar{B})).$$

Since we are assuming that the selection process is done with replacement so that the 10 student selections are independent of one another, we obtain

$$\begin{aligned} P(\bar{A}) &= P(\text{no French speaking student is on the committee}) = P(\bar{F})^{10} = (0.85)^{10}, \\ P(\bar{B}) &= P(\text{no woman student is on the committee}) = P(\bar{W})^{10} = (0.55)^{10}, \text{ and} \\ P(\bar{A} \cap \bar{B}) &= P(\text{student on the committee is neither French speaking nor a woman}) \\ &= P(\bar{F} \cap \bar{W})^{10} \\ &= (0.49)^{10}. \end{aligned}$$

Therefore, we get

$$P(A \cap B) = 1 - (0.85)^{10} - (0.55)^{10} + (0.49)^{10} \approx 0.801391.$$

- 2.4.7 Let D be the event representing that a person is susceptible for a disease. Let A , B , and C represent the three genetic types that a person in this population could have. The probability that a randomly selected person from this population is susceptible for the disease is given by

$$\begin{aligned} P(D) &= P(A)P(D|A) + P(B)P(D|B) + P(C)P(D|C) \text{ by the Law of Total Probability} \\ &= (0.3)(0.05) + (0.6)(0.04) + (0.1)(0.02) \\ &= 0.041. \end{aligned}$$

Therefore, $P(\bar{D}) = 1 - 0.041 = 0.959$. Since persons are independently selected from this very large population, we have that

$$\begin{aligned} &P(\text{at least one of the 9 is susceptible for the disease}) \\ &= 1 - P(\text{none of the 9 is susceptible for the disease}) \\ &= 1 - (0.959)^9 \\ &\approx 0.313932. \end{aligned}$$

- 2.4.8 Let A , B , and C denote the events that the test question is answered correctly by Abby, Barry, and Cindy, respectively. We also define the event $D =$ “two of the three students get the correct answer”. Since students answer the test question independently of one another, we have that

$$P(D) = P(AB\bar{C}) + P(A\bar{B}C) + P(\bar{A}BC) = (0.9)(0.7)(0.6) + (0.9)(0.3)(0.4) + (0.1)(0.7)(0.4) = 0.514.$$

Therefore, it follows that

$$P(\bar{C}|D) = \frac{P(\bar{C} \cap D)}{P(D)} = \frac{P(AB\bar{C})}{P(D)} = \frac{(0.9)(0.7)(0.6)}{0.514} = \frac{189}{257}.$$

- 2.4.9 (a) Let A and B denote the events that the student gets test question A and B, respectively. Moreover, let Y be the event that the student answers “yes” to the question. Based on the given information, we obtain via the Law of Total Probability:

$$P(Y) = P(A)P(Y|A) + P(B)P(Y|B) = \frac{20}{100} \cdot \frac{1}{6} + \frac{80}{100} \cdot p = \frac{1}{30} + \frac{4p}{5} = \frac{24p+1}{30}.$$

- (b) If x of n students answer “yes”, then to get an estimate for p , we would naturally set

$$P(Y) = \frac{x}{n} \implies \frac{24p+1}{30} = \frac{x}{n} \implies p = \frac{\frac{30x}{n} - 1}{24} = \frac{30x - n}{24n}.$$

- (c) We wish to calculate

$$P(B|Y) = \frac{P(B \cap Y)}{P(Y)} = \frac{P(B)P(Y|B)}{P(Y)} = \frac{\frac{4p}{5}}{\frac{24p+1}{30}} = \frac{24p}{24p+1}.$$

- 2.4.10 (a) Since the three wheels spin independently and each has probability 0.1 of landing at any position, it follows that

$$P(\text{all three positions show a flower}) = 0.2 \cdot 0.6 \cdot 0.2 = 0.024.$$

- (b) Suppose that there are $m \geq 1$ flowers on wheel 1, $n \geq 1$ flowers on wheel 2, and $10 - m - n \geq 1$ flowers on wheel 3. The probability that all three positions show a flower is given by

$$0.1m \cdot 0.1n \cdot 0.1(10 - m - n) = \frac{mn(10 - m - n)}{1000}.$$

Let $f(m, n) = mn(10 - m - n)$. We wish to minimize $f(m, n)$ subject to the restrictions $m \geq 1$, $n \geq 1$, and $10 - m - n \geq 1$. For each value of m , $f(m, n) = mn(10 - m - n)$ is a quadratic function of n which is minimized for $n = 1$ or $n = 9 - m$. Note that

$$f(m, 1) = f(m, 9 - m) = m(9 - m),$$

and this function is minimized for $m = 1$ or $m = 8$. This leads to observing that

$$f(1, 1) = f(1, 8) = f(8, 1) = 8.$$

Hence, the values of $(m, n, 10 - m - n)$ which minimize the probability that all three positions show a flower are $(1, 1, 8)$, $(1, 8, 1)$, and $(8, 1, 1)$.

2.4.11 (a) Using the Law of Total Probability, we wish to calculate

$$P(A) = P(\text{Spam})P(A|\text{Spam}) + P(\text{Not Spam})P(A|\text{Not Spam}) = (0.5)(0.2) + (0.5)(0.001) = 0.1005.$$

(b) Using Bayes' Rule, we obtain

$$P(\text{Spam}|A) = \frac{P(\text{Spam})P(A|\text{Spam})}{P(A)} = \frac{(0.5)(0.2)}{0.1005} = \frac{200}{201}.$$

Therefore, it immediately follows that $P(\text{Not Spam}|A) = 1 - P(\text{Spam}|A) = \frac{1}{201}$.

(c) Since any email message containing the word Viagra is declared as spam, it immediately follows that

$$P(\text{declare message as spam}|\text{Spam}) = P(A|\text{Spam}) = 0.2, \text{ as given.}$$

2.4.12 (a) Note that $A_1A_2A_3$ denotes the event that a message has all three features. Using the Law of Total Probability, we wish to calculate

$$\begin{aligned} & P(A_1A_2A_3) \\ &= P(\text{Spam})P(A_1A_2A_3|\text{Spam}) + P(\text{Not Spam})P(A_1A_2A_3|\text{Not Spam}) \\ &= 0.5P(A_1|\text{Spam})P(A_2|\text{Spam})P(A_3|\text{Spam}) + 0.5P(A_1|\text{Not Spam})P(A_2|\text{Not Spam})P(A_3|\text{Not Spam}) \\ & \quad \text{due to the conditional independence of } A_1, A_2, \text{ and } A_3 \\ &= (0.5)(0.2)(0.1)(0.1) + (0.5)(0.005)(0.004)(0.005) \\ &= 0.00100005. \end{aligned}$$

(b) Using Bayes' Rule, we obtain

$$P(\text{Spam}|A_1A_2A_3) = \frac{P(\text{Spam})P(A_1A_2A_3|\text{Spam})}{P(A_1A_2A_3)} = \frac{(0.5)(0.2)(0.1)(0.1)}{0.00100005} = \frac{20000}{20001}.$$

(c) Using Bayes' Rule, we obtain

$$\begin{aligned} P(\text{Spam}|A_1A_2\bar{A}_3) &= \frac{P(\text{Spam})P(A_1A_2\bar{A}_3|\text{Spam})}{P(\text{Spam})P(A_1A_2\bar{A}_3|\text{Spam}) + P(\text{Not Spam})P(A_1A_2\bar{A}_3|\text{Not Spam})} \\ &= \frac{(0.5)(0.2)(0.1)(0.9)}{(0.5)(0.2)(0.1)(0.9) + (0.5)(0.005)(0.004)(0.995)} \\ &= \frac{0.009}{0.00900995} \\ &= \frac{180000}{180199}. \end{aligned}$$

- (d) If any email message with at least one of the features present is declared as spam, then we would obtain

$$\begin{aligned}
P(\text{declare message as spam}|\text{Spam}) &= P(A_1 \cup A_2 \cup A_3|\text{Spam}) \\
&= P(\overline{\bar{A}_1 \cap \bar{A}_2 \cap \bar{A}_3}|\text{Spam}) \text{ by De Morgan's Law} \\
&= 1 - P(\bar{A}_1 \cap \bar{A}_2 \cap \bar{A}_3|\text{Spam}) \\
&= 1 - P(\bar{A}_1|\text{Spam})P(\bar{A}_2|\text{Spam})P(\bar{A}_3|\text{Spam}) \\
&= 1 - (0.8)(0.9)(0.9) \\
&= 0.352.
\end{aligned}$$

- (e) We first calculate

$$\begin{aligned}
P(\text{declare message as spam}|\text{Not Spam}) &= P(A_1 \cup A_2 \cup A_3|\text{Not Spam}) \\
&= P(\overline{\bar{A}_1 \cap \bar{A}_2 \cap \bar{A}_3}|\text{Not Spam}) \text{ by De Morgan's Law} \\
&= 1 - P(\bar{A}_1 \cap \bar{A}_2 \cap \bar{A}_3|\text{Not Spam}) \\
&= 1 - P(\bar{A}_1|\text{Not Spam})P(\bar{A}_2|\text{Not Spam})P(\bar{A}_3|\text{Not Spam}) \\
&= 1 - (0.995)(0.996)(0.995) \\
&= 0.0139351.
\end{aligned}$$

Using the Law of Total Probability, we next obtain

$$\begin{aligned}
&P(\text{declare message as spam}) \\
&= P(\text{Spam})P(\text{declare message as spam}|\text{Spam}) \\
&\quad + P(\text{Not Spam})P(\text{declare message as spam}|\text{Not Spam}) \\
&= (0.5)(0.352) + (0.5)(0.0139351) \\
&= 0.18296755.
\end{aligned}$$

Therefore, applying Bayes' Rule, we get

$$\begin{aligned}
P(\text{Spam}|\text{declare message as spam}) &= \frac{P(\text{Spam})P(\text{declare message as spam}|\text{Spam})}{P(\text{declare message as spam})} \\
&= \frac{(0.5)(0.352)}{0.18296755} \\
&= \frac{0.176}{0.18296755} \\
&\approx 0.961919.
\end{aligned}$$

(f) Using the same rule as in part (d), we now wish to calculate

$$\begin{aligned}
P(A_1|\text{declare message as spam}) &= \frac{P(A_1)P(\text{declare message as spam}|A_1)}{P(\text{declare message as spam})} \\
&= \frac{P(A_1)P(A_1 \cup A_2 \cup A_3|A_1)}{P(\text{declare message as spam})} \\
&= \frac{P(A_1)}{0.18296755} \text{ since } P(A_1 \cup A_2 \cup A_3|A_1) = 1 \\
&= \frac{P(\text{Spam})P(A_1|\text{Spam}) + P(\text{Not Spam})P(A_1|\text{Not Spam})}{0.18296755} \\
&= \frac{(0.5)(0.2) + (0.5)(0.005)}{0.18296755} \\
&= \frac{0.1025}{0.18296755} \\
&\approx 0.560209.
\end{aligned}$$

Solutions to End of Section Chapter 3 Problems

Section 3.1:

3.1.1 (a) Note that $A = \{0, 1, 2\}$. To determine the value of c , we make use of the condition that

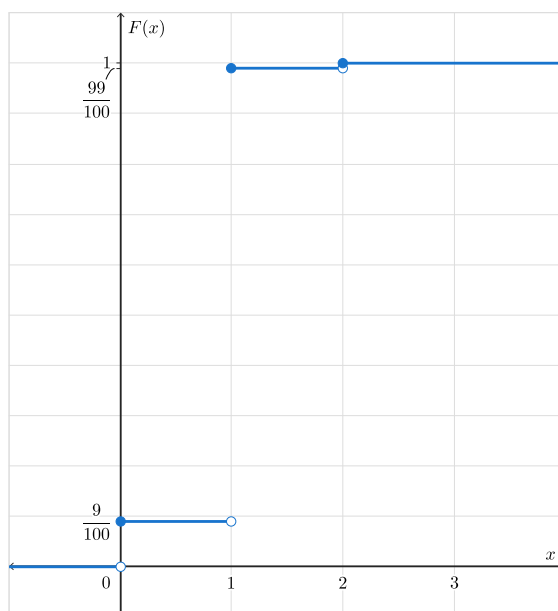
$$\begin{aligned}\sum_{x \in A} f(x) &= 1 \\ 9c^2 + 9c + c^2 &= 1 \\ 10c^2 + 9c - 1 &= 0 \\ (10c - 1)(c + 1) &= 0 \\ c &= \frac{1}{10} \text{ or } -1.\end{aligned}$$

However, $c \neq -1$ since $f(x) \geq 0$ for all $x \in A$. Hence, we must have that $c = \frac{1}{10}$.

(b) Using the given table for the pmf with $c = \frac{1}{10}$, we can readily construct a similar table for the cdf:

x	0	1	2
$F(x)$	$\frac{9}{100}$	$\frac{99}{100}$	1

Thus, the plot of $F(x)$ is as follows:



3.1.2 (a) We have that $A = \{0, 1, 2, 3, 4, 5\}$. To determine the value of k , we make use of the condition that

$$1 = \lim_{x \rightarrow \infty} F(x) = F(5) \implies 4k^2 = 1 \implies k = \pm \frac{1}{2}.$$

However, $k \neq -\frac{1}{2}$ since $F(x) \geq 0$ for all $x \in \mathbb{R}$. Hence, we must have that $k = \frac{1}{2}$.

(b) With $k = \frac{1}{2}$, the given table for the cdf becomes:

x	1	2	3	4	5
$F(x)$	0.05	0.2	0.25	0.5	1

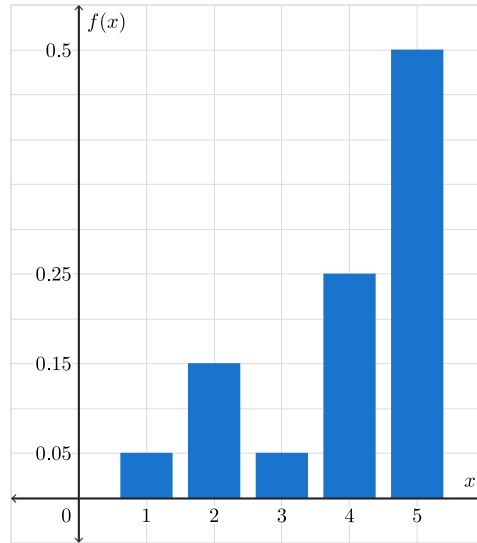
Therefore, we wish to calculate

$$P(2 < X \leq 4) = P(X \leq 4) - P(X \leq 2) = F(4) - F(2) = 0.5 - 0.2 = 0.3.$$

(c) Using the above table for the cdf, we can readily determine the pmf of X :

$$\begin{aligned} f(1) &= F(1) = 0.05, \\ f(2) &= F(2) - F(1) = 0.2 - 0.05 = 0.15, \\ f(3) &= F(3) - F(2) = 0.25 - 0.2 = 0.05, \\ f(4) &= F(4) - F(3) = 0.5 - 0.25 = 0.25, \text{ and} \\ f(5) &= F(5) - F(4) = 1 - 0.5 = 0.5. \end{aligned}$$

Thus, the plot of $f(x)$ is as follows:



3.1.3 (a) We wish to calculate

$$P(X \geq 5) = 1 - P(X < 5) = 1 - P(X \leq 4) = 1 - F(4) = 1 - (1 - 2^{-4}) = \frac{1}{16}.$$

(b) With $A = \{1, 2, 3, \dots\}$, it follows that

$$f(1) = F(1) = 1 - 2^{-1} = \frac{1}{2}$$

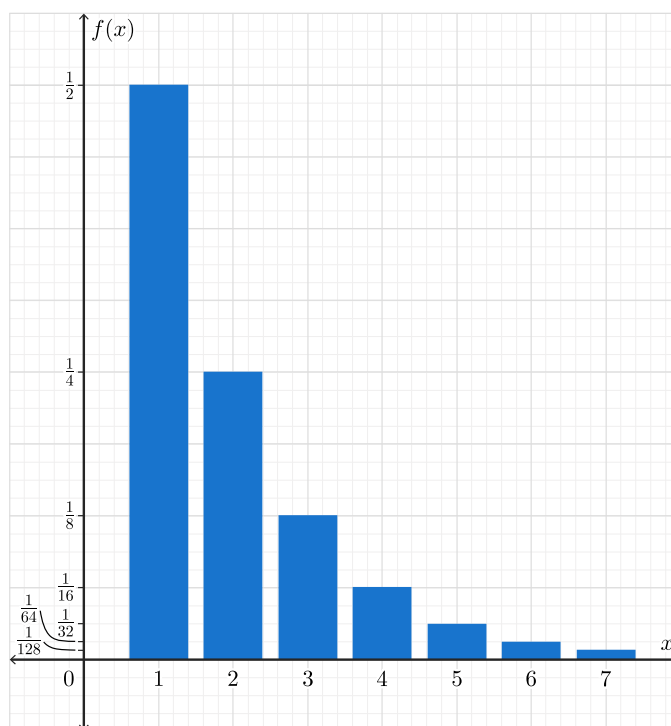
and

$$\begin{aligned} f(x) &= F(x) - F(x-1) \\ &= (1 - 2^{-x}) - (1 - 2^{-(x-1)}) \\ &= \left(\frac{1}{2}\right)^{x-1} - \left(\frac{1}{2}\right)^x \\ &= \left(\frac{1}{2}\right)^{x-1} \left(1 - \frac{1}{2}\right) \\ &= \left(\frac{1}{2}\right)^x \text{ for } x = 2, 3, 4, \dots \end{aligned}$$

Therefore, we ultimately obtain

$$f(x) = \left(\frac{1}{2}\right)^x \text{ for } x = 1, 2, 3, \dots,$$

and its histogram looks like:



(c) Note that

$$\sum_{x=1}^{\infty} f(x) = \underbrace{\sum_{x=1}^{\infty} \left(\frac{1}{2}\right)^x}_{\text{infinite geometric series}} = \frac{\frac{1}{2}}{1 - \frac{1}{2}} = 1.$$

3.1.4 A possible sample space to use for this experiment is

$$S = \{(1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (1, 7), (1, 8), (2, 3), (2, 4), (2, 5), (2, 6), (2, 7), (2, 8), (3, 4), (3, 5), (3, 6), (3, 7), (3, 8), (4, 5), (4, 6), (4, 7), (4, 8), (5, 6), (5, 7), (5, 8), (6, 7), (6, 8), (7, 8)\},$$

in which outcome (i, j) represents the position in line of myself (position i) and of my friend (position j). Note that S contains $\binom{8}{2} = 28$ equally likely outcomes. The following table shows how each numerical value of X , the number of people standing between myself and my friend, is matched to the outcomes in S :

Outcomes	Definition of the event
$X = 0$	$\{(1, 2), (2, 3), (3, 4), (4, 5), (5, 6), (6, 7), (7, 8)\}$
$X = 1$	$\{(1, 3), (2, 4), (3, 5), (4, 6), (5, 7), (6, 8)\}$
$X = 2$	$\{(1, 4), (2, 5), (3, 6), (4, 7), (5, 8)\}$
$X = 3$	$\{(1, 5), (2, 6), (3, 7), (4, 8)\}$
$X = 4$	$\{(1, 6), (2, 7), (3, 8)\}$
$X = 5$	$\{(1, 7), (2, 8)\}$
$X = 6$	$\{(1, 8)\}$

With the use of the above table, the pmf of X is easily calculated:

$$\begin{aligned} f(0) &= P(X = 0) = P(\{(1, 2), (2, 3), (3, 4), (4, 5), (5, 6), (6, 7), (7, 8)\}) = \frac{7}{28} = \frac{1}{4}, \\ f(1) &= P(X = 1) = P(\{(1, 3), (2, 4), (3, 5), (4, 6), (5, 7), (6, 8)\}) = \frac{6}{28} = \frac{3}{14}, \\ f(2) &= P(X = 2) = P(\{(1, 4), (2, 5), (3, 6), (4, 7), (5, 8)\}) = \frac{5}{28}, \\ f(3) &= P(X = 3) = P(\{(1, 5), (2, 6), (3, 7), (4, 8)\}) = \frac{4}{28} = \frac{1}{7}, \\ f(4) &= P(X = 4) = P(\{(1, 6), (2, 7), (3, 8)\}) = \frac{3}{28}, \\ f(5) &= P(X = 5) = P(\{(1, 7), (2, 8)\}) = \frac{2}{28} = \frac{1}{14}, \\ f(6) &= P(X = 6) = P(\{(1, 8)\}) = \frac{1}{28}. \end{aligned}$$

Note that we can succinctly represent the pmf of X as follows:

$$f(x) = \frac{7-x}{28} \text{ for } x = 0, 1, 2, 3, 4, 5, 6.$$

Therefore, for $x \in \{0, 1, 2, 3, 4, 5, 6\}$, the cdf of X is given by

$$\begin{aligned}
 F(x) &= \sum_{u \leq x} f(u) \\
 &= \sum_{u=0}^x \frac{7-u}{28} \\
 &= \frac{7(x+1)}{28} - \frac{1}{28} \sum_{u=0}^x u \\
 &= \frac{x+1}{4} - \frac{1}{28} \cdot \frac{x(x+1)}{2} \\
 &= \frac{14(x+1) - x(x+1)}{56} \\
 &= \frac{(x+1)(14-x)}{56}.
 \end{aligned}$$

- 3.1.5 (a) If the draws are made without replacement, then a possible sample space to use for this experiment is

$$\begin{aligned}
 S = \{ &(1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (1, 7), (2, 3), (2, 4), (2, 5), (2, 6), (2, 7), \\
 &(3, 4), (3, 5), (3, 6), (3, 7), (4, 5), (4, 6), (4, 7), (5, 6), (5, 7), (6, 7) \},
 \end{aligned}$$

consisting of the possible (unordered) pairs of size 2 chosen from the 7 numbered balls. In other words, S contains $\binom{7}{2} = 21$ equally likely outcomes. The following table shows how each numerical value of X is matched to the outcomes in S :

<u>Outcomes</u>	<u>Definition of the event</u>
$X = 1$	$\{(1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (1, 7)\}$
$X = 2$	$\{(2, 3), (2, 4), (2, 5), (2, 6), (2, 7)\}$
$X = 3$	$\{(3, 4), (3, 5), (3, 6), (3, 7)\}$
$X = 4$	$\{(4, 5), (4, 6), (4, 7)\}$
$X = 5$	$\{(5, 6), (5, 7)\}$
$X = 6$	$\{(6, 7)\}$

Counting the number of outcomes in each of the above events, we obtain the following pmf of X :

x	1	2	3	4	5	6
$f_X(x)$	$\frac{2}{7}$	$\frac{5}{21}$	$\frac{4}{21}$	$\frac{1}{7}$	$\frac{2}{21}$	$\frac{1}{21}$

In a similar fashion, the following table shows how each numerical value of Y is matched to the outcomes in S :

<u>Outcomes</u>	<u>Definition of the event</u>
$Y = 3$	$\{(1, 2)\}$
$Y = 4$	$\{(1, 3)\}$
$Y = 5$	$\{(1, 4), (2, 3)\}$
$Y = 6$	$\{(1, 5), (2, 4)\}$
$Y = 7$	$\{(1, 6), (2, 5), (3, 4)\}$
$Y = 8$	$\{(1, 7), (2, 6), (3, 5)\}$
$Y = 9$	$\{(2, 7), (3, 6), (4, 5)\}$
$Y = 10$	$\{(3, 7), (4, 6)\}$
$Y = 11$	$\{(4, 7), (5, 6)\}$
$Y = 12$	$\{(5, 7)\}$
$Y = 13$	$\{(6, 7)\}$

This immediately leads to the following pmf of Y :

y	3	4	5	6	7	8	9	10	11	12	13
$f_Y(y)$	$\frac{1}{21}$	$\frac{1}{21}$	$\frac{2}{21}$	$\frac{2}{21}$	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{2}{21}$	$\frac{2}{21}$	$\frac{1}{21}$	$\frac{1}{21}$

(b) If the draws are made with replacement, then

$$S = \{(x, y) : x = 1, 2, \dots, 7 \text{ and } y = 1, 2, \dots, 7\}$$

is a possible sample space (consisting of $7^2 = 49$ outcomes) to use for this experiment. To obtain $f_X(x)$, we first remark that $A = \{1, 2, 3, 4, 5, 6, 7\}$ represents the range of X . For $x \in A$, note that

$$\begin{aligned}
& f_X(x) \\
&= P(x \text{ is the smaller of the numbers on the two drawn balls}) \\
&= P(x \text{ appears on both drawn balls}) \\
&\quad + 2P(x \text{ appears on 1}^{\text{st}} \text{ drawn ball and a number } > x \text{ appears on 2}^{\text{nd}} \text{ drawn ball}) \\
&= \frac{1}{49} + 2 \cdot \frac{1 \times (7 - x)}{49} \text{ since there are } 7 - x \text{ numbers greater than } x \text{ to choose from} \\
&= \frac{15 - 2x}{49}.
\end{aligned}$$

Moving next to the distribution of Y , we have the following table indicating how each numerical value of Y is matched to the outcomes in S :

<u>Outcomes</u>	<u>Definition of the event</u>
$Y = 2$	$\{(1, 1)\}$
$Y = 3$	$\{(1, 2), (2, 1)\}$
$Y = 4$	$\{(1, 3), (2, 2), (3, 1)\}$
$Y = 5$	$\{(1, 4), (2, 3), (3, 2), (4, 1)\}$
$Y = 6$	$\{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\}$
$Y = 7$	$\{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$
$Y = 8$	$\{(1, 7), (2, 6), (3, 5), (4, 4), (5, 3), (6, 2), (7, 1)\}$
$Y = 9$	$\{(2, 7), (3, 6), (4, 5), (5, 4), (6, 3), (7, 2)\}$
$Y = 10$	$\{(3, 7), (4, 6), (5, 5), (6, 4), (7, 3)\}$
$Y = 11$	$\{(4, 7), (5, 6), (6, 5), (7, 4)\}$
$Y = 12$	$\{(5, 7), (6, 6), (7, 5)\}$
$Y = 13$	$\{(6, 7), (7, 6)\}$
$Y = 14$	$\{(7, 7)\}$

This immediately leads to the following pmf of Y :

y	2	3	4	5	6	7	8	9	10	11	12	13	14
$f_Y(y)$	$\frac{1}{49}$	$\frac{2}{49}$	$\frac{3}{49}$	$\frac{4}{49}$	$\frac{5}{49}$	$\frac{6}{49}$	$\frac{1}{7}$	$\frac{6}{49}$	$\frac{5}{49}$	$\frac{4}{49}$	$\frac{3}{49}$	$\frac{2}{49}$	$\frac{1}{49}$

3.1.6 First of all, we observe that the smallest value that X can take on is 0, resulting when the first 8 selections from the bin are all sixty-watt bulbs. On the other hand, the largest value that X can take on is 35, which clearly arises if all 35 forty-watt bulbs are selected before any sixty-watt bulb is even picked. Therefore, the range of X is $A = \{0, 1, \dots, 35\}$.

Next, assume that $x \in A$. If x forty-watt bulbs were drawn from the bin, then a total of $x + 8$ bulbs must have been selected, and the last selection must necessarily have been the 8th sixty-watt bulb (of which there would be $70 - 7 = 63$ to choose from at that point). In other words, in the first $x + 7$ selections taken from the bin containing a total of $35 + 70 = 105$ bulbs, we must have 7 sixty-watt bulbs and x forty-watt bulbs. Thus, it follows that

$$\begin{aligned}
f(x) &= P(x \text{ forty-watt bulbs drawn from the bin}) \\
&= \frac{\binom{35}{x} \binom{70}{7}}{\binom{105}{x+7}} \cdot \frac{63}{105 - (x + 7)} \\
&= \frac{\binom{35}{x} \binom{70}{7}}{\binom{105}{x+7}} \cdot \frac{63}{98 - x} \text{ for } x = 0, 1, \dots, 35.
\end{aligned}$$

Section 3.2:

3.2.1 (a) Note that $A = \{1, 2, \dots, 9\}$. To determine the value of c , we make use of the condition that

$$\begin{aligned}\sum_{x \in A} f_X(x) &= 1 \\ c \sum_{x=1}^9 x^2 &= 1 \\ c \frac{9(10)[2(9) + 1]}{6} &= 1 \quad \text{since} \quad \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6} \quad \text{for positive integer } n \\ 285c &= 1 \\ c &= \frac{1}{285}.\end{aligned}$$

(b) Using the given pmf with $c = \frac{1}{285}$, the following table provides a helpful summary of how the values of X map to the random variable $Y = 6 \pmod{X}$:

x	$y = 6 \pmod{x}$	$f_X(x)$
1	0	$\frac{1}{285}$
2	0	$\frac{4}{285}$
3	0	$\frac{3}{95}$
4	2	$\frac{16}{285}$
5	1	$\frac{5}{57}$
6	0	$\frac{12}{95}$
7	6	$\frac{49}{285}$
8	6	$\frac{64}{285}$
9	6	$\frac{27}{95}$

With the use of the above table, the pmf of Y is readily calculated as follows:

$$\begin{aligned}f_Y(0) &= f_X(1) + f_X(2) + f_X(3) + f_X(6) = \frac{1}{285} + \frac{4}{285} + \frac{3}{95} + \frac{12}{95} = \frac{50}{285} = \frac{10}{57}, \\ f_Y(1) &= f_X(5) = \frac{5}{57}, \\ f_Y(2) &= f_X(4) = \frac{16}{285}, \\ f_Y(6) &= f_X(7) + f_X(8) + f_X(9) = \frac{49}{285} + \frac{64}{285} + \frac{27}{95} = \frac{194}{285}.\end{aligned}$$

3.2.2 First of all, the range of the random variable Y is given by

$$A_Y = \left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\right\}.$$

Therefore, for $y \in A_Y$, the cdf of Y is given by

$$\begin{aligned} F_Y(y) &= P(Y \leq y) \\ &= P\left(\frac{1}{X} \leq y\right) \\ &= P\left(X \geq \frac{1}{y}\right) \\ &= \sum_{x=1/y}^{\infty} f_X(x) \\ &= \underbrace{\sum_{x=1/y}^{\infty} \left(\frac{1}{2}\right)^x}_{\text{infinite geometric series}} \\ &= \frac{\left(\frac{1}{2}\right)^{\frac{1}{y}}}{1 - \frac{1}{2}} \\ &= \left(\frac{1}{2}\right)^{\frac{1}{y}-1}. \end{aligned}$$

3.2.3 (a) First of all, X clearly has range $A = \{0, 1, 2, 3, 4\}$. Coin tosses are treated independently of each other, with a probability of $\frac{1}{2}$ that a toss is heads and a probability of $\frac{1}{2}$ that a toss is tails. Therefore, for $x \in A$, we obtain the following pmf of X :

$$f_X(x) = P(x \text{ tails are obtained}) = \binom{4}{x} \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{4-x} = \frac{\binom{4}{x}}{16}.$$

(b) If X represents the number of tails in 4 tosses of the fair coin, then $4 - X$ represents the number of heads obtained. Therefore, the random variable Y is given by

$$Y = X - (4 - X) = 2X - 4.$$

Since Y is a linear function of X , it immediately follows that the pmf of Y is given by

$$f_Y(y) = f_X\left(\frac{y+4}{2}\right) = \frac{\binom{4}{\frac{y+4}{2}}}{16} \text{ for } y = -4, -2, 0, 2, 4.$$

- 3.2.4 (a) First of all, the number of face cards chosen, X , clearly has range $A_X = \{0, 1, 2, 3\}$. For each $x \in A$, we calculate:

$$f_X(0) = P(X = 0) = \frac{\binom{8}{0}\binom{6}{3}}{\binom{14}{3}} = \frac{20}{364} = \frac{5}{91},$$

$$f_X(1) = P(X = 1) = \frac{\binom{8}{1}\binom{6}{2}}{\binom{14}{3}} = \frac{120}{364} = \frac{30}{91},$$

$$f_X(2) = P(X = 2) = \frac{\binom{8}{2}\binom{6}{1}}{\binom{14}{3}} = \frac{168}{364} = \frac{6}{13},$$

$$f_X(3) = P(X = 3) = \frac{\binom{8}{3}\binom{6}{0}}{\binom{14}{3}} = \frac{56}{364} = \frac{2}{13}.$$

This leads to the following pmf of X :

x	0	1	2	3
$f_X(x)$	$\frac{5}{91}$	$\frac{30}{91}$	$\frac{6}{13}$	$\frac{2}{13}$

- (b) If X represents the number of face cards obtained in 3 selections, then $3 - X$ represents the number of non-face cards selected. Based on how the winnings of the game work, the random variable Y is given by

$$Y = 3X - (3 - X) = 4X - 3.$$

Since Y is a linear function of X , the set $A_X = \{0, 1, 2, 3\}$ maps (one-to-one) to the set $A_Y = \{-3, 1, 5, 9\}$, and this immediately yields

y	-3	1	5	9
$f_Y(y)$	$\frac{5}{91}$	$\frac{30}{91}$	$\frac{6}{13}$	$\frac{2}{13}$

- 3.2.5 Let X be the random variable representing the total number of draws made until the last blue ball is obtained. We note that the smallest value that X can take on is 2, resulting when the first 2 selections from the box are the 2 blue balls. On the other hand, the largest value that X can take on is 7, which happens when all 5 yellow balls are selected before any blue ball is even picked. Therefore, the range of X is $A_X = \{2, 3, 4, 5, 6, 7\}$.

Therefore, for $x \in A_X$, we can determine the pmf of X in the following manner:

$$\begin{aligned}
f_X(x) &= P(x \text{ total draws are made}) \\
&= P(\underbrace{1 \text{ blue ball drawn in first } x-1 \text{ draws}}_{\text{call this event } B} \text{ and } \underbrace{2^{\text{nd}} \text{ blue ball obtained on } x^{\text{th}} \text{ draw}}_{\text{call this event } C}) \\
&= P(B)P(C|B) \\
&= \frac{\binom{2}{1}\binom{5}{x-2}}{\binom{7}{x-1}} \cdot \frac{1}{7-(x-1)} \\
&= \frac{5!}{(x-2)!(7-x)!} \cdot \frac{(x-1)!(8-x)!}{7!} \cdot \frac{2}{8-x} \\
&= \frac{x-1}{21}.
\end{aligned}$$

Now, let Y be the player's net winnings, which is given by $Y = 2X - 9$. Since Y is a linear function of X , it immediately follows that the pmf of Y is given by

$$f_Y(y) = f_X\left(\frac{y+9}{2}\right) = \frac{\frac{y+9}{2}-1}{21} = \frac{y+7}{42} \text{ for } y = -5, -3, -1, 1, 3, 5.$$

Hence, $P(\text{player makes a profit}) = f_Y(1) + f_Y(3) + f_Y(5) = \frac{8}{42} + \frac{10}{42} + \frac{12}{42} = \frac{30}{42} = \frac{5}{7}$.

3.2.6

Let X be the random variable indicating which integer is randomly chosen from the set D . The pmf of X is clearly given by $f_X(x) = \frac{1}{10}$ for $x = 0, 1, \dots, 9$. For a given value of $x \in \{0, 1, \dots, 9\}$, note that the subsets D_1 and D_2 consist of $x+1$ and $9-x$ elements, respectively. Therefore, the random variable R , representing the ratio of the number of elements in the smaller subset to the larger subset, is defined by

$$R = \frac{\min(X+1, 9-X)}{\max(X+1, 9-X)}.$$

To help determine the probability distribution of R , we consider the following table:

x	$x+1$	$9-x$	$r = \frac{\min(x+1, 9-x)}{\max(x+1, 9-x)}$	$f_X(x)$
0	1	9	$\frac{1}{9}$	$\frac{1}{10}$
1	2	8	$\frac{2}{8}$	$\frac{1}{10}$
2	3	7	$\frac{3}{7}$	$\frac{1}{10}$
3	4	6	$\frac{4}{6}$	$\frac{1}{10}$
4	5	5	$\frac{5}{5}$	$\frac{1}{10}$
5	6	4	$\frac{4}{6}$	$\frac{1}{10}$
6	7	3	$\frac{3}{7}$	$\frac{1}{10}$
7	8	2	$\frac{2}{8}$	$\frac{1}{10}$
8	9	1	$\frac{1}{9}$	$\frac{1}{10}$
9	10	0	$\frac{0}{10}$	$\frac{1}{10}$

Based on the results of the above table, we determine the pmf of R as follows:

r	0	$\frac{1}{9}$	$\frac{1}{4}$	$\frac{3}{7}$	$\frac{2}{3}$	1
$f_R(r)$	$\frac{1}{10}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{10}$

Section 3.3:

3.3.1 We calculate the following quantities:

$$\begin{aligned}\mu &= \sum_{x=1}^6 x f(x) \\&= 1 \left(\frac{11}{40} \right) + \sum_{i=2}^6 x \frac{1}{2x} \\&= \frac{11}{40} + \frac{1}{2} \sum_{i=2}^6 1 \\&= \frac{11}{40} + \frac{1}{2} \cdot 5 \\&= \frac{11 + 100}{40} \\&= \frac{111}{40},\end{aligned}$$

$$\begin{aligned}E(X^2) &= \sum_{x=1}^6 x^2 f(x) \\&= 1^2 \left(\frac{11}{40} \right) + \sum_{i=2}^6 x^2 \frac{1}{2x} \\&= \frac{11}{40} + \frac{1}{2} \sum_{i=2}^6 x \\&= \frac{11}{40} + \frac{1}{2} (20) \\&= \frac{11 + 400}{40} \\&= \frac{411}{40},\end{aligned}$$

$$\sigma^2 = E(X^2) - \mu^2 = \frac{411}{40} - \left(\frac{111}{40} \right)^2 = \frac{4119}{1600} = 2.574375,$$

and

$$\sigma = \sqrt{\frac{4119}{1600}} = \frac{\sqrt{4199}}{40} \approx 1.604486.$$

3.3.2 Let X be the random variable representing the number of tosses needed to obtain the first occurrence of heads. Since the coin is fair and tosses are independent of each other, the pmf of X is given by

$$f(x) = \left(\frac{1}{2}\right)^x \text{ for } x = 1, 2, 3, \dots$$

Let Y be the random variable representing the person's winnings. Based on the given information, we have that

$$Y = g(X) = \begin{cases} 2^X & \text{if } X = 1, 2, 3, 4, 5, \\ -256 & \text{if } X = 6, 7, 8, \dots \end{cases}$$

Thus, it follows that

$$\begin{aligned} \mu_Y &= E(g(X)) \\ &= \sum_{x=1}^{\infty} g(x)f(x) \\ &= \sum_{x=1}^5 2^x \left(\frac{1}{2}\right)^x + \sum_{x=6}^{\infty} (-256) \left(\frac{1}{2}\right)^x \\ &= \sum_{x=1}^5 1 - 256 \cdot \frac{\left(\frac{1}{2}\right)^6}{1 - \frac{1}{2}} \\ &= 5 - 2^8 \left(\frac{1}{2}\right)^5 \\ &= 5 - 8 \\ &= -3 \end{aligned}$$

and

$$\begin{aligned} E(Y^2) &= E(g(X)^2) \\ &= \sum_{x=1}^{\infty} g(x)^2 f(x) \\ &= \sum_{x=1}^5 (2^x)^2 \left(\frac{1}{2}\right)^x + \sum_{x=6}^{\infty} (-256)^2 \left(\frac{1}{2}\right)^x \\ &= \sum_{x=1}^5 2^x + 2^{16} \cdot \frac{\left(\frac{1}{2}\right)^6}{1 - \frac{1}{2}} \\ &= \frac{2 - 2^6}{1 - 2} + 2^{11} \text{ using the finite geometric series formula } \sum_{i=1}^n r^i = \frac{r - r^{n+1}}{1 - r} \text{ for } r \neq 1 \\ &= 62 + 2048 \\ &= 2110. \end{aligned}$$

Therefore, we obtain $\sigma_Y^2 = E(Y^2) - \mu_Y^2 = 2110 - (-3)^2 = 2110 - 9 = 2101$.

- 3.3.3 (a) Let X_n be the random variable representing the number of times the roulette wheel stops on a red number in n independent plays. On a single play, the probability of stopping on a red number is $\frac{18}{37}$ and the probability of not stopping on a red number is $1 - \frac{18}{37} = \frac{19}{37}$. It follows that the pmf of X_n is given by

$$\begin{aligned} f_{X_n}(x) &= P(\text{red number obtained } x \text{ times out of } n \text{ independent plays}) \\ &= \binom{n}{x} \left(\frac{18}{37}\right)^x \left(\frac{19}{37}\right)^{n-x} \quad \text{for } x = 0, 1, \dots, n. \end{aligned}$$

Note that

$$\begin{aligned} E(X_n) &= \sum_{x=0}^n x \binom{n}{x} \left(\frac{18}{37}\right)^x \left(\frac{19}{37}\right)^{n-x} \\ &= \sum_{x=1}^n x \frac{n!}{x!(n-x)!} \left(\frac{18}{37}\right)^x \left(\frac{19}{37}\right)^{n-x} \\ &= \sum_{x=1}^n \frac{n(n-1)!}{(x-1)!((n-1)-(x-1))!} \left(\frac{18}{37}\right)^{x-1+1} \left(\frac{19}{37}\right)^{(n-1)-(x-1)} \\ &= n \left(\frac{18}{37}\right) \sum_{k=0}^{n-1} \underbrace{\frac{(n-1)!}{k!((n-1)-k)!}}_{= \binom{n-1}{k}} \left(\frac{18}{37}\right)^k \left(\frac{19}{37}\right)^{(n-1)-k} \quad \text{if we let } k = x - 1 \\ &= \frac{18n}{37} \left(\frac{18}{37} + \frac{19}{37}\right)^{n-1} \quad \text{using the binomial series formula} \\ &= \frac{18n}{37}. \end{aligned}$$

If b is the bet amount per play and we let $Y_{b,n}$ be the random variable denoting net winnings from n plays, then it follows that $Y_{b,n} = 2bX_n - bn$. Therefore, we wish to calculate

$$E(Y_{1,10}) = 2(1)E(X_{10}) - (1)(10) = 2 \left(\frac{18(10)}{37}\right) - 10 = \frac{360 - 370}{37} = -\frac{10}{37} \approx -\$0.27.$$

and

$$E(Y_{10,1}) = 2(10)E(X_1) - (10)(1) = 20 \left(\frac{18}{37}\right) - 10 = \frac{360 - 370}{37} = -\frac{10}{37} \approx -\$0.27.$$

- (b) For each of the two scenarios in part (a), we wish to calculate

$$P(\text{profit is made}) = P(Y_{b,n} > 0) = P(2bX_n - bn > 0) = P\left(X_n > \frac{bn}{2b}\right) = P\left(X_n > \frac{n}{2}\right).$$

In the case when $b = 1$ and $n = 10$, we obtain

$$P(\text{profit is made}) = P\left(X_{10} > \frac{10}{2}\right) = P(X_{10} \geq 6) = \sum_{x=6}^{10} \binom{10}{x} \left(\frac{18}{37}\right)^x \left(\frac{19}{37}\right)^{10-x} \approx 0.3442.$$

In the case when $b = 10$ and $n = 1$, we obtain

$$P(\text{profit is made}) = P\left(X_1 > \frac{1}{2}\right) = P(X_1 = 1) = \binom{1}{1} \left(\frac{18}{37}\right)^1 \left(\frac{19}{37}\right)^{1-1} = \frac{18}{37} = 0.4865.$$

3.3.4 (a) Let W represent the net winnings random variable. The sample space S of all 2-card selections with replacement is given by

$$S = \{\underbrace{(1, 1)}_{W=-3}, \underbrace{(1, 5)}_{W=1}, \underbrace{(1, 8)}_{W=4}, \underbrace{(5, 1)}_{W=1}, \underbrace{(5, 5)}_{W=5}, \underbrace{(5, 8)}_{W=8}, \underbrace{(8, 1)}_{W=4}, \underbrace{(8, 5)}_{W=8}, \underbrace{(8, 8)}_{W=11}\}.$$

Since S consists of $3^2 = 9$ equally likely outcomes, the pmf of W is given by

w	-3	1	4	5	8	11
$f(w)$	$\frac{1}{9}$	$\frac{2}{9}$	$\frac{2}{9}$	$\frac{1}{9}$	$\frac{2}{9}$	$\frac{1}{9}$

Therefore, it follows that

$$\begin{aligned} \mu_W &= (-3) \left(\frac{1}{9}\right) + (1) \left(\frac{2}{9}\right) + (4) \left(\frac{2}{9}\right) + (5) \left(\frac{1}{9}\right) + (8) \left(\frac{2}{9}\right) + (11) \left(\frac{1}{9}\right) \\ &= \frac{-3 + 2 + 8 + 5 + 16 + 11}{9} \\ &= \frac{39}{9} \\ &= \frac{13}{3}, \end{aligned}$$

$$\begin{aligned} E(W^2) &= (9) \left(\frac{1}{9}\right) + (1) \left(\frac{2}{9}\right) + (16) \left(\frac{2}{9}\right) + (25) \left(\frac{1}{9}\right) + (64) \left(\frac{2}{9}\right) + (121) \left(\frac{1}{9}\right) \\ &= \frac{9 + 2 + 32 + 25 + 128 + 121}{9} \\ &= \frac{317}{9}, \end{aligned}$$

$$\text{and } SD(W) = \sqrt{\frac{317}{9} - \left(\frac{13}{3}\right)^2} = \sqrt{\frac{148}{9}} = \frac{2\sqrt{37}}{3} \approx 4.0552.$$

- (b) We seek the value of k which now gives $\mu_W = 7$. With 3 cards labelled \$1, \$5, and \$ k , the possible net winnings are now $\{-3, 1, k-4, 5, k, 2k-5\}$. Therefore, the value of k must satisfy

$$(-3) \left(\frac{1}{9}\right) + (1) \left(\frac{2}{9}\right) + (k-4) \left(\frac{2}{9}\right) + (5) \left(\frac{1}{9}\right) + (k) \left(\frac{2}{9}\right) + (2k-5) \left(\frac{1}{9}\right) = 7$$

$$-3 + 2 + 2(k-4) + 5 + 2k + 2k - 5 = 63$$

$$6k - 9 = 63$$

$$k = \frac{63+9}{6} = \$12.$$

- 3.3.5 Since the three wheels spin independently and each has probability 0.1 of landing at any position, we have that

$$P(\text{all 3 wheels stop on a flower}) = (0.2)(0.6)(0.2) = 0.024,$$

$$P(\text{all 3 wheels stop on a dog}) = (0.4)(0.3)(0.3) = 0.036, \text{ and}$$

$$P(\text{all 3 wheels stop on a house}) = (0.4)(0.1)(0.5) = 0.02.$$

Let W be the random variable representing the winnings per dollar spent. The pmf of W is therefore given by

w	0	5	10	20
$f_W(w)$	0.92	0.02	0.036	0.024

As a result, we now calculate

$$\mu_W = (0)(0.92) + (5)(0.02) + (10)(0.036) + (20)(0.024) = 0.94,$$

$$E(W^2) = (0)(0.92) + (25)(0.02) + (100)(0.36) + (400)(0.024) = 13.7,$$

$$\text{and } \sigma_W = \sqrt{E(W^2) - \mu_W^2} = \sqrt{13.7 - (0.94)^2} = \sqrt{12.8164} = 3.58.$$

- 3.3.6 In Problem 3.2.3, we found that the pmf of Y is given by

$$f_Y(y) = \frac{\binom{4}{\frac{y+4}{2}}}{16} \text{ for } y = -4, -2, 0, 2, 4.$$

Therefore, we wish to calculate

$$\mu_Y = (-4)\frac{\binom{4}{0}}{16} + (-2)\frac{\binom{4}{1}}{16} + (0)\frac{\binom{4}{2}}{16} + (2)\frac{\binom{4}{3}}{16} + (4)\frac{\binom{4}{4}}{16} = 0$$

and

$$\begin{aligned}
\text{Var}(Y) &= E(Y^2) \text{ since } \mu_Y = 0 \\
&= (-4)^2 \frac{\binom{4}{0}}{16} + (-2)^2 \frac{\binom{4}{1}}{16} + (0)^2 \frac{\binom{4}{2}}{16} + (2)^2 \frac{\binom{4}{3}}{16} + (4)^2 \frac{\binom{4}{4}}{16} \\
&= (16) \frac{2}{16} + (4) \frac{8}{16} \\
&= 2 + 2 \\
&= 4.
\end{aligned}$$

3.3.7 In Problem 3.2.5, we found that the pmf of Y is given by

$$f_Y(y) = \frac{y+7}{42} \text{ for } y = -5, -3, -1, 1, 3, 5.$$

Therefore, we wish to calculate

$$\begin{aligned}
\mu_Y &= (-5) \frac{2}{42} + (-3) \frac{4}{42} + (-1) \frac{6}{42} + (1) \frac{8}{42} + (3) \frac{10}{42} + (5) \frac{12}{42} \\
&= \frac{-10 - 12 - 6 + 8 + 30 + 60}{42} \\
&= \frac{70}{42} \\
&= \frac{5}{3},
\end{aligned}$$

$$\begin{aligned}
E(Y^2) &= (-5)^2 \frac{2}{42} + (-3)^2 \frac{4}{42} + (-1)^2 \frac{6}{42} + (1)^2 \frac{8}{42} + (3)^2 \frac{10}{42} + (5)^2 \frac{12}{42} \\
&= \frac{50 + 36 + 6 + 8 + 90 + 300}{42} \\
&= \frac{490}{42} \\
&= \frac{35}{3},
\end{aligned}$$

$$\text{and } \sigma_Y = \sqrt{E(Y^2) - \mu_Y^2} = \sqrt{\frac{35}{3} - \frac{25}{9}} = \sqrt{\frac{80}{9}} = \frac{4\sqrt{5}}{3} \approx 2.9814.$$

3.3.8 Let Y be the number of cups of coffee that Abbott drinks in a week. Since $Y = 2X^2$, it follows that

$$\begin{aligned}
\mu_Y &= \sum_{x=0}^5 (2x^2) f(x) \\
&= (0)(0.09) + (2)(0.1) + (8)(0.25) + (18)(0.4) + (32)(0.15) + (50)(0.01) \\
&= 14.7
\end{aligned}$$

and

$$\begin{aligned}
Var(Y) &= E(Y^2) - \mu_Y^2 \\
&= \sum_{x=0}^5 (2x^2)^2 f(x) - (14.7)^2 \\
&= (0)(0.09) + (4)(0.1) + (64)(0.25) + (324)(0.4) + (1024)(0.15) + (2500)(0.01) - 216.09 \\
&= 324.6 - 216.09 \\
&= 108.51.
\end{aligned}$$

On the other hand, let Z be the number of cups of coffee that Costello drinks in a week. Since $Z = |2X - 1|$, it follows that

$$\begin{aligned}
\mu_Z &= \sum_{x=0}^5 |2x - 1| f(x) \\
&= (1)(0.09) + (1)(0.1) + (3)(0.25) + (5)(0.4) + (7)(0.15) + (9)(0.01) \\
&= 4.08
\end{aligned}$$

and

$$\begin{aligned}
Var(Z) &= E(Z^2) - \mu_Z^2 \\
&= \sum_{x=0}^5 (2x - 1)^2 f(x) - (4.08)^2 \\
&= (1)(0.09) + (1)(0.1) + (9)(0.25) + (25)(0.4) + (49)(0.15) + (81)(0.01) - 16.6464 \\
&= 20.6 - 16.6464 \\
&= 3.9536.
\end{aligned}$$

3.3.9 (a) Let X be the contestant's winnings when question A is answered first. Based on the given information, we have that

$$\begin{aligned}
P(X = 0) &= P(A \text{ answered incorrectly}) = 0.2, \\
P(X = 100) &= P(A \text{ answered correctly and } B \text{ answered incorrectly}) = (0.8)(0.4) = 0.32, \text{ and} \\
P(X = 300) &= P(A \text{ answered correctly and } B \text{ answered correctly}) = (0.8)(0.6) = 0.48.
\end{aligned}$$

Therefore,

$$E(X) = (0)(0.2) + (100)(0.32) + (300)(0.48) = 176.$$

On the other hand, let Y be the contestant's winnings when question B is answered first. Based on the given information, we have that

$$\begin{aligned}
P(Y = 0) &= P(B \text{ answered incorrectly}) = 0.4, \\
P(Y = 200) &= P(B \text{ answered correctly and } A \text{ answered incorrectly}) = (0.6)(0.2) = 0.12, \text{ and} \\
P(Y = 300) &= P(B \text{ answered correctly and } A \text{ answered correctly}) = (0.6)(0.8) = 0.48.
\end{aligned}$$

In this case,

$$E(Y) = (0)(0.4) + (200)(0.12) + (300)(0.48) = 168.$$

In order to maximize expected winnings, we see that question A should be attempted first.

- (b) Now, with a \$50 penalty incurred if the first question is answered incorrectly, the pmf of X is modified to become

x	-50	100	300
$f_X(x)$	0.2	0.32	0.48

Similarly, the pmf of Y is modified to become

y	-50	200	300
$f_Y(y)$	0.4	0.12	0.48

Therefore, we now obtain

$$E(X) = (-50)(0.2) + (100)(0.32) + (300)(0.48) = 166$$

and

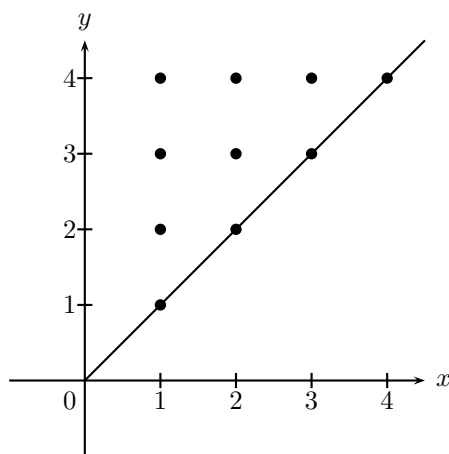
$$E(Y) = (-50)(0.4) + (200)(0.12) + (300)(0.48) = 148.$$

We see that question A should still be attempted first.

3.3.10 If X is a discrete random variable with pmf $f_X(x)$ and range $A = \{0, 1, 2, \dots\}$, then

$$\sum_{x=0}^{\infty} P(X > x) = \sum_{x=1}^{\infty} P(X \geq x) = \sum_{x=1}^{\infty} \sum_{y=x}^{\infty} f_X(y).$$

A graphical display of the region to sum over is as follows:



Therefore, interchanging the order of summation, we get

$$\begin{aligned}
\sum_{x=0}^{\infty} P(X > x) &= \sum_{y=1}^{\infty} f_X(y) \sum_{x=1}^y 1 \\
&= \sum_{y=1}^{\infty} f_X(y) \cdot y \\
&= \sum_{y=0}^{\infty} y f_X(y) \\
&= E(X).
\end{aligned}$$

In a similar fashion, note that

$$\begin{aligned}
2 \sum_{x=1}^{\infty} x P(X > x) &= 2 \sum_{x=0}^{\infty} x P(X > x) \\
&= 2 \sum_{x=0}^{\infty} x P(X \geq x+1) \\
&= 2 \sum_{y=1}^{\infty} (y-1) P(X \geq y) \text{ if we let } y = x+1 \\
&= 2 \sum_{y=1}^{\infty} (y-1) \sum_{z=y}^{\infty} f_X(z) \\
&= 2 \sum_{z=1}^{\infty} f_X(z) \sum_{y=1}^z (y-1) \text{ by interchanging the order of summation} \\
&= 2 \sum_{z=1}^{\infty} f_X(z) \cdot \frac{(z-1)z}{2} \\
&= \sum_{z=0}^{\infty} z(z-1) f_X(z) \\
&= E(X(X-1)).
\end{aligned}$$

Section 3.4:

3.4.1 Recall that the mgf of any random variable X must satisfy

$$M_X(0) = E(e^{0X}) = E(1) = 1.$$

With the given mgf, however, we see that $M_X(0) = \frac{0}{1-0} = 0 \neq 1$. Hence, there can be no random variable X for which $M_X(t) = \frac{t}{1-t}$ for $|t| < 1$.

3.4.2 (a) Let X be a discrete random variable with range A and pmf $f(x)$. Let us further assume that $0 < t < a$. By definition, we have that

$$\begin{aligned} M(t) &= \sum_{x \in A} e^{tx} f(x) \\ &\geq \sum_{x \geq a} e^{tx} f(x) \\ &\geq \sum_{x \geq a} e^{ta} f(x) \text{ since } e^{tx} \geq e^{ta} \text{ for } x \geq a \text{ and } 0 < t < a \\ &= e^{at} \sum_{x \geq a} f(x) \\ &= e^{at} P(X \geq a). \end{aligned}$$

Using the above inequality, we immediately obtain $P(X \geq a) \leq e^{-at} M(t)$, as required.

(b) Suppose now that $-a < t < 0$. Note that

$$\begin{aligned} M(t) &= \sum_{x \in A} e^{tx} f(x) \\ &\geq \sum_{x \leq a} e^{tx} f(x) \\ &\geq \sum_{x \leq a} e^{ta} f(x) \text{ since } e^{tx} \geq e^{ta} \text{ for } x \leq a \text{ and } -a < t < 0 \\ &= e^{at} \sum_{x \leq a} f(x) \\ &= e^{at} P(X \leq a). \end{aligned}$$

Using the above inequality, we immediately obtain $P(X \leq a) \leq e^{-at} M(t)$, as required.

3.4.3 Based on the form of the mgf, we propose the range of X to be $A = \{-3, 0, 2\}$ and the pmf of X to be:

$$\begin{array}{c|ccc} x & -3 & 0 & 2 \\ \hline f_X(x) & \frac{1}{3} & \frac{1}{2} & \frac{1}{6} \end{array}$$

With this choice of pmf, we clearly see that

$$\sum_{x \in A} e^{tx} f_X(x) = e^{-3t} \left(\frac{1}{3} \right) + e^{0t} \left(\frac{1}{2} \right) + e^{2t} \left(\frac{1}{6} \right) = M_X(t) \text{ for } t \in \mathbb{R}.$$

Therefore, by the mgf uniqueness property, this must be the pmf of X . Hence, the cdf of X is immediately given by

$$F_X(x) = \begin{cases} 0 & \text{if } x < -3, \\ \frac{1}{3} & \text{if } -3 \leq x < 0, \\ \frac{5}{6} & \text{if } 0 \leq x < 2, \\ 1 & \text{if } x \geq 2. \end{cases}$$

3.4.4 First of all, since $t < \ln 3$, we can write

$$M_Y(t) = \frac{2}{3 - e^t} = \frac{\frac{2}{3}}{1 - \frac{e^t}{3}} = \frac{2}{3} \sum_{y=0}^{\infty} \left(\frac{e^t}{3} \right)^y = \sum_{y=0}^{\infty} e^{ty} \left(\frac{1}{3} \right)^y \frac{2}{3}.$$

By the mgf uniqueness property, it follows that the pmf of Y is given by

$$f_Y(y) = \left(\frac{1}{3} \right)^y \frac{2}{3} \text{ for } y = 0, 1, 2, \dots$$

3.4.5 (a) With $C(t) = \ln M(t)$, note that

$$C^{(1)}(t) = \frac{d}{dt} C(t) = \frac{d}{dt} \ln M(t) = \frac{M^{(1)}(t)}{M(t)}$$

and

$$C^{(2)}(t) = \frac{d}{dt} C^{(1)}(t) = \frac{d}{dt} \left(\frac{M^{(1)}(t)}{M(t)} \right) = \frac{M(t)M^{(2)}(t) - (M^{(1)}(t))^2}{(M(t))^2}.$$

Therefore, evaluating the above derivatives at $t = 0$, we obtain

$$C^{(1)}(0) = \frac{M^{(1)}(0)}{M(0)} = \frac{E(X)}{1} = E(X)$$

and

$$C^{(2)}(0) = \frac{M(0)M^{(2)}(0) - (M^{(1)}(0))^2}{(M(0))^2} = \frac{1 \cdot E(X^2) - (E(X))^2}{1^2} = E(X^2) - (E(X))^2 = \text{Var}(X).$$

(b) Suppose that $M(t) = e^{3(e^{4t}-1)}$ for $t \in \mathbb{R}$. As a result, we have that

$$C(t) = \ln M(t) = \ln e^{3(e^{4t}-1)} = 3(e^{4t} - 1) \text{ for } t \in \mathbb{R}.$$

Using the results in part (a), we obtain

$$E(X) = C^{(1)}(0) = \frac{d}{dt} (3(e^{4t} - 1)) \big|_{t=0} = (12e^{4t}) \big|_{t=0} = 12$$

and

$$Var(X) = C^{(2)}(0) = \frac{d}{dt} (12e^{4t}) \big|_{t=0} = (48e^{4t}) \big|_{t=0} = 48.$$

3.4.6 Note that $Y = \frac{1}{2}(X - 4)$ is of the form $aX + b$ with $a = \frac{1}{2}$ and $b = -2$. Therefore, we immediately have that

$$M_Y(t) = e^{bt} M(at) = \frac{e^{-2t}}{125} \left(4 + e^{\frac{t}{2}}\right)^3 \text{ for } t \in \mathbb{R}.$$

Using both the product rule and chain rule for differentiation, let us take the first two derivatives of $M_Y(t)$:

$$\begin{aligned} M_Y^{(1)}(t) &= \frac{e^{-2t}}{125} \cdot 3 \left(4 + e^{\frac{t}{2}}\right)^2 \left(\frac{e^{\frac{t}{2}}}{2}\right) - \frac{2e^{-2t}}{125} \left(4 + e^{\frac{t}{2}}\right)^3 \\ &= \frac{3e^{-\frac{3t}{2}}}{250} \left(4 + e^{\frac{t}{2}}\right)^2 - \frac{2e^{-2t}}{125} \left(4 + e^{\frac{t}{2}}\right)^3, \end{aligned}$$

$$\begin{aligned} M_Y^{(2)}(t) &= \frac{3e^{-\frac{3t}{2}}}{250} \cdot 2 \left(4 + e^{\frac{t}{2}}\right) \left(\frac{e^{\frac{t}{2}}}{2}\right) - \frac{9e^{-\frac{3t}{2}}}{500} \left(4 + e^{\frac{t}{2}}\right)^2 \\ &\quad - \frac{2e^{-2t}}{125} \cdot 3 \left(4 + e^{\frac{t}{2}}\right)^2 \left(\frac{e^{\frac{t}{2}}}{2}\right) + \frac{4e^{-2t}}{125} \left(4 + e^{\frac{t}{2}}\right)^3 \\ &= \frac{3e^{-t}}{250} \left(4 + e^{\frac{t}{2}}\right) - \frac{21e^{-\frac{3t}{2}}}{500} \left(4 + e^{\frac{t}{2}}\right)^2 + \frac{4e^{-2t}}{125} \left(4 + e^{\frac{t}{2}}\right)^3. \end{aligned}$$

Therefore, it follows that

$$\mu_Y = M_Y^{(1)}(0) = \frac{3}{250}(4+1)^2 - \frac{2}{125}(4+1)^3 = \frac{3}{10} - 2 = -\frac{17}{10},$$

$$E(Y^2) = M_Y^{(2)}(0) = \frac{3}{250}(4+1) - \frac{21}{500}(4+1)^2 + \frac{4}{125}(4+1)^3 = \frac{3}{50} - \frac{21}{20} + 4 = \frac{6 - 105 + 400}{100} = \frac{301}{100},$$

and

$$\sigma_Y^2 = E(Y^2) - \mu_Y^2 = \frac{301}{100} - \frac{289}{100} = \frac{12}{100} = \frac{3}{25}.$$

3.4.7 The mgf of X is given by

$$\begin{aligned}
 M(t) &= \sum_{x=0}^{\infty} e^{tx} 2 \left(\frac{1}{3}\right)^{x+1} \\
 &= 2 \left(\frac{1}{3}\right) \underbrace{\sum_{x=0}^{\infty} \left(\frac{e^t}{3}\right)^x}_{\text{infinite geometric series}} \\
 &= \frac{2}{3} \cdot \frac{1}{1 - \frac{e^t}{3}} \text{ provided that } \frac{e^t}{3} < 1 \\
 &= \frac{2}{3} \left(1 - \frac{e^t}{3}\right)^{-1} \text{ for } t < \ln 3.
 \end{aligned}$$

Let us now take the first two derivatives of $M(t)$:

$$M^{(1)}(t) = -\frac{2}{3} \left(1 - \frac{e^t}{3}\right)^{-2} \left(-\frac{e^t}{3}\right) = \frac{2e^t}{9} \left(1 - \frac{e^t}{3}\right)^{-2},$$

$$M^{(2)}(t) = \frac{2e^t}{9}(-2) \left(1 - \frac{e^t}{3}\right)^{-3} \left(-\frac{e^t}{3}\right) + \frac{2e^t}{9} \left(1 - \frac{e^t}{3}\right)^{-2} = \frac{4e^{2t}}{27} \left(1 - \frac{e^t}{3}\right)^{-3} + \frac{2e^t}{9} \left(1 - \frac{e^t}{3}\right)^{-2}.$$

Therefore, it follows that

$$\mu_X = M^{(1)}(0) = \frac{2}{9} \left(1 - \frac{1}{3}\right)^{-2} = \frac{2}{9} \cdot \frac{9}{4} = \frac{1}{2},$$

$$E(X^2) = M^{(2)}(0) = \frac{4}{27} \left(1 - \frac{1}{3}\right)^{-3} + \frac{2}{9} \left(1 - \frac{1}{3}\right)^{-2} = \frac{4}{27} \cdot \frac{27}{8} + \frac{2}{9} \cdot \frac{9}{4} = \frac{1}{2} + \frac{1}{2} = 1,$$

and

$$\sigma_X^2 = E(X^2) - \mu_X^2 = 1 - \frac{1}{4} = \frac{3}{4}.$$

3.4.8 Recall that in Example 3.4.2, we showed that

$$\frac{e^t}{2} \left(1 - \frac{e^t}{2}\right)^{-1} = \sum_{x=1}^{\infty} e^{tx} \left(\frac{1}{2}\right)^x \text{ for } t < \ln 2. \quad (7)$$

Looking at the given mgf $M_Y(t)$ for $t < \ln 2$, note that

$$\begin{aligned} & M_Y(t) \\ &= \frac{e^{2t}}{4} \left(1 - \frac{e^t}{2}\right)^{-2} \\ &= \left(\frac{e^t}{2} \left(1 - \frac{e^t}{2}\right)^{-1}\right) \left(\frac{e^t}{2} \left(1 - \frac{e^t}{2}\right)^{-1}\right) \\ &= \left(\sum_{x=1}^{\infty} e^{tx} \left(\frac{1}{2}\right)^x\right) \left(\sum_{z=1}^{\infty} e^{tz} \left(\frac{1}{2}\right)^z\right) \text{ using (7)} \\ &= \left(e^t \left(\frac{1}{2}\right)^1 + e^{2t} \left(\frac{1}{2}\right)^2 + e^{3t} \left(\frac{1}{2}\right)^3 + \dots\right) \left(e^t \left(\frac{1}{2}\right)^1 + e^{2t} \left(\frac{1}{2}\right)^2 + e^{3t} \left(\frac{1}{2}\right)^3 + \dots\right) \\ &= e^{2t} \left(\frac{1}{2}\right)^2 + e^{3t} \left(\left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^1\right) + e^{4t} \left(\left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^1\right) + \dots \\ &= e^{2t} \cdot 1 \left(\frac{1}{2}\right)^2 + e^{3t} \cdot 2 \left(\frac{1}{2}\right)^3 + e^{4t} \cdot 3 \left(\frac{1}{2}\right)^4 + \dots \\ &= \sum_{y=2}^{\infty} e^{ty} (y-1) \left(\frac{1}{2}\right)^y. \end{aligned}$$

Therefore, by the mgf uniqueness property, it follows that the pmf of Y is given by

$$f_Y(y) = (y-1) \left(\frac{1}{2}\right)^y \text{ for } y = 2, 3, 4, \dots$$

Section 3.5:

- 3.5.1 We wish to find the pmf of X , representing the smallest digit in the four-digit number that is randomly created. To do so, we first focus our attention on determining the quantity $P(X > x) = 1 - P(X \leq x) = 1 - F_X(x)$, where $x \in \{1, 2, \dots, 9\}$. We have that

$$\begin{aligned}
 P(X > x) &= P(\text{smallest digit is } > x) \\
 &= P(\text{all four digits are } > x) \\
 &= P(\{1^{\text{st}} \text{ digit is } > x\} \cap \{2^{\text{nd}} \text{ digit is } > x\} \cap \{3^{\text{rd}} \text{ digit is } > x\} \cap \{4^{\text{th}} \text{ digit is } > x\}) \\
 &= P(1^{\text{st}} \text{ digit is } > x)P(2^{\text{nd}} \text{ digit is } > x)P(3^{\text{rd}} \text{ digit is } > x)P(4^{\text{th}} \text{ digit is } > x) \\
 &\hspace{15em} \text{since digit selections are independent} \\
 &= P(Y > x)^4,
 \end{aligned}$$

where Y is the random variable indicating which number is selected for whichever digit of the four-digit number. Clearly, $Y \sim DU(1, 9)$, from which it follows that

$$P(Y > x) = 1 - P(Y \leq x) = 1 - F_Y(x) = 1 - \frac{x - 1 + 1}{9 - 1 + 1} = 1 - \frac{x}{9} = \frac{9 - x}{9}.$$

Substituting this result into the above expression yields

$$P(X > x) = \left(\frac{9 - x}{9}\right)^4,$$

and so

$$F_X(x) = 1 - P(X > x) = 1 - \left(\frac{9 - x}{9}\right)^4 \text{ for } x = 1, 2, \dots, 9.$$

Therefore, we now obtain

$$f_X(1) = F_X(1) = 1 - \left(\frac{9 - 1}{9}\right)^4 = \frac{9^4 - 8^4}{9^4},$$

and for $x = 2, 3, \dots, 9$,

$$f_X(x) = F_X(x) - F_X(x-1) = 1 - \left(\frac{9 - x}{9}\right)^4 - \left(1 - \left(\frac{9 - (x-1)}{9}\right)^4\right) = \frac{(10 - x)^4 - (9 - x)^4}{9^4}.$$

Putting everything together and expanding, we can write

$$f_X(x) = \frac{3439 - 1084x + 114x^2 - 4x^3}{6561} \text{ for } x = 1, 2, \dots, 9.$$

- 3.5.2 (a) Let X be the random variable counting the number of inspected tins which are tainted. Based on the given information, we have that $X \sim HG(12, d, 7)$. Therefore, it follows that

$$P(\text{none of the 7 cans are tainted}) = P(X = 0) = \frac{\binom{d}{0} \binom{12-d}{7}}{\binom{12}{7}} = \frac{\binom{12-d}{7}}{\binom{12}{7}}.$$

We can now calculate the following probabilities for each value of d :

- When $d = 0$: $P(\text{none of the 7 cans are tainted}) = \frac{\binom{12}{7}}{\binom{12}{7}} = 1$,
- When $d = 1$: $P(\text{none of the 7 cans are tainted}) = \frac{\binom{11}{7}}{\binom{12}{7}} = \frac{330}{792} = \frac{5}{12}$,
- When $d = 2$: $P(\text{none of the 7 cans are tainted}) = \frac{\binom{10}{7}}{\binom{12}{7}} = \frac{120}{792} = \frac{5}{33}$,
- When $d = 3$: $P(\text{none of the 7 cans are tainted}) = \frac{\binom{9}{7}}{\binom{12}{7}} = \frac{36}{792} = \frac{1}{22}$.

- (b) Although it is possible we could find no tainted tins if d is as big as 3, it is not likely to happen based on the probabilities obtained in part (a). Therefore, the box is not likely to have as many as three tainted tins.

- 3.5.3 (a) Let X be the number of clients picked that have irregularities. Based on the given information, we have that $X \sim HG(130, 26, 6)$. Therefore, we wish to calculate

$$P(X = 2) = \frac{\binom{26}{2} \binom{104}{4}}{\binom{130}{6}} = \frac{2299063}{9174480} \approx 0.25059.$$

- (b) Since $N = 130$ is large and $n = 6$ is relatively small, we will apply a binomial approximation to the probability in part (a). In particular, we have that

$$P(X = 2) \approx \binom{6}{2} \left(\frac{26}{130} \right)^2 \left(1 - \frac{26}{130} \right)^4 = \frac{768}{3125} \approx 0.24576.$$

- 3.5.4 Let X be the number of individuals who do not test positive for type O^+ blood before the 5th person with type O^+ blood is found. Based on the given information, we have that $X \sim NB(5, 0.38)$. We wish to calculate

$$\begin{aligned} P(\text{over 10 people have to be tested}) &= P(X > 5) \\ &= 1 - P(X \leq 5) \\ &= 1 - \sum_{x=0}^5 \binom{x+4}{x} (0.38)^5 (0.62)^x \\ &\approx 0.682313. \end{aligned}$$

- 3.5.5 Let X be the number of non-leaky windshields the inspector checks before finding the first one that is leaky. We have that $X \sim Geo(0.3)$. We wish to find the value of n such that $P(X \geq n - 1) = 0.05$. However, we note that

$$P(X \geq n - 1) = 1 - P(X \leq n - 2) = 1 - (1 - (1 - 0.3)^{n-2+1}) = (0.7)^{n-1}.$$

With this result, we can solve the necessary equation for n :

$$\begin{aligned} (0.7)^{n-1} &= 0.05 \\ (n-1)\ln(0.7) &= \ln(0.05) \\ n &= \frac{\ln(0.05)}{\ln(0.7)} + 1 = 9.39905. \end{aligned}$$

Since at least 9.39905 cars means at least 10 cars must be checked, we therefore conclude that $n = 10$.

- 3.5.6 Let X be the number of passengers who show up on time for the flight. Based on the given information, we have that $X \sim Bin(122, 0.97)$. We wish to calculate

$$\begin{aligned} &P(\text{more people show up than can be carried on the flight}) \\ &= P(X > 120) \\ &= P(X = 121) + P(X = 122) \\ &= \binom{122}{121} (0.97)^{121} (0.03)^1 + \binom{122}{122} (0.97)^{122} (0.03)^0 \\ &\approx 0.116134. \end{aligned}$$

In order to use the Poisson approximation, let us define Y to be the number of passengers who do not show up on time for the flight. It follows that $Y \sim Bin(122, 0.03)$. Since $n = 122$ is large and $p = 0.03$ is small, the Poisson approximation with $\mu = np = 122(0.03) = 3.66$ should perform reasonably well. In particular, we obtain

$$P(X > 120) = P(Y < 2) = P(Y \leq 1) \approx \frac{e^{-3.66}(3.66)^0}{0!} + \frac{e^{-3.66}(3.66)^1}{1!} \approx 0.119914.$$

3.5.7 (a) We have that $X \sim \text{Geo}(p)$ so that its pmf is given $f(x) = p(1-p)^x$ for $x = 0, 1, 2, \dots$.
We wish to calculate

$$\begin{aligned}
 P(X \text{ is an odd number}) &= P(X \in \{1, 3, 5, \dots\}) \\
 &= \sum_{k=1}^{\infty} P(X = 2k-1) \\
 &= \sum_{k=1}^{\infty} p(1-p)^{2k-1} \\
 &= \frac{p}{1-p} \underbrace{\sum_{k=1}^{\infty} ((1-p)^2)^k}_{\text{infinite geometric series}} \\
 &= \frac{p}{1-p} \cdot \frac{(1-p)^2}{1-(1-p)^2} \\
 &= \frac{p(1-p)}{(1-(1-p))(1+(1-p))} \\
 &= \frac{1-p}{2-p}.
 \end{aligned}$$

(b) We wish to calculate

$$\begin{aligned}
 P(X \text{ is divisible by } 3) &= P(X \in \{0, 3, 6, \dots\}) \\
 &= \sum_{k=0}^{\infty} P(X = 3k) \\
 &= \sum_{k=0}^{\infty} p(1-p)^{3k} \\
 &= p \underbrace{\sum_{k=0}^{\infty} ((1-p)^3)^k}_{\text{infinite geometric series}} \\
 &= \frac{p}{1-(1-p)^3}.
 \end{aligned}$$

- (c) First of all, we note that the random variable $R = X \pmod{4}$ takes on values in the set $\{0, 1, 2, 3\}$. For $r \in \{0, 1, 2, 3\}$, we have that

$$\begin{aligned}
 P(R = r) &= P(X \pmod{4} = r) \\
 &= P(X \in \{r, 4 + r, 8 + r, \dots\}) \\
 &= \sum_{k=0}^{\infty} P(X = 4k + r) \\
 &= \sum_{k=0}^{\infty} p(1 - p)^{4k+r} \\
 &= p(1 - p)^r \underbrace{\sum_{k=0}^{\infty} ((1 - p)^4)^k}_{\text{infinite geometric series}} \\
 &= \frac{p(1 - p)^r}{1 - (1 - p)^4}.
 \end{aligned}$$

- 3.5.8 (a) If a group of 40 persons out of 150 are to be selected to form the jury, then it follows that

$$P(\text{you are chosen}) = \frac{\binom{1}{1} \binom{149}{39}}{\binom{150}{40}} = \frac{149!}{39! 110!} \cdot \frac{40! 110!}{150!} = \frac{40}{150} = \frac{4}{15}.$$

- (b) Assuming that men and women are equally likely to be picked, it follows that $Y \sim HG(150, 74, 12)$. Therefore, the pmf of Y is given by

$$f_Y(y) = \frac{\binom{74}{y} \binom{76}{12-y}}{\binom{150}{12}} \text{ for } y = 0, 1, \dots, 12.$$

- (c) Using the pmf from part (b), we see that

$$P(Y \leq 2) = \sum_{y=0}^2 \frac{\binom{74}{y} \binom{76}{12-y}}{\binom{150}{12}} \approx 0.0175908.$$

With such a low probability, we would view the event that two men were selected for the jury as a rather rare event to occur, and this would raise the question as to whether men and women were truly equally likely to be picked.

- 3.5.9 (a) Let X be the number of prizes the motorist wins in ten fill-ups. Based on the given information, it follows that $X \sim HG(500000, 500, 10)$. Therefore, we wish to calculate

$$P(X \geq 1) = 1 - P(X = 0) = 1 - \frac{\binom{500}{0} \binom{499500}{10}}{\binom{500000}{10}} \approx 0.00995521.$$

- (b) Let Y be the number of prizes won among the gas bar's customers. Based on the given information, it follows that $Y \sim HG(500000, 500, 2000)$. Therefore, we wish to calculate

$$P(Y \geq 1) = 1 - P(Y = 0) = 1 - \frac{\binom{500}{0} \binom{499500}{2000}}{\binom{500000}{2000}} \approx 0.865342.$$

Since $N = 500000$ is large and $n = 2000$ is relatively small, we could employ a $Bin(2000, \frac{500}{500000})$ approximation as follows:

$$P(Y \geq 1) \approx 1 - \binom{2000}{0} \left(\frac{1}{1000}\right)^0 \left(\frac{999}{1000}\right)^{2000} \approx 0.8648.$$

In the above binomial calculation, note that $n = 2000$ is large and $p = \frac{1}{1000}$ is small. Therefore, we can apply a Poisson approximation with $\mu = np = 2$ to get

$$P(Y \geq 1) \approx 1 - \frac{e^{-2} 2^0}{0!} = 1 - e^{-2} \approx 0.864665.$$

- 3.5.10 First of all, we have the key linear relationship $Y = X + k$ which links the two random variables X and Y . Using this connection, the pmf of Y is given by

$$\begin{aligned} f_Y(y) &= P(Y = y) \\ &= P(X + k = y) \\ &= P(X = y - k) \\ &= f_X(y - k) \\ &= \binom{y - k + k - 1}{y - k} p^k (1 - p)^{y - k} \\ &= \binom{y - 1}{y - k} p^k (1 - p)^{y - k} \text{ for } y = k, k + 1, k + 2, \dots \end{aligned}$$

Moreover, the mgf, mean, and variance of Y are readily obtained as follows:

$$M_Y(t) = E(e^{t(X+k)}) = e^{kt} M_X(t) = e^{kt} \left(\frac{p}{1 - (1-p)e^t} \right)^k = \left(\frac{pe^t}{1 - (1-p)e^t} \right)^k \text{ for } t < \ln(1-p)^{-1},$$

$$E(Y) = E(X + k) = E(X) + k = \frac{k(1-p)}{p} + k = \frac{k}{p},$$

$$Var(Y) = Var(X + k) = Var(X) = \frac{k(1-p)}{p^2}.$$

Solutions to End of Section Chapter 4 Problems

Section 4.1:

- 4.1.1 (a) The marginal probability distributions are obtained by summing across the rows and columns of the given table:

$f(x, y)$		x			
		0	1	2	
y	0	0.15	0.1	0.05	0.3
	1	0.35	0.2	0.15	0.7
		0.5	0.3	0.2	1

Therefore, the marginal pmf of X is given by

x	0	1	2
$f_X(x)$	0.5	0.3	0.2

and the marginal pmf of Y is given by

y	0	1
$f_Y(y)$	0.3	0.7

- (b) X and Y are not independent random variables since

$$0.1 = P(X = 1, Y = 0) \neq P(X = 1)P(Y = 0) = (0.3)(0.3) = 0.09.$$

- (c) Note that

$$P(X > Y) = f(1, 0) + f(2, 0) + f(2, 1) = 0.1 + 0.05 + 0.15 = 0.3.$$

- (d) We wish to find the pmf of $T = X + Y$. The following table specifies the (x, y) combinations which produce the values that T can take on:

$t = x + y$		x		
		0	1	2
y	0	0	1	2
	1	1	2	3

With the use of this table, the pmf of T immediately follows:

t	0	1	2	3
$f_T(t)$	0.15	0.45	0.25	0.15

- 4.1.2 (a) Note that the event $\{X^2 < 2Y\}$ can be represented in terms of the points (x, y) taken from the range of (X, Y) :

$$\{X^2 < 2Y\} = \{(1, 2), (1, 4), (1, 6), (2, 4), (2, 6), (3, 6)\}.$$

Therefore,

$$\begin{aligned} P(X^2 < 2Y) &= f(1, 2) + f(1, 4) + f(1, 6) + f(2, 4) + f(2, 6) + f(3, 6) \\ &= 0.09 + 0.15 + 0.06 + 0.05 + 0.09 + 0.15 \\ &= 0.59. \end{aligned}$$

- (b) The marginal probability distributions are obtained by summing across the rows and columns of the given table:

$f(x, y)$		x			
		1	2	3	
y	2	0.09	0.06	0.15	0.30
	4	0.15	0.05	0.20	0.40
	6	0.06	0.09	0.15	0.30
		0.30	0.20	0.50	1

Note that

$$0.05 = P(X = 2, Y = 4) \neq P(X = 2)P(Y = 4) = (0.20)(0.40) = 0.08.$$

By definition, X and Y are not independent random variables.

- (c) We wish to find the pmf of $D = |X - Y|$. The following table specifies the (x, y) combinations which produce the values that D can take on:

$d = x - y $		x		
		1	2	3
y	2	1	0	1
	4	3	2	1
	6	5	4	3

With the use of the above table, the pmf of D immediately follows:

d	0	1	2	3	4	5
$f_D(d)$	0.06	0.44	0.05	0.3	0.09	0.06

4.1.3 (a) We wish to calculate

$$P(X + Y \geq 5) = f(1, 4) + f(2, 3) + f(2, 4) = 0.1 + 0.15 + 0.25 = 0.5.$$

- (b) The marginal probability distributions are obtained by summing across the rows and columns of the given table:

$f(x, y)$		y			
		2	3	4	
x	1	0.2	0.15	0.1	0.45
	2	0.15	0.15	0.25	0.55
		0.35	0.3	0.35	1

Therefore, the marginal pmf of Y is given by

y	2	3	4
$f_Y(y)$	0.35	0.3	0.35

Hence, the mean number of chocolate chips in a random scoop is given by

$$E(Y) = 2(0.35) + 3(0.3) + 4(0.35) = 3.$$

- (c) The marginal pmf of X is given by

x	1	2
$f_X(x)$	0.45	0.55

Note that

$$0.2 = f(1, 2) \neq f_X(1)f_Y(2) = (0.45)(0.35) = 0.1575,$$

and this implies that X and Y are not independent random variables.

- (d) Let C represent the production cost (in cents) of a scoop of ice cream. We have that $C = 3 + Z$ where we define $Z = 5X + Y$. To find the pmf of Z , we use the following table which specifies the (x, y) combinations which produce the values that Z can take on:

$z = 5x + y$		y		
		2	3	4
x	1	7	8	9
	2	12	13	14

With the use of the above table, the pmf of Z immediately follows:

z	7	8	9	12	13	14
$f_Z(z)$	0.2	0.15	0.1	0.15	0.15	0.25

Therefore, we can now calculate

$$\begin{aligned}
 E(C) &= 3 + E(Z) \\
 &= 3 + 7(0.2) + 8(0.15) + 9(0.1) + 12(0.15) + 13(0.15) + 14(0.25) \\
 &= 3 + 10.75 \\
 &= 13.75.
 \end{aligned}$$

4.1.4 (a) To find the value of k , we apply the condition that $\sum_{all (x,y)} f(x,y) = 1$, which immediately leads to

$$\begin{aligned}
 f(1,1) + f(1,2) + f(2,1) + f(2,2) + f(3,1) + f(3,2) &= 1 \\
 k + 4k + 2k + 8k + 3k + 12k &= 1 \\
 30k &= 1 \\
 k &= \frac{1}{30}.
 \end{aligned}$$

(b) The marginal pmf of X is obtained as follows:

$$f_X(x) = \sum_{all y} f(x,y) = \sum_{y=1}^2 \frac{xy^2}{30} = \frac{x + 4x}{30} = \frac{x}{6} \text{ for } x = 1, 2, 3.$$

Similarly, the marginal pmf of Y is given by

$$f_Y(y) = \sum_{all x} f(x,y) = \sum_{x=1}^3 \frac{xy^2}{30} = \frac{y^2 + 2y^2 + 3y^2}{30} = \frac{y^2}{5} \text{ for } y = 1, 2.$$

(c) Since

$$f_X(x)f_Y(y) = \frac{x}{6} \cdot \frac{y^2}{5} = \frac{xy^2}{30} = f(x,y) \text{ for } x = 1, 2, 3 \text{ and } y = 1, 2,$$

the random variables X and Y are independent by definition.

4.1.5 (a) The joint pmf of X and Y is given by

$$f(x,y) = \frac{\binom{2}{x}\binom{1}{y}\binom{7}{3-x-y}}{\binom{10}{3}} \text{ for } x = 0, 1, 2 \text{ and } y = 0, 1.$$

- (b) Since X represents the number of items selected (in a sample of size 3 without replacement) having a minor defect from a population of 10 items in which 2 have a minor defect and 8 do not, it follows that $X \sim HG(10, 2, 3)$, so that its pmf is given by

$$f_X(x) = \frac{\binom{2}{x} \binom{8}{3-x}}{\binom{10}{3}} \text{ for } x = 0, 1, 2.$$

Applying a similar line of logic, it follows that $Y \sim HG(10, 1, 3)$. The marginal pmf of Y is therefore given by

$$f_Y(y) = \frac{\binom{1}{y} \binom{9}{3-y}}{\binom{10}{3}} = \frac{\binom{9}{3-y}}{\binom{10}{3}} \text{ for } y = 0, 1.$$

- (c) Define the event $B = \{X = Y\}$. Therefore, we have that

$$P(X = Y) = \sum_{(x,y) \in B} f(x, y) = f(0, 0) + f(1, 1) = \frac{\binom{2}{0} \binom{1}{0} \binom{7}{3}}{\binom{10}{3}} + \frac{\binom{2}{1} \binom{1}{1} \binom{7}{1}}{\binom{10}{3}} = \frac{7}{24} + \frac{7}{60} = \frac{49}{120}.$$

4.1.6 (a) First of all, note that the region

$$y = 0, 1, 2, \dots \text{ and } x = 0, 1, \dots, y$$

is equivalent to the region

$$x = 0, 1, 2, \dots \text{ and } y = x, x + 1, x + 2, \dots$$

With this in mind, the marginal pmf of X is given by

$$f_X(x) = \sum_{\text{all } y} f(x, y) = \sum_{y=x}^{\infty} \frac{e^{-2}}{x!(y-x)!} = \frac{e^{-2}}{x!} \sum_{y=x}^{\infty} \frac{1^{y-x}}{(y-x)!} = \frac{e^{-2}}{x!} \cdot e^1 = \frac{e^{-1} 1^x}{x!} \text{ for } x = 0, 1, 2, \dots$$

Likewise, the marginal pmf of Y is given by

$$f_Y(y) = \sum_{\text{all } x} f(x, y) = \sum_{x=0}^y \frac{e^{-2}}{x!(y-x)!} = \frac{e^{-2}}{y!} \sum_{x=0}^y \binom{y}{x} 1^x 1^{y-x} = \frac{e^{-2}}{y!} (1+1)^y = \frac{e^{-2} 2^y}{y!} \text{ for } y = 0, 1, 2, \dots$$

Note that $X \sim Poi(1)$ and $Y \sim Poi(2)$.

- (b) Note that $f(1, 1) = e^{-2}$, but

$$f_X(1)f_Y(1) = \frac{e^{-1}}{1!} \cdot \frac{e^{-2} 2^1}{1!} = 2e^{-3}.$$

Since they are not equal, X and Y are not independent random variables.

4.1.7 We first need to find $f(x, y)$. Since X and Y are independent, we know that

$$f(x, y) = f_X(x)f_Y(y) \text{ for all } (x, y).$$

Using the form of the $Geo(p)$ pmf, we have that

$$f(x, y) = p(1-p)^x \cdot p(1-p)^y = p^2(1-p)^{x+y} \text{ for } x = 0, 1, 2, \dots \text{ and } y = 0, 1, 2, \dots$$

Now, the pmf of $T = X + Y$ is given by

$$\begin{aligned} f_T(t) &= P(X + Y = t) \\ &= \sum_{\text{all } x} f(x, t-x) \\ &= \sum_{x=0}^t p^2(1-p)^{x+t-x} \\ &= p^2(1-p)^t \sum_{x=0}^t 1 \\ &= (t+1)p^2(1-p)^t \text{ for } t = 0, 1, 2, \dots \end{aligned}$$

Interestingly, if we take a closer look at the form of $f_T(t)$, then we see that

$$f_T(t) = (t+1)p^2(1-p)^t = \frac{(t+2-1)!}{t!(2-1)!} p^2(1-p)^t = \binom{t+2-1}{t} p^2(1-p)^t \text{ for } t = 0, 1, 2, \dots$$

In other words, we have that $T \sim NB(2, p)$.

4.1.8 Let X_A and X_B be random variables which represent the number of heads flipped by players A and B , respectively. The maximum number of heads that both players could achieve is limited by the smaller of the total number of attempts for either player. Since the coin is fair, it follows that $X_A \sim \text{Bin}(j, 1/2)$ and $X_B \sim \text{Bin}(m - j, 1/2)$ are independent random variables. Therefore, we obtain

$$\begin{aligned}
P(X_A = X_B) &= \sum_{i=0}^{\min(j, m-j)} P(X_A = i, X_B = i) \\
&= \sum_{i=0}^{\min(j, m-j)} P(X_A = i)P(X_B = i) \text{ by independence} \\
&= \sum_{i=0}^{\min(j, m-j)} \binom{j}{i} \left(\frac{1}{2}\right)^j \binom{m-j}{i} \left(\frac{1}{2}\right)^{m-j} \\
&= \sum_{i=0}^{\min(j, m-j)} \binom{j}{i} \binom{m-j}{i} \left(\frac{1}{2}\right)^m \\
&= \left(\frac{1}{2}\right)^m \sum_{i=0}^{\min(j, m-j)} \binom{m-j}{i} \binom{j}{j-i} \text{ since } \binom{j}{i} = \binom{j}{j-i} \\
&= \left(\frac{1}{2}\right)^m \binom{m}{j} \sum_{i=0}^{\min(j, m-j)} \frac{\binom{m-j}{i} \binom{j}{j-i}}{\binom{m}{j}}.
\end{aligned}$$

However,

$$\sum_{i=0}^{\min(j, m-j)} \frac{\binom{m-j}{i} \binom{j}{j-i}}{\binom{m}{j}} = 1,$$

as it is the sum of a $HG(m, m - j, j)$ pmf. Therefore, we simply get

$$P(X_A = X_B) = \binom{m}{j} \left(\frac{1}{2}\right)^m = \binom{m}{j} \left(\frac{1}{2}\right)^j \left(\frac{1}{2}\right)^{m-j},$$

which is the probability of observing j heads from m total flips of a fair coin.

Section 4.2:

4.2.1 (a) Let X_i be the number of class- i policy holders chosen, $i = A, B, C, D$. Based on the given information, we have that $(X_A, X_B, X_C, X_D) \sim MN(25; 0.1, 0.4, 0.3, 0.2)$. We wish to provide an expression for

$$\begin{aligned} P(3As, 11Bs, 7Cs, \text{ and } 4Ds) &= P(X_A = 3, X_B = 11, X_C = 7, X_D = 4) \\ &= \frac{25!}{3! 11! 7! 4!} (0.1)^3 (0.4)^{11} (0.3)^7 (0.2)^4. \end{aligned}$$

(b) We wish to provide an expression for

$$P(3As \text{ and } 11Bs) = P(X_A = 3, X_B = 11) = \frac{25!}{3! 11! 11!} (0.1)^3 (0.4)^{11} (0.5)^{11}.$$

4.2.2 (a) Note that $(X_1, X_2, X_3) \sim MN(9; 0.5, 0.4, 0.1)$. We wish to calculate

$$\begin{aligned} P(\text{all 3 sprinters win the same number of races}) &= P(X_1 = 3, X_2 = 3, X_3 = 3) \\ &= \frac{9!}{3! 3! 3!} (0.5)^3 (0.4)^3 (0.1)^3 \\ &= 0.01344. \end{aligned}$$

(b) Note that $X_1 + X_3$ represents the number of races won by Alvin or Charlie. Moreover, we know that $X_1 + X_3 \sim \text{Bin}(9, \underbrace{0.5 + 0.1}_{= 0.6})$, and so this leads to

$$\begin{aligned} P(\text{at most 4 races are won by Alvin or Charlie}) &= P(X_1 + X_3 \leq 4) \\ &= \sum_{i=0}^4 \binom{9}{i} (0.6)^i (0.4)^{9-i} \\ &\approx 0.266568. \end{aligned}$$

(c) We wish to calculate

$$\begin{aligned}
& P(\text{exactly 2 of the 3 sprinters win the same number of races}) \\
&= \sum_{\substack{i=0 \\ i \neq 3}}^4 P(X_1 = i, X_2 = i, X_3 = 9 - 2i) + \sum_{\substack{i=0 \\ i \neq 3}}^4 P(X_1 = i, X_2 = 9 - 2i, X_3 = i) \\
&\quad + \sum_{\substack{i=0 \\ i \neq 3}}^4 P(X_1 = 9 - 2i, X_2 = i, X_3 = i) \\
&= \sum_{\substack{i=0 \\ i \neq 3}}^4 \frac{9!}{i! i! (9 - 2i)!} (0.5)^i (0.4)^i (0.1)^{9-2i} + \sum_{\substack{i=0 \\ i \neq 3}}^4 \frac{9!}{i! (9 - 2i)! i!} (0.5)^i (0.4)^{9-2i} (0.1)^i \\
&\quad + \sum_{\substack{i=0 \\ i \neq 3}}^4 \frac{9!}{(9 - 2i)! i! i!} (0.5)^{9-2i} (0.4)^i (0.1)^i \\
&\approx 0.101104 + 0.027089 + 0.0630595 \\
&\approx 0.1913.
\end{aligned}$$

4.2.3 (a) Let X_i be the number of genetic type- i offspring observed, $i = 1, 2, 3, 4$. Based on the given information, we have that

$$(X_1, X_2, X_3, X_4) \sim MN \left(40; \frac{3}{16}, \frac{5}{16}, \frac{5}{16}, \frac{3}{16} \right).$$

We wish to provide an expression for

$$\begin{aligned}
P(10 \text{ offspring of each type}) &= P(X_1 = 10, X_2 = 10, X_3 = 10, X_4 = 10) \\
&= \frac{40!}{10! 10! 10! 10!} \left(\frac{3}{16} \right)^{10} \left(\frac{5}{16} \right)^{10} \left(\frac{5}{16} \right)^{10} \left(\frac{3}{16} \right)^{10} \\
&= \frac{40!}{10! 10! 10! 10!} \left(\frac{3}{16} \right)^{20} \left(\frac{5}{16} \right)^{20}.
\end{aligned}$$

(b) We are interested in the total number of type 1 or type 2 offspring observed, which is modelled by the random variable $X_1 + X_2$. Since $X_1 + X_2 \sim \text{Bin} \left(40, \frac{3}{16} + \frac{5}{16} \right)$, it follows that the desired expression is given by

$$P(X_1 + X_2 = 16) = \binom{40}{16} \left(\frac{1}{2} \right)^{16} \left(\frac{1}{2} \right)^{40-16} = \binom{40}{16} \left(\frac{1}{2} \right)^{40}.$$

4.2.4 (a) Let $X \sim \text{Geo}(p)$ represent the lifetime (in days) of a certain battery. To determine the value of p , we are given that

$$100 = E(X) = \frac{1-p}{p} \implies 100p = 1-p \implies p = \frac{1}{101}.$$

With this value of p , we can calculate the following proportions through the use of the formula

$$P(X > x) = 1 - F_X(x) = 1 - \left(1 - \left(1 - \frac{1}{101}\right)^{x+1}\right) = \left(\frac{100}{101}\right)^{x+1} \quad \text{for } x = 0, 1, 2, \dots$$

- (i) Define $p_1 = P(X \leq 50) = 1 - P(X > 50) = 1 - \left(\frac{100}{101}\right)^{51}$.
 - (ii) Define $p_2 = P(50 < X \leq 100) = P(X > 50) - P(X > 100) = \left(\frac{100}{101}\right)^{51} - \left(\frac{100}{101}\right)^{101}$.
 - (iii) Define $p_3 = P(100 < X \leq 150) = P(X > 100) - P(X > 150) = \left(\frac{100}{101}\right)^{101} - \left(\frac{100}{101}\right)^{151}$.
 - (iv) Define $p_4 = P(X > 150) = \left(\frac{100}{101}\right)^{151}$.
- (b) In a small shipment of 20 batteries, let X_1 be the number of batteries that last less than or equal to 50 days, X_2 the number of batteries that last more than 50 days but less than or equal to 100 days, X_3 the number of batteries that last more than 100 days but less than or equal to 150 days, and X_4 the number of batteries that last more than 150 days. Since $(X_1, X_2, X_3, X_4) \sim MN(20; p_1, p_2, p_3, p_4)$, we wish to provide an expression for

$$P(X_1 = 5, X_2 = 8, X_4 = 2) = P(X_1 = 5, X_2 = 8, X_3 = 5, X_4 = 2) = \frac{20!}{5!8!5!2!} p_1^5 p_2^8 p_3^5 p_4^2.$$

- (c) In a large shipment of 50 batteries, we now have that $(X_1, X_2, X_3, X_4) \sim MN(50; p_1, p_2, p_3, p_4)$. Therefore, it follows that $X_4 \sim Bin(50, p_4)$. Using the value of p_4 obtained in part (a), we wish to calculate

$$P(X_4 \geq 4) = 1 - P(X_4 \leq 3) = 1 - \sum_{i=0}^3 \binom{50}{i} p_4^i (1 - p_4)^{50-i} \approx 0.998036.$$

- 4.2.5 (a) Since $(X_1, X_2, X_3, X_4, X_5, X_6) \sim MN(10; 0.1, 0.05, 0.05, 0.15, 0.15, 0.5)$, the joint pmf of $(X_1, X_2, X_3, X_4, X_5, X_6)$ is given by

$$f(x_1, x_2, x_3, x_4, x_5, x_6) = \frac{10!}{x_1!x_2!x_3!x_4!x_5!x_6!} (0.1)^{x_1} (0.05)^{x_2} (0.05)^{x_3} (0.15)^{x_4} (0.15)^{x_5} (0.5)^{x_6},$$

where $x_i = 0, 1, \dots, 10$ for each $i = 1, 2, 3, 4, 5, 6$ subject to $\sum_{i=1}^6 x_i = 10$.

- (b) Note that $X_1 + X_2 + X_3 + X_4 \sim Bin(10, \underbrace{0.1 + 0.05 + 0.05 + 0.15}_{= 0.35})$. Therefore, we wish to calculate

$$P(X_1 + X_2 + X_3 + X_4 > 5) = \sum_{i=6}^{10} \binom{10}{i} (0.35)^i (0.65)^{10-i} \approx 0.0949341.$$

(c) If we define the events

$$A = \{X_5 \geq 2\} \text{ and } B = \{X_6 \geq 2\},$$

then we wish to calculate $P(A \cap B)$. We have that

$$\begin{aligned} P(A \cap B) &= 1 - P(\overline{A \cap B}) \\ &= 1 - P(\bar{A} \cup \bar{B}) \text{ by De Morgan's Law} \\ &= 1 - (P(\bar{A}) + P(\bar{B}) - P(\bar{A} \cap \bar{B})). \end{aligned} \tag{8}$$

Note that

$$P(\bar{A}) = P(X_5 \leq 1) = \sum_{i=0}^1 \binom{10}{i} (0.15)^i (0.85)^{10-i} = (0.85)^{10} + 10(0.15)(0.85)^9 = 2.35(0.85)^9,$$

$$P(\bar{B}) = P(X_6 \leq 1) = \sum_{i=0}^1 \binom{10}{i} (0.5)^i (0.5)^{10-i} = (0.5)^{10} + 10(0.5)(0.5)^9 = 11(0.5)^{10}, \text{ and}$$

$$\begin{aligned} P(\bar{A} \cap \bar{B}) &= P(X_5 \leq 1, X_6 \leq 1) \\ &= P(X_5 = 0, X_6 = 0) + P(X_5 = 0, X_6 = 1) + P(X_5 = 1, X_6 = 0) + P(X_5 = 1, X_6 = 1) \\ &= \frac{10!}{0!0!10!} (0.15)^0 (0.5)^0 (0.35)^{10} + \frac{10!}{0!1!9!} (0.15)^0 (0.5)^1 (0.35)^9 \\ &\quad + \frac{10!}{1!0!9!} (0.15)^1 (0.5)^0 (0.35)^9 + \frac{10!}{1!1!8!} (0.15)^1 (0.5)^1 (0.35)^8 \\ &= 9.1475(0.35)^8. \end{aligned}$$

Substituting the above probabilities into (8) therefore yields

$$P(A \cap B) = 1 - 2.35(0.85)^9 - 11(0.5)^{10} + 9.1475(0.35)^8 \approx 0.447018.$$

Section 4.3:

4.3.1 (a) Setting $g(X, Y) = \frac{Y}{X} = Z$, we wish to calculate

$$E(Z) = \sum_{all (x,y)} g(x, y)f(x, y) = \sum_{x=1}^2 \sum_{y=2}^3 \frac{y}{x} \cdot \frac{4}{5xy} = \sum_{x=1}^2 \sum_{y=2}^3 \frac{4}{5x^2} = \frac{8}{5} \sum_{x=1}^2 \frac{1}{x^2} = \frac{8}{5} \cdot \frac{5}{4} = 2,$$

$$E(Z^2) = \sum_{all (x,y)} (g(x, y))^2 f(x, y) = \sum_{x=1}^2 \sum_{y=2}^3 \frac{y^2}{x^2} \cdot \frac{4}{5xy} = \sum_{x=1}^2 \sum_{y=2}^3 \frac{4y}{5x^3} = \frac{4}{5} \sum_{x=1}^2 \frac{1}{x^3} \sum_{y=2}^3 y = \frac{4}{5} \cdot \frac{9}{8} \cdot 5 = \frac{9}{2},$$

and

$$Var(Z) = E(Z^2) - (E(Z))^2 = \frac{9}{2} - 4 = \frac{1}{2}.$$

(b) To calculate the correlation coefficient of X and Y , we first need to calculate $Cov(X, Y)$. Therefore, we need to find

$$E(XY) = \sum_{all (x,y)} xyf(x, y) = \sum_{x=1}^2 \sum_{y=2}^3 xy \cdot \frac{4}{5xy} = \sum_{x=1}^2 \sum_{y=2}^3 \frac{4}{5} = \frac{4}{5} \cdot 4 = \frac{16}{5},$$

$$E(X) = \sum_{all (x,y)} xf(x, y) = \sum_{x=1}^2 \sum_{y=2}^3 x \cdot \frac{4}{5xy} = \sum_{x=1}^2 \sum_{y=2}^3 \frac{4}{5y} = \frac{8}{5} \sum_{y=2}^3 \frac{1}{y} = \frac{8}{5} \cdot \frac{5}{6} = \frac{4}{3},$$

and

$$E(Y) = \sum_{all (x,y)} yf(x, y) = \sum_{x=1}^2 \sum_{y=2}^3 y \cdot \frac{4}{5xy} = \sum_{x=1}^2 \sum_{y=2}^3 \frac{4}{5x} = \frac{8}{5} \sum_{x=1}^2 \frac{1}{x} = \frac{8}{5} \cdot \frac{3}{2} = \frac{12}{5}.$$

Therefore, we have that

$$Cov(X, Y) = E(XY) - E(X)E(Y) = \frac{16}{5} - \left(\frac{4}{3}\right) \left(\frac{12}{5}\right) = \frac{16}{5} - \frac{16}{5} = 0.$$

Hence, $\rho = 0$ and the random variables X and Y are uncorrelated. In fact, though, it is straightforward to verify that X and Y are actually independent random variables.

4.3.2 Note that

$$\begin{aligned} Cov(X + a, Y + b) &= E((X + a)(Y + b)) - E(X + a)E(Y + b) \\ &= E(XY + bX + aY + ab) - (E(X) + a)(E(Y) + b) \\ &= E(XY) + bE(X) + aE(Y) + ab - E(X)E(Y) - bE(X) - aE(Y) - ab \\ &= E(XY) - E(X)E(Y) \\ &= Cov(X, Y). \end{aligned}$$

4.3.3 To calculate the correlation coefficient of X and Y , we first calculate

$$\begin{aligned}
 E(XY) &= \sum_{all (x,y)} xyf(x,y) \\
 &= (0)(0)(0.06) + (0)(1)(0.14) + (1)(0)(0.15) + (1)(1)(0.35) + (2)(0)(0.09) + (2)(1)(0.21) \\
 &= 0.35 + 0.42 \\
 &= 0.77.
 \end{aligned}$$

Next, we note that $Y \sim Bin(1, 0.7)$ and the marginal pmf of X is given by

x	0	1	2
$f_X(x)$	0.2	0.5	0.3

As a result, $E(Y) = 0.7$ and $Var(Y) = 0.7(1 - 0.7) = 0.21$. Also,

$$E(X) = (0)(0.2) + (1)(0.5) + (2)(0.3) = 1.1$$

and

$$Var(X) = (0)^2(0.2) + (1)^2(0.5) + (2)^2(0.3) - (1.1)^2 = 1.7 - 1.21 = 0.49.$$

Therefore, we obtain

$$\rho = \frac{Cov(X, Y)}{\sigma_X \sigma_Y} = \frac{0.77 - (1.1)(0.7)}{\sqrt{0.49}\sqrt{0.21}} = \frac{0}{\sqrt{0.1029}} = 0,$$

implying that the random variables X and Y are uncorrelated. In fact, though, it is straightforward to verify that X and Y are actually independent random variables.

4.3.4 (a) Looking at the given joint pmf table, we see that

$$f_X(2) + f_X(4) + f_X(6) = \frac{3}{8} + \frac{3}{8} + \frac{1}{4} = 1$$

and

$$f_Y(-1) + f_Y(1) = \left(\frac{3}{8} + p\right) + \left(\frac{5}{8} - p\right) = 1.$$

Hence, the only restriction on the value of p is to ensure that the individual entries p and $\frac{1}{4} - p$ in the table are legitimate probabilities. Hence, we require that $0 \leq p \leq \frac{1}{4}$.

(b) In order for X and Y to be uncorrelated, we require $Cov(X, Y) = 0$. Note that

$$\begin{aligned}
& E(XY) \\
&= \sum_{all (x,y)} xyf(x,y) \\
&= (2)(-1) \left(\frac{1}{8}\right) + (4)(-1) \left(\frac{1}{4}\right) + (6)(-1)p + (2)(1) \left(\frac{1}{4}\right) + (4)(1) \left(\frac{1}{8}\right) + (6)(1) \left(\frac{1}{4} - p\right) \\
&= 2 \left(\frac{1}{4} - \frac{1}{8}\right) + 4 \left(\frac{1}{8} - \frac{1}{4}\right) + 6 \left(\frac{1}{4} - 2p\right) \\
&= \frac{1}{4} - \frac{1}{2} + \frac{3}{2} - 12p \\
&= \frac{5}{4} - 12p,
\end{aligned}$$

$$E(X) = \sum_{all x} xf_X(x) = (2) \left(\frac{3}{8}\right) + (4) \left(\frac{3}{8}\right) + (6) \left(\frac{1}{4}\right) = \frac{15}{4},$$

and

$$E(Y) = \sum_{all y} yf_Y(y) = (-1) \left(\frac{3}{8} + p\right) + (1) \left(\frac{5}{8} - p\right) = \frac{1}{4} - 2p.$$

Let us now set $Cov(X, Y) = E(XY) - E(X)E(Y) = 0$, which leads to

$$\begin{aligned}
\frac{5}{4} - 12p - \frac{15}{4} \left(\frac{1}{4} - 2p\right) &= 0 \\
\frac{5}{4} - \frac{15}{16} - 12p + \frac{15}{2}p &= 0 \\
\frac{5}{16} - \frac{9}{2}p &= 0 \\
p &= \frac{5}{16} \cdot \frac{2}{9} \\
p &= \frac{5}{72}.
\end{aligned}$$

(c) If X and Y are to be independent random variables, then $Cov(X, Y)$ must be equal to 0, which implies that $p = \frac{5}{72}$ from part (b). However, if $p = \frac{5}{72}$, then note that

$$f_X(2)f_Y(1) = \left(\frac{3}{8}\right) \left(\frac{5}{8} - \frac{5}{72}\right) = \left(\frac{3}{8}\right) \left(\frac{5}{9}\right) = \frac{5}{24} \neq f(2, 1) = \frac{1}{4}.$$

Therefore, X and Y cannot be independent for any value of p .

4.3.5

Let X_1 be the number of ones that occur in n rolls of a fair six-sided die. Let X_2 be the number of twos that occur in n rolls of a fair six-sided die. Let X_3 be the number

of threes, fours, fives, or sixes that occur in n rolls of a fair six-sided die. Based on the given information, we have that

$$(X_1, X_2, X_3) \sim MN\left(n; \frac{1}{6}, \frac{1}{6}, \frac{2}{3}\right).$$

As a result of this fact, we also know that $X_i \sim \text{Bin}(n, \frac{1}{6})$ for $i = 1, 2$. Therefore, for $i = 1, 2$, we have that

$$\sigma_i = \sqrt{\text{Var}(X_i)} = \sqrt{n \left(\frac{1}{6}\right) \left(1 - \frac{1}{6}\right)} = \frac{\sqrt{5n}}{6}.$$

Applying the result of Example 4.3.5, we immediately obtain

$$\text{Cov}(X_1, X_2) = -n \left(\frac{1}{6}\right) \left(\frac{1}{6}\right) = -\frac{n}{36}.$$

Therefore, the correlation coefficient of X and Y is given by

$$\rho = \frac{\text{Cov}(X_1, X_2)}{\sigma_1 \sigma_2} = \frac{-\frac{n}{36}}{\frac{\sqrt{5n}}{6} \cdot \frac{\sqrt{5n}}{6}} = -\frac{n}{36} \cdot \frac{36}{5n} = -\frac{1}{5}.$$

4.3.6 From part (a) of Problem 4.1.5, the joint pmf of X and Y is given by

$$f(x, y) = \frac{\binom{2}{x} \binom{1}{y} \binom{7}{3-x-y}}{\binom{10}{3}} \text{ for } x = 0, 1, 2 \text{ and } y = 0, 1.$$

In order to calculate $E(XY)$, we use this joint pmf (only considering the non-zero values of x and y) to obtain

$$E(XY) = \sum_{\text{all } (x,y)} xyf(x, y) = (1)(1)f(1, 1) + (2)(1)f(2, 1) = \frac{\binom{2}{1} \binom{1}{1} \binom{7}{1}}{\binom{10}{3}} + 2 \frac{\binom{2}{2} \binom{1}{1} \binom{7}{0}}{\binom{10}{3}} = \frac{14}{120} + \frac{2}{120} = \frac{2}{15}.$$

From part (b) of Problem 4.1.5, $X \sim HG(10, 2, 3)$ and $Y \sim HG(10, 1, 3)$. As a result, we know the following means and variances:

$$\begin{aligned} E(X) &= \frac{3(2)}{10} = \frac{3}{5}, & \text{Var}(X) &= \frac{3(2)(10-2)(10-3)}{10^2(9)} = \frac{28}{75}, \\ E(Y) &= \frac{3(1)}{10} = \frac{3}{10}, & \text{Var}(Y) &= \frac{3(1)(10-1)(10-3)}{10^2(9)} = \frac{21}{100}. \end{aligned}$$

Therefore, the correlation coefficient of X and Y is given by

$$\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\frac{2}{15} - \frac{3}{5} \cdot \frac{3}{10}}{\sqrt{\frac{28}{75} \cdot \frac{21}{100}}} = -\frac{7}{150} \cdot \frac{25}{7} = -\frac{1}{6}.$$

4.3.7 Recall from Problem 4.1.6 that the joint pmf of X and Y is given by

$$f(x, y) = \frac{e^{-2}}{x!(y-x)!} \text{ for } y = 0, 1, 2, \dots \text{ and } x = 0, 1, \dots, y.$$

We use this joint pmf to calculate $E(XY)$ as follows:

$$\begin{aligned} E(XY) &= \sum_{\text{all } (x,y)} xyf(x, y) \\ &= \sum_{y=0}^{\infty} \sum_{x=0}^y xy \frac{e^{-2}}{x!(y-x)!} \\ &= \sum_{y=1}^{\infty} \sum_{x=1}^y \frac{ye^{-2}}{(x-1)!(y-x)!} \\ &= \sum_{x=1}^{\infty} \sum_{y=x}^{\infty} \frac{ye^{-2}}{(x-1)!(y-x)!} \text{ by interchanging the order of summation} \\ &= \sum_{x=1}^{\infty} \frac{e^{-1}}{(x-1)!} \sum_{y=x}^{\infty} y \frac{e^{-1}}{(y-x)!} \\ &= \sum_{x=1}^{\infty} \frac{e^{-1}}{(x-1)!} \underbrace{\sum_{z=0}^{\infty} (z+x) \frac{e^{-1}1^z}{z!}}_{= E(Z+x) \text{ where } Z \sim Poi(1)} \text{ if we let } z = y - x \\ &= \sum_{x=1}^{\infty} \frac{e^{-1}}{(x-1)!} (1+x) \\ &= \sum_{w=0}^{\infty} (w+2) \frac{e^{-1}1^w}{w!} \text{ if we let } w = x - 1 \\ &= \underbrace{E(Z+2) \text{ where } Z \sim Poi(1)} \\ &= 1 + 2 \\ &= 3. \end{aligned}$$

From part (a) of Problem 4.1.6, $X \sim Poi(1)$ and $Y \sim Poi(2)$. As a result, we know the following means and variances:

$$E(X) = Var(X) = 1 \text{ and } E(Y) = Var(Y) = 2.$$

Therefore, the correlation coefficient of X and Y is given by

$$\rho = \frac{Cov(X, Y)}{\sigma_X \sigma_Y} = \frac{3 - (1)(2)}{\sqrt{1}\sqrt{2}} = \frac{1}{\sqrt{2}}.$$

- 4.3.8 (a) For $i = 1, 2, \dots, n+1$, let X_i be the number of plays in which outcome A_i occurs. Based on the given information, it follows that $(X_1, X_2, \dots, X_{n+1}) \sim MN(m; p_1, p_2, \dots, p_{n+1})$. If N represents the net profit random variable from m independent plays of the slot machine, it follows that

$$N = \sum_{i=1}^n (a_i - d)X_i - dX_{n+1} = \sum_{i=1}^n a_i X_i - d \underbrace{\sum_{i=1}^{n+1} X_i}_{=m} = \sum_{i=1}^n a_i X_i - dm.$$

We wish to determine

$$\begin{aligned} E(N) &= E\left(\sum_{i=1}^n a_i X_i - dm\right) \\ &= \sum_{i=1}^n a_i E(X_i) - dm \\ &= \sum_{i=1}^n a_i (mp_i) - dm \quad \text{since } X_i \sim \text{Bin}(m, p_i) \text{ for each } i = 1, 2, \dots, n \\ &= m \left(\sum_{i=1}^n a_i p_i - d \right). \end{aligned}$$

- (b) Suppose that $n = 4$, $p_1 = 0.1$, and $p_2 = p_3 = p_4 = 0.04$. In addition, let $a_1 = 10$ and $a_2 = a_3 = a_4 = 25$. With these parameter values and the result we derived in part (a), we wish to determine the value of d such that

$$\begin{aligned} E(N) &< 0 \\ m \left(\sum_{i=1}^4 a_i p_i - d \right) &< 0 \\ 10(0.1) + 25(0.04) + 25(0.04) + 25(0.04) - d &< 0 \\ 4 - d &< 0 \\ d &> 4. \end{aligned}$$

Therefore, the owner should charge more than \$4 per play.

Section 4.4:

4.4.1 Using the given joint pmf, we first calculate:

$$\begin{aligned}E(X) &= 0(0.5) + 1(0.3) + 2(0.2) = 0.7, \\E(X^2) &= 0(0.5) + 1(0.3) + 4(0.2) = 1.1, \\Var(X) &= 1.1 - (0.7)^2 = 0.61, \\E(Y) &= 0(0.3) + 1(0.7) = 0.7, \\E(Y^2) &= 0(0.3) + 1(0.7) = 0.7, \\Var(Y) &= 0.7 - (0.7)^2 = 0.21, \\E(XY) &= (0)(0)(0.15) + (0)(1)(0.35) + (1)(0)(0.1) + (1)(1)(0.2) + (2)(0)(0.05) + (2)(1)(0.15) \\&= 0.5, \\Cov(X, Y) &= 0.5 - (0.7)(0.7) = 0.01.\end{aligned}$$

Therefore, it follows that

$$\begin{aligned}Var(U) &= Var(3X - 2Y) \\&= 9Var(X) + 4Var(Y) + 2(3)(-2)Cov(X, Y) \\&= 9(0.61) + 4(0.21) - 12(0.01) \\&= 6.21.\end{aligned}$$

4.4.2 Note that

$$0.5 = \rho_{XY} = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}} = \frac{Cov(X, Y)}{\sqrt{(1.69)(4)}} = \frac{Cov(X, Y)}{2.6} \implies Cov(X, Y) = 2.6(0.5) = 1.3.$$

Therefore, we have that

$$\begin{aligned}Var(Z) &= Var(2X + Y) \\&= 4Var(X) + Var(Y) + 2(2)(1)Cov(X, Y) \\&= 4(1.69) + 4 + 4(1.3) \\&= 15.96.\end{aligned}$$

Thus, $SD(Z) = \sqrt{15.96} \approx 3.995$.

4.4.3 Note that

$$-0.7 = \rho_{XY} = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}} = \frac{Cov(X, Y)}{\sqrt{(13)(34)}} = \frac{Cov(X, Y)}{\sqrt{442}} \implies Cov(X, Y) = -0.7\sqrt{442}.$$

Therefore, we have that

$$\begin{aligned}Var(X - 2Y) &= Var(X) + 4Var(Y) + 2(1)(-2)Cov(X, Y) \\&= 13 + 4(34) - 4(-0.7\sqrt{442}) \\&= 149 + 2.8\sqrt{442} \\&\approx 207.867.\end{aligned}$$

4.4.4 Since X and Y are independent random variables, $Cov(X, Y) = 0$. As a result, this simply leads to

$$\begin{aligned} Var(2X + Y) &= 4Var(X) + Var(Y) = 4(1) + 2 = 6, \\ Var(X - 2Y) &= Var(X) + 4Var(Y) = 1 + 4(2) = 9, \text{ and} \\ Cov(2X + Y, X - 2Y) &= (2)(1)Cov(X, X) + (1)(-2)Cov(Y, Y) \\ &= 2Var(X) - 2Var(Y) \\ &= 2(1) - 2(2) \\ &= -2. \end{aligned}$$

Therefore, the desired correlation coefficient is given by

$$Corr(2X + Y, X - 2Y) = \frac{Cov(2X + Y, X - 2Y)}{\sqrt{Var(2X + Y)Var(X - 2Y)}} = \frac{-2}{\sqrt{6 \cdot 9}} = -\frac{2}{3\sqrt{6}} \approx -0.272.$$

4.4.5 (a) First of all, since each of Jack and Jill independently toss a fair coin three times, it follows that X and Y are independent random variables. Moreover, we have that $X \sim Bin(3, 0.5)$ and $Y \sim Bin(3, 0.5)$. As a result, it follows that

$$\begin{aligned} E(U) &= E(X + Y) = E(X) + E(Y) = 3(0.5) + 3(0.5) = 3, \\ E(V) &= E(X - Y) = E(X) - E(Y) = 3(0.5) - 3(0.5) = 0, \\ Var(U) &= Var(X + Y) = Var(X) + Var(Y) = 3(0.5)(0.5) + 3(0.5)(0.5) = 1.5, \text{ and} \\ Var(V) &= Var(X - Y) = Var(X) + Var(Y) = 3(0.5)(0.5) + 3(0.5)(0.5) = 1.5. \end{aligned}$$

(b) Since X and Y are independent random variables, $Cov(X, Y) = 0$. Therefore, we have that

$$\begin{aligned} Cov(U, V) &= Cov(X + Y, X - Y) \\ &= (1)(1)Cov(X, X) + (1)(-1)Cov(Y, Y) \\ &= Var(X) - Var(Y) \\ &= 0 \text{ since } X \text{ and } Y \text{ are identically distributed.} \end{aligned}$$

(c) Let us consider $P(U = 0, V = 2)$, which is given by

$$\begin{aligned} P(U = 0, V = 2) &= P(X + Y = 0, X - Y = 2) \\ &= P(X = 1, Y = -1) \\ &= 0 \text{ since it is impossible for } Y \text{ to be negative.} \end{aligned}$$

However, note that

$$P(U = 0) = P(X + Y = 0) = \binom{6}{0}(0.5)^6 = 0.015625 \text{ since } X + Y \sim Bin(6, 0.5)$$

and

$$\begin{aligned}
P(V = 2) &= P(X - Y = 2) \\
&= P(X = 2, Y = 0) + P(X = 3, Y = 1) \\
&= P(X = 2)P(Y = 0) + P(X = 3)P(Y = 1) \quad \text{since } X \text{ and } Y \text{ are independent} \\
&= \binom{3}{2}(0.5)^3 \binom{3}{0}(0.5)^3 + \binom{3}{3}(0.5)^3 \binom{3}{1}(0.5)^3 \\
&= 0.09375.
\end{aligned}$$

Since $P(U = 0, V = 2) \neq P(U = 0)P(V = 2)$, U and V are not independent random variables.

4.4.6 (a) For $i = 1, 2, \dots, 100$, define the following events:

$$\begin{aligned}
A_i &= \text{“the student knows the correct answer to the } i^{\text{th}} \text{ question”}, \\
B_i &= \text{“the student gets the } i^{\text{th}} \text{ question correct”}.
\end{aligned}$$

If the student does not refrain from guessing the correct answer, it follows that

$$P(B_i) = P(A_i)P(B_i|A_i) + P(\bar{A}_i)P(B_i|\bar{A}_i) = p_i \cdot 1 + (1 - p_i) \cdot \frac{1}{5} = \frac{1 + 4p_i}{5}.$$

For $i = 1, 2, \dots, 100$, let X_i be an indicator random variable defined as follows:

$$\begin{aligned}
X_i &= 0 \text{ if the } i^{\text{th}} \text{ question is answered incorrectly, and} \\
X_i &= 1 \text{ if the } i^{\text{th}} \text{ question is answered correctly.}
\end{aligned}$$

Note that $X_i \sim \text{Bin}(1, P(B_i))$. Moreover, since questions are answered independently of each other, the random variables $X_1, X_2, \dots, X_{100}$ are independent. If we now let T represent the student's total mark, then it follows that

$$T = \sum_{i=1}^{100} X_i - \frac{1}{4} \sum_{i=1}^{100} (1 - X_i) = \sum_{i=1}^{100} X_i - 25 + \frac{1}{4} \sum_{i=1}^{100} X_i = \frac{5}{4} \sum_{i=1}^{100} X_i - 25.$$

We wish to find

$$E(T) = \frac{5}{4} \sum_{i=1}^{100} E(X_i) - 25 = \frac{5}{4} \sum_{i=1}^{100} \frac{1 + 4p_i}{5} - 25 = 25 + \sum_{i=1}^{100} p_i - 25 = \sum_{i=1}^{100} p_i$$

and

$$\begin{aligned}
Var(T) &= \left(\frac{5}{4}\right)^2 \sum_{i=1}^{100} Var(X_i) \\
&= \frac{25}{16} \sum_{i=1}^{100} \frac{1+4p_i}{5} \left(1 - \frac{1+4p_i}{5}\right) \text{ since } X_i \sim Bin(1, P(B_i)) \\
&= \frac{25}{16} \sum_{i=1}^{100} \frac{4(1+4p_i)(1-p_i)}{25} \\
&= \frac{1}{4} \sum_{i=1}^{100} (4p_i(1-p_i) + 1-p_i) \\
&= \sum_{i=1}^{100} p_i(1-p_i) + \frac{1}{4} \sum_{i=1}^{100} (1-p_i) \\
&= \sum_{i=1}^{100} p_i(1-p_i) + \frac{1}{4} \left(100 - \sum_{i=1}^{100} p_i\right).
\end{aligned}$$

- (b) If the student refrains from guessing the correct answer (i.e., the student simply does not answer a question they do not know and are therefore not penalized for giving an incorrect answer), then $P(B_i) = p_i$ and the student's total mark is simply given by $T = \sum_{i=1}^{100} X_i$. As a result, we immediately obtain

$$E(T) = \sum_{i=1}^{100} E(X_i) = \sum_{i=1}^{100} p_i$$

and

$$Var(T) = \sum_{i=1}^{100} Var(X_i) = \sum_{i=1}^{100} p_i(1-p_i).$$

When (i) $p_i = 0.9$ for $i = 1, 2, \dots, 100$, we calculate:

$$Var(T) = 11.5 \text{ for part (a) and } Var(T) = 9 \text{ for part (b).}$$

When (ii) $p_i = 0.5$ for $i = 1, 2, \dots, 100$, we calculate:

$$Var(T) = 37.5 \text{ for part (a) and } Var(T) = 25 \text{ for part (b).}$$

4.4.7 (a) Consider the sample variance S^2 given by

$$\begin{aligned}
S^2 &= \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \\
&= \frac{1}{n-1} \sum_{i=1}^n (X_i^2 - 2\bar{X}X_i + \bar{X}^2) \\
&= \frac{1}{n-1} \left(\sum_{i=1}^n X_i^2 - 2\bar{X} \underbrace{\sum_{i=1}^n X_i}_{=n\bar{X}} + n\bar{X}^2 \right) \\
&= \frac{1}{n-1} \left(\sum_{i=1}^n X_i^2 - 2n\bar{X}^2 + n\bar{X}^2 \right) \\
&= \frac{1}{n-1} \left(\sum_{i=1}^n X_i^2 - n(\bar{X})^2 \right).
\end{aligned}$$

(b) Using the representation of S^2 in part (a), we wish to determine

$$\begin{aligned}
E(S^2) &= \frac{1}{n-1} E \left(\sum_{i=1}^n X_i^2 - n(\bar{X})^2 \right) \\
&= \frac{1}{n-1} \left(\sum_{i=1}^n E(X_i^2) - nE(\bar{X}^2) \right) \\
&= \frac{1}{n-1} \left(\sum_{i=1}^n \left(\text{Var}(X_i) + E(X_i)^2 \right) - n \left(\text{Var}(\bar{X}) + E(\bar{X})^2 \right) \right) \\
&= \frac{1}{n-1} \left(\sum_{i=1}^n (\sigma^2 + \mu^2) - n \left(\frac{\sigma^2}{n} + \mu^2 \right) \right) \text{ using the results of Example 4.4.1} \\
&= \frac{1}{n-1} (n\sigma^2 + n\mu^2 - \sigma^2 - n\mu^2) \\
&= \frac{1}{n-1} (n-1)\sigma^2 \\
&= \sigma^2.
\end{aligned}$$

4.4.8 (a) For $i = 2, 3, \dots, n$, we wish to calculate

$$\begin{aligned}
\text{Cov}(X_{i-1}, X_i) &= \text{Cov}(Y_{i-2} + Y_{i-1}, Y_{i-1} + Y_i) \\
&= \text{Cov}(Y_{i-2}, Y_{i-1}) + \text{Cov}(Y_{i-2}, Y_i) + \text{Cov}(Y_{i-1}, Y_{i-1}) + \text{Cov}(Y_{i-1}, Y_i) \\
&= 0 + 0 + \text{Var}(Y_{i-1}) + 0 \text{ since } Y_0, Y_1, \dots, Y_n \text{ are uncorrelated} \\
&= \sigma^2.
\end{aligned}$$

(b) First of all, if $j \geq i + 2$, we see that

$$\begin{aligned}
Cov(X_i, X_j) &= Cov(Y_{i-1} + Y_i, Y_{j-1} + Y_j) \\
&= Cov(Y_{i-1}, Y_{j-1}) + Cov(Y_{i-1}, Y_j) + Cov(Y_i, Y_{j-1}) + Cov(Y_i, Y_j) \\
&= 0 + 0 + 0 + 0 \quad \text{since } Y_0, Y_1, \dots, Y_n \text{ are uncorrelated} \\
&= 0.
\end{aligned}$$

As a result of this observation, it follows that

$$\begin{aligned}
Var\left(\sum_{i=1}^n X_i\right) &= \sum_{i=1}^n Var(X_i) + 2 \sum_{i=1}^{n-1} \sum_{j=i+1}^n Cov(X_i, X_j) \\
&= \sum_{i=1}^n Var(Y_{i-1} + Y_i) + 2 \underbrace{\sum_{i=1}^{n-1} Cov(X_i, X_{i+1})}_{\text{Let } k = i + 1} \\
&= \sum_{i=1}^n 2\sigma^2 + 2 \sum_{k=2}^n Cov(X_{k-1}, X_k) \quad \text{since } Y_0, Y_1, \dots, Y_n \text{ are uncorrelated} \\
&= 2n\sigma^2 + 2 \sum_{k=2}^n \sigma^2 \quad \text{by the result of part (a)} \\
&= 2n\sigma^2 + 2(n-1)\sigma^2 \\
&= (4n-2)\sigma^2.
\end{aligned}$$

4.4.9 For $i = 1, 2, \dots, 24$, let X_i be an indicator random variable defined as follows:

$$\begin{aligned}
X_i &= 0 \text{ if item } i \text{ is not completely separate, and} \\
X_i &= 1 \text{ if item } i \text{ is completely separate.}
\end{aligned}$$

Therefore, $Y = \sum_{i=1}^{24} X_i$ represents the number of items which are completely separate. Letting $p_i = P(X_i = 1)$, note that

$$p_i = P(1 \text{ cut is successfully made}) = 0.9 \text{ for } i = 1, 24$$

and

$$p_i = P(2 \text{ independent cuts are successfully made}) = (0.9)^2 = 0.81 \text{ for } i = 2, 3, \dots, 23.$$

Since $X_i \sim \text{Bin}(1, p_i)$, we have that $\mu_i = p_i$ and $\sigma_i^2 = p_i(1-p_i)$ for each $i = 1, 2, \dots, 24$.

Next, we deduce that X_i and X_j are independent random variables whenever $j \geq i + 2$ since there are no common connecting pieces and cuts are made independently.

However, when $j = i + 1$, we observe that

$$\begin{aligned} E(X_i X_{i+1}) &= P(X_i = 1, X_{i+1} = 1) \\ &= P(2 \text{ independent cuts are successfully made}) \\ &= 0.81 \text{ for } i = 1, 23 \end{aligned}$$

and

$$\begin{aligned} E(X_i X_{i+1}) &= P(X_i = 1, X_{i+1} = 1) \\ &= P(3 \text{ independent cuts are successfully made}) \\ &= (0.9)^3 \\ &= 0.729 \text{ for } i = 2, 3, \dots, 22. \end{aligned}$$

This leads to

$$Cov(X_i, X_{i+1}) = E(X_i X_{i+1}) - E(X_i)E(X_{i+1}) = 0.81 - (0.9)(0.81) = 0.081 \text{ for } i = 1, 23, \text{ and}$$

$$Cov(X_i, X_{i+1}) = E(X_i X_{i+1}) - E(X_i)E(X_{i+1}) = 0.729 - (0.81)(0.81) = 0.0729 \text{ for } i = 2, 3, \dots, 22.$$

Therefore, we can now obtain

$$E(Y) = \sum_{i=1}^{24} \mu_i = 0.9 + 22(0.81) + 0.9 = 19.62,$$

$$\begin{aligned} Var(Y) &= \sum_{i=1}^{24} \sigma_i^2 + 2 \sum_{i=1}^{23} \sum_{j=i+1}^{24} Cov(X_i, X_j) \\ &= (0.9)(1 - 0.9) + 22(0.81)(1 - 0.81) + (0.9)(1 - 0.9) + 2 \sum_{i=1}^{23} Cov(X_i, X_{i+1}) \\ &= 3.5658 + 2(0.081) + 2(21)(0.0729) + 2(0.081) \\ &= 3.5658 + 3.3858 \\ &= 6.9516, \end{aligned}$$

and

$$SD(Y) = \sqrt{6.9516} \approx 2.637.$$

4.4.10 For $i = 1, 2, 3, 4, 5$, let X_i be an indicator random variable defined as follows:

$$\begin{aligned} X_i &= 0 \text{ if island } i \text{ is not completely cut off, and} \\ X_i &= 1 \text{ if island } i \text{ is completely cut off.} \end{aligned}$$

As a result, $Y = \sum_{i=1}^5 X_i$ represents the number of islands which are completely cut off. Let $p_i = P(X_i = 1)$ for $i = 1, 2, 3, 4, 5$. Based on the layout of the islands, it follows that

$$p_5 = P(4 \text{ bridges are independently closed}) = p^4$$

and

$$p_i = P(3 \text{ bridges are independently closed}) = p^3 \text{ for } i = 1, 2, 3, 4.$$

Since $X_i \sim \text{Bin}(1, p_i)$, we have that $\mu_i = p_i$ and $\sigma_i^2 = p_i(1 - p_i)$ for each $i = 1, 2, 3, 4, 5$.

We deduce that X_1 and X_3 are independent random variables since there are no common connecting bridges and bridge closures are independent of each other. The same reasoning applies to X_2 and X_4 . Next, in looking at X_1 and X_2 , note that

$$\begin{aligned} \text{Cov}(X_1, X_2) &= E(X_1 X_2) - E(X_1)E(X_2) \\ &= P(X_1 = 1, X_2 = 1) - (p^3)(p^3) \\ &= P(5 \text{ bridges are independently closed}) - p^6 \\ &= p^5 - p^6 \\ &= p^5(1 - p). \end{aligned}$$

In fact, we also have that $\text{Cov}(X_2, X_3) = \text{Cov}(X_3, X_4) = \text{Cov}(X_1, X_4) = p^5(1 - p)$. Finally, in looking at X_1 and X_5 , note that

$$\begin{aligned} \text{Cov}(X_1, X_5) &= E(X_1 X_5) - E(X_1)E(X_5) \\ &= P(X_1 = 1, X_5 = 1) - (p^3)(p^4) \\ &= P(6 \text{ bridges are independently closed}) - p^7 \\ &= p^6 - p^7 \\ &= p^6(1 - p). \end{aligned}$$

By the same logic, $\text{Cov}(X_2, X_5) = \text{Cov}(X_3, X_5) = \text{Cov}(X_4, X_5) = p^6(1 - p)$.

Putting these pieces together, we can now determine

$$E(Y) = \sum_{i=1}^5 \mu_i = 4p^3 + p^4 = p^3(4 + p)$$

and

$$\begin{aligned} \text{Var}(Y) &= \sum_{i=1}^5 \sigma_i^2 + 2 \sum_{i=1}^4 \sum_{j=i+1}^5 \text{Cov}(X_i, X_j) \\ &= 4p^3(1 - p^3) + p^4(1 - p^4) + 2(4p^5(1 - p) + 4p^6(1 - p)) \\ &= 4p^3(1 - p^3) + p^4(1 - p^4) + 8p^5(1 - p) + 8p^6(1 - p) \\ &= 4p^3(1 - p^3) + p^4(1 - p^4) + 8p^5(1 - p)(1 + p) \\ &= 4p^3(1 - p^3) + p^4(1 - p^4) + 8p^5(1 - p^2). \end{aligned}$$

Section 4.5:

4.5.1 (a) We wish to determine the conditional pmf of X given $Y = 6$, which is given by

$$f_{X|Y}(x|6) = \frac{f(x, 6)}{f_Y(6)}.$$

Using the marginal pmf of Y found in part (b) of Problem 4.1.2, we have:

$$\begin{aligned} f_{X|Y}(1|6) &= \frac{f(1, 6)}{f_Y(6)} = \frac{0.06}{0.3} = \frac{1}{5}, \\ f_{X|Y}(2|6) &= \frac{f(2, 6)}{f_Y(6)} = \frac{0.09}{0.3} = \frac{3}{10}, \\ f_{X|Y}(3|6) &= \frac{f(3, 6)}{f_Y(6)} = \frac{0.15}{0.3} = \frac{1}{2}. \end{aligned}$$

Therefore, the conditional pmf of X given $Y = 6$ can be summarized as follows:

x	1	2	3
$f_{X Y}(x 6)$	$\frac{1}{5}$	$\frac{3}{10}$	$\frac{1}{2}$

Moreover, its conditional mean is given by

$$E(X|Y = 6) = \sum_{all\ x} x f_{X|Y}(x|6) = 1 \left(\frac{1}{5} \right) + 2 \left(\frac{3}{10} \right) + 3 \left(\frac{1}{2} \right) = \frac{23}{10} = 2.3.$$

(b) We wish to determine the conditional pmf of Y given $X = 2$, which is given by

$$f_{Y|X}(y|2) = \frac{f(2, y)}{f_X(2)}.$$

Using the marginal pmf of X found in part (b) of Problem 4.1.2, we have:

$$\begin{aligned} f_{Y|X}(2|2) &= \frac{f(2, 2)}{f_X(2)} = \frac{0.06}{0.2} = \frac{3}{10}, \\ f_{Y|X}(4|2) &= \frac{f(2, 4)}{f_X(2)} = \frac{0.05}{0.2} = \frac{1}{4}, \\ f_{Y|X}(6|2) &= \frac{f(2, 6)}{f_X(2)} = \frac{0.09}{0.2} = \frac{9}{20}. \end{aligned}$$

Therefore, the conditional pmf of Y given $X = 2$ can be summarized as follows:

y	2	4	6
$f_{Y X}(y 2)$	$\frac{3}{10}$	$\frac{1}{4}$	$\frac{9}{20}$

Furthermore, its conditional mean is given by

$$E(Y|X = 2) = \sum_{all\ y} y f_{Y|X}(y|2) = 2 \left(\frac{3}{10} \right) + 4 \left(\frac{1}{4} \right) + 6 \left(\frac{9}{20} \right) = \frac{43}{10} = 4.3.$$

- (c) We wish to determine the conditional pmf of Y given $D = |X - Y| = 1$, which is given by

$$\begin{aligned}
 f_{Y|D}(y|1) &= \frac{P(D = 1, Y = y)}{f_D(1)} \\
 &= \frac{P(|X - y| = 1, Y = y)}{f_D(1)} \\
 &= \frac{P(X = y - 1, Y = y) + P(X = y + 1, Y = y)}{f_D(1)} \\
 &= \frac{f(y - 1, y) + f(y + 1, y)}{f_D(1)}.
 \end{aligned}$$

From part (c) of Problem 4.1.2, $f_D(1) = 0.44$. Therefore, we have:

$$\begin{aligned}
 f_{Y|D}(2|1) &= \frac{f(1, 2) + f(3, 2)}{0.44} = \frac{0.09 + 0.15}{0.44} = \frac{0.24}{0.44} = \frac{6}{11}, \\
 f_{Y|D}(4|1) &= \frac{f(3, 4) + f(5, 4)}{0.44} = \frac{0.2 + 0}{0.44} = \frac{5}{11}, \\
 f_{Y|D}(6|1) &= \frac{f(5, 6) + f(7, 6)}{0.44} = \frac{0 + 0}{0.44} = 0.
 \end{aligned}$$

From these results, the conditional pmf of Y given $D = 1$ is simply given by

$$\begin{array}{c|cc}
 y & 2 & 4 \\
 \hline
 f_{Y|D}(y|1) & \frac{6}{11} & \frac{5}{11}
 \end{array}$$

Thus, its conditional mean is given by

$$E(Y|D = 1) = \sum_{\text{all } y} y f_{Y|D}(y|1) = 2 \left(\frac{6}{11} \right) + 4 \left(\frac{5}{11} \right) = \frac{32}{11}.$$

- 4.5.2 (a) We wish to determine the conditional pmf of X given $Y = y$ where $y \in \{0, 1, 2, \dots\}$. This conditional pmf is given by

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}.$$

Using the given joint pmf as well as the marginal pmf of Y found in part (a) of Problem 4.1.6, we have:

$$\begin{aligned}
 f_{X|Y}(x|y) &= \frac{\frac{e^{-2}}{x!(y-x)!}}{\frac{e^{-2}2^y}{y!}} \\
 &= \frac{y!}{x!(y-x)!} \left(\frac{1}{2} \right)^y \\
 &= \binom{y}{x} \left(\frac{1}{2} \right)^x \left(\frac{1}{2} \right)^{y-x} \quad \text{for } x = 0, 1, \dots, y.
 \end{aligned}$$

Looking at the derived formula, we see that the conditional random variable X given $Y = y$ has a $\text{Bin}(y, \frac{1}{2})$ distribution.

- (b) We wish to determine the conditional pmf of Y given $X = x$ where $x \in \{0, 1, 2, \dots\}$. This conditional pmf is given by

$$f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)}.$$

Using the given joint pmf as well as the marginal pmf of X found in part (a) of Problem 4.1.6, we have:

$$\begin{aligned} f_{Y|X}(y|x) &= \frac{\frac{e^{-2}}{x!(y-x)!}}{\frac{e^{-1}1^x}{x!}} \\ &= \frac{e^{-2}}{x!(y-x)!} \cdot \frac{x!}{e^{-1}} \\ &= \frac{e^{-1}}{(y-x)!} \\ &= \frac{e^{-1}1^{y-x}}{(y-x)!} \quad \text{for } y = x, x+1, x+2, \dots \end{aligned}$$

Looking at the above formula, we see that the conditional random variable Y given $X = x$ has the same probability distribution as $W + x$, where $W \sim \text{Poi}(1)$.

- (c) First of all, by making use of the formula for the mean of a $\text{Bin}(y, \frac{1}{2})$ distribution, we simply obtain

$$E(X|Y = y) = \frac{y}{2}.$$

In addition, from the observation we made in part (b), it immediately follows that

$$E(Y|X = x) = E(W + x) = E(W) + x = 1 + x.$$

- 4.5.3 (a) From Problem 4.2.1, $(X_A, X_B, X_C, X_D) \sim MN(25; 0.1, 0.4, 0.3, 0.2)$. Therefore, applying the result of Example 4.5.4, we know that (X_A, X_B, X_C) given $X_D = 4$ has a $MN(21; \frac{0.1}{0.8}, \frac{0.4}{0.8}, \frac{0.3}{0.8})$ distribution. As a result, it follows that

$$\begin{aligned} P(X_A = 3, X_C = 11|X_D = 4) &= P(X_A = 3, X_B = 7, X_C = 11|X_D = 4) \\ &= \frac{21!}{3! 7! 11!} \left(\frac{1}{8}\right)^3 \left(\frac{1}{2}\right)^7 \left(\frac{3}{8}\right)^{11} \\ &\approx 0.013319. \end{aligned}$$

- (b) Let Y be the number of policy holders chosen that are either class B or class D . Therefore, it immediately follows that $(X_A, X_C, Y) \sim MN(25; 0.1, 0.3, 0.6)$. Once again, by

the result of Example 4.5.4, we know that (X_A, X_C) given $Y = 9$ has a $MN(16; \frac{0.1}{0.4}, \frac{0.3}{0.4})$ distribution. As a result, we obtain

$$P(X_C = 11|Y = 9) = P(X_A = 5, X_C = 11|Y = 9) = \frac{16!}{5!11!} \left(\frac{1}{4}\right)^5 \left(\frac{3}{4}\right)^{11} \approx 0.180159.$$

4.5.4 (a) We wish to find the joint pmf of (X, Y) , which is given by

$$f(x, y) = P(X = x, Y = y) = P(X = x)P(Y = y|X = x). \quad (9)$$

Based on the given information, it follows that $X \sim HG(8, 5, 2)$, and so

$$P(X = x) = \frac{\binom{5}{x}\binom{3}{2-x}}{\binom{8}{2}} \text{ for } x = 0, 1, 2. \quad (10)$$

Next, we turn our attention to the quantity $P(Y = y|X = x)$. For convenience, let $\mathcal{Y}$ = “yellow balls” and $\mathcal{R}$ = “red balls”. Given $X = x$ where $x \in \{0, 1, 2\}$, this means that after the first two draws are made, the box contains 6 balls consisting of $5 - x$ $\mathcal{Y}$ and $1 + x$ $\mathcal{R}$. We are interested in what happens over the next two draws. Note that when $x = 0$, the conditional random variable Y given $X = 0$ is either equal to 1 or 2. However, when $x = 1$ or 2, the conditional random variable Y given $X = x$ must either be equal to $x, x + 1$, or $x + 2$. With this in mind, it therefore follows that

$$\begin{aligned} P(Y = y|X = x) &= P(y - x \text{ } \mathcal{Y} \text{ selected in next 2 draws from } 5 - x \text{ } \mathcal{Y} \text{ and } 1 + x \text{ } \mathcal{R}) \\ &= \frac{\binom{5-x}{y-x}\binom{1+x}{2+x-y}}{\binom{6}{2}} \text{ for } y = \max(x, 1), \dots, x + 2. \end{aligned} \quad (11)$$

Substituting (10) and (11) into (9) yields

$$\begin{aligned} f(x, y) &= \frac{\binom{5}{x}\binom{3}{2-x}}{\binom{8}{2}} \cdot \frac{\binom{5-x}{y-x}\binom{1+x}{2+x-y}}{\binom{6}{2}} \\ &= \frac{5!}{x!(5-x)!} \cdot \frac{3!}{(2-x)!(1+x)!} \cdot \frac{(5-x)!}{(y-x)!(5-y)!} \cdot \frac{(1+x)!}{(2+x-y)!(y-1)!} \cdot \frac{2!6!2!4!}{8!6!} \\ &= \frac{2!}{x!(2-x)!} \cdot \frac{4!}{(y-1)!(5-y)!} \cdot \frac{2!}{(y-x)!(2+x-y)!} \\ &= \frac{8!}{3!5!} \\ &= \frac{\binom{2}{x}\binom{4}{y-1}\binom{2}{y-x}}{\binom{8}{3}} \text{ for } x = 0, 1, 2 \text{ and } y = \max(x, 1), \dots, x + 2. \end{aligned}$$

(b) We wish to determine the conditional pmf of X given $Y = y$ where $y \in \{1, 2, 3, 4\}$. This conditional pmf is given by

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}.$$

However, $Y \sim HG(8, 5, 4)$, and so

$$\begin{aligned}
f_{X|Y}(x|y) &= \frac{\binom{2}{x} \binom{4}{y-1} \binom{2}{y-x}}{\binom{8}{3}} \cdot \frac{\binom{8}{4}}{\binom{5}{y} \binom{3}{4-y}} \\
&= \frac{\binom{2}{x} \binom{4}{y-1} \binom{2}{y-x} \frac{8!}{4! 4!}}{\frac{8!}{3! 5!} \cdot \frac{5!}{y!(5-y)!} \cdot \frac{3!}{(4-y)!(y-1)!}} \\
&= \frac{\binom{2}{x} \binom{4}{y-1} \binom{2}{y-x}}{\frac{4}{5} \cdot \frac{5 \times 4!}{y(y-1)!(5-y)!} \cdot \frac{3!}{(4-y)!(y-1)!}} \\
&= \frac{\binom{2}{x} \binom{4}{y-1} \binom{2}{y-x}}{\binom{4}{y-1} \binom{4}{y}} \\
&= \frac{\binom{2}{x} \binom{2}{y-x}}{\binom{4}{y}} \text{ for } x = \max(0, y-2), \dots, \min(2, y).
\end{aligned}$$

Looking at the above formula, we see that the conditional random variable X given $Y = y$ has a $HG(4, 2, y)$ distribution.

- (c) Making use of the formula for the mean of a $HG(4, 2, y)$ distribution, we immediately obtain

$$E(X|Y = y) = \frac{y \cdot 2}{4} = \frac{y}{2}.$$

- 4.5.5 (a) Since the selection process is performed without replacement, it immediately follows that $Y \sim HG(14, 5, 6)$. Using this fact, it follows that the conditional pmf of X given $Y = 1$ is given by

$$f_{X|Y}(x|1) = P(X = x|Y = 1) = \frac{P(X = x, Y = 1)}{P(Y = 1)} = \frac{\frac{\binom{3}{x} \binom{6}{5-x} \binom{5}{1}}{\binom{14}{6}}}{\frac{\binom{5}{1} \binom{9}{5}}{\binom{14}{6}}} = \frac{\binom{3}{x} \binom{6}{5-x}}{\binom{9}{5}} \text{ for } x = 0, 1, 2, 3.$$

Note that the above conditional pmf is that of a $HG(9, 3, 5)$ distribution.

- (b) Making use of the distributional observation in part (a), it immediately follows that

$$E(X|Y = 1) = \frac{5 \cdot 3}{9} = \frac{5}{3}.$$

- (c) Let Z denote the number of red balls selected. If we are now selecting balls with replacement, then the joint probability distribution of (X, Y, Z) is $MN(6; \frac{3}{14}, \frac{5}{14}, \frac{6}{14})$. By the result of Example 4.5.4, we have that (X, Z) given $Y = 1$ has a $MN(5; \frac{1}{3}, \frac{2}{3})$

distribution. Therefore, the conditional pmf of X given $Y = 1$ is given by

$$\begin{aligned} f_{X|Y}(x|1) &= P(X = x|Y = 1) \\ &= P(X = x, Z = 5 - x|Y = 1) \\ &= \frac{5!}{x!(5-x)!} \left(\frac{1}{3}\right)^x \left(\frac{2}{3}\right)^{5-x} \text{ for } x = 0, 1, 2, 3, 4, 5. \end{aligned}$$

Since we recognize this conditional pmf as that of a $Bin(5, \frac{1}{3})$ distribution, it immediately follows that

$$E(X|Y = 1) = 5 \cdot \frac{1}{3} = \frac{5}{3}.$$

4.5.6 Let $T = X_1 + X_2$. We are interested in determining the conditional pmf of X_1 given $T = m$, to be denoted by

$$g(x_1|m) = P(X_1 = x_1|T = m) = \frac{P(X_1 = x_1, T = m)}{P(T = m)} = \frac{P(X_1 = x_1, X_1 + X_2 = m)}{P(T = m)}.$$

First of all, we investigate the numerator:

$$\begin{aligned} P(X_1 = x_1, X_1 + X_2 = m) &= P(X_1 = x_1, x_1 + X_2 = m) \\ &= P(X_1 = x_1, X_2 = m - x_1) \\ &= P(X_1 = x_1)P(X_2 = m - x_1) \text{ since } X_1 \text{ and } X_2 \text{ are independent.} \end{aligned}$$

Since $X_i \sim Poi(\mu_i)$ for $i = 1, 2$ and X_1 and X_2 are independent, we know from Example 4.1.4 that

$$T = X_1 + X_2 \sim Poi(\mu_1 + \mu_2).$$

As a result, we now obtain

$$\begin{aligned} g(x_1|m) &= \frac{\frac{e^{-\mu_1} \mu_1^{x_1}}{x_1!} \cdot \frac{e^{-\mu_2} \mu_2^{m-x_1}}{(m-x_1)!}}{\frac{e^{-(\mu_1+\mu_2)} (\mu_1+\mu_2)^m}{m!}} \text{ provided that } x_1 \geq 0 \text{ and } m - x_1 \geq 0 \\ &= \frac{m!}{x_1!(m-x_1)!} \cdot \frac{\mu_1^{x_1} \mu_2^{m-x_1}}{(\mu_1 + \mu_2)^{x_1} (\mu_1 + \mu_2)^{m-x_1}} \\ &= \binom{m}{x_1} \left(\frac{\mu_1}{\mu_1 + \mu_2}\right)^{x_1} \left(1 - \frac{\mu_1}{\mu_1 + \mu_2}\right)^{m-x_1} \text{ for } x_1 = 0, 1, \dots, m. \end{aligned}$$

We remark that X_1 given $X_1 + X_2 = m$ is distributed according to the $Bin(m, \frac{\mu_1}{\mu_1 + \mu_2})$ distribution. This immediately implies that

$$E(X_1|X_1 + X_2 = m) = m \cdot \frac{\mu_1}{\mu_1 + \mu_2} = \frac{m\mu_1}{\mu_1 + \mu_2}.$$

4.5.7 (a) First of all, define $Y_i = X_i - 1$ for $i = 1, 2, 3$. Note that the pmf of Y_i is given by

$$P(Y_i = y) = P(X_i - 1 = y_i) = f_i(y_i + 1) = p(1-p)^{y_i+1-1} = p(1-p)^{y_i} \text{ for } y_i = 0, 1, 2, \dots$$

In other words, $Y_i \sim \text{Geo}(p)$ for each $i = 1, 2, 3$. Since Y_1 , Y_2 , and Y_3 are also independent, we know from Example 4.3.3 that $Y_1 + Y_2 \sim \text{NB}(2, p)$, and so

$$P(Y_1 + Y_2 = y) = \binom{y+2-1}{y} p^2(1-p)^y = (y+1)p^2(1-p)^y \text{ for } y = 0, 1, 2, \dots$$

Therefore, we easily obtain the pmf of $X_1 + X_2$ as follows:

$$\begin{aligned} P(X_1 + X_2 = n) &= P(Y_1 + 1 + Y_2 + 1 = n) \\ &= P(Y_1 + Y_2 = n - 2) \\ &= (n - 2 + 1)p^2(1-p)^{n-2} \\ &= (n - 1)p^2(1-p)^{n-2} \text{ for } n = 2, 3, 4, \dots \end{aligned}$$

With this result in place, we now turn our attention to finding the conditional pmf of X_1 given $X_1 + X_2 = n$, which is given by

$$\begin{aligned} P(X_1 = x | X_1 + X_2 = n) &= \frac{P(X_1 = x, X_1 + X_2 = n)}{P(X_1 + X_2 = n)} \\ &= \frac{P(X_1 = x, X_2 = n - x)}{P(X_1 + X_2 = n)} \\ &= \frac{P(X_1 = x)P(X_2 = n - x)}{P(X_1 + X_2 = n)} \text{ since } X_1 \text{ and } X_2 \text{ are independent} \\ &= \frac{p(1-p)^{x-1} \cdot p(1-p)^{n-x-1}}{(n-1)p^2(1-p)^{n-2}} \text{ provided that } x \geq 1 \text{ and } n-x \geq 1 \\ &= \frac{1}{n-1} \text{ for } x = 1, 2, \dots, n-1. \end{aligned}$$

Note that the above function is the pmf of a $DU(1, n-1)$ distribution.

(b) In a similar fashion, we know from Example 4.3.3 that $Y_1 + Y_2 + Y_3 \sim \text{NB}(3, p)$, so that its pmf is

$$P(Y_1 + Y_2 + Y_3 = y) = \binom{y+3-1}{y} p^3(1-p)^y = \frac{(y+2)(y+1)}{2} p^3(1-p)^y \text{ for } y = 0, 1, 2, \dots$$

As a result, the pmf of $X_1 + X_2 + X_3$ is given by

$$\begin{aligned} P(X_1 + X_2 + X_3 = n) &= P(Y_1 + 1 + Y_2 + 1 + Y_3 + 1 = n) \\ &= P(Y_1 + Y_2 + Y_3 = n - 3) \\ &= \frac{(n-3+2)(n-3+1)}{2} p^3(1-p)^{n-3} \\ &= \frac{(n-1)(n-2)}{2} p^3(1-p)^{n-3} \text{ for } n = 3, 4, 5, \dots \end{aligned}$$

Therefore, the conditional pmf of X_3 given $X_1 + X_2 + X_3 = n$ is given by

$$\begin{aligned}
& P(X_3 = x | X_1 + X_2 + X_3 = n) \\
&= \frac{P(X_3 = x, X_1 + X_2 + X_3 = n)}{P(X_1 + X_2 + X_3 = n)} \\
&= \frac{P(X_3 = x, X_1 + X_2 = n - x)}{P(X_1 + X_2 + X_3 = n)} \\
&= \frac{P(X_3 = x)P(X_1 + X_2 = n - x)}{P(X_1 + X_2 + X_3 = n)} \text{ due to the independence of } X_1, X_2, \text{ and } X_3 \\
&= \frac{p(1-p)^{x-1} \cdot (n-x-1)p^2(1-p)^{n-x-2}}{\frac{(n-1)(n-2)}{2}p^3(1-p)^{n-3}} \text{ provided that } x \geq 1 \text{ and } n-x \geq 2 \\
&= \frac{2(n-x-1)}{(n-1)(n-2)} \text{ for } x = 1, 2, \dots, n-2. \tag{12}
\end{aligned}$$

Since X_3 has the same probability distribution as X_1 and X_2 , it follows that (12) also represents the conditional pmf of X_1 given $X_1 + X_2 + X_3 = n$ and of X_2 given $X_1 + X_2 + X_3 = n$.

(c) Using the conditional pmf determined in part (b), we obtain

$$\begin{aligned}
E(X_i | X_1 + X_2 + X_3 = n) &= \sum_{\text{all } x} xP(X_i = x | X_1 + X_2 + X_3 = n) \\
&= \sum_{x=1}^{n-2} x \cdot \frac{2(n-x-1)}{(n-1)(n-2)} \\
&= \frac{2}{(n-1)(n-2)} \sum_{x=1}^{n-2} ((n-1)x - x^2) \\
&= \frac{2}{n-2} \sum_{x=1}^{n-2} x - \frac{2}{(n-1)(n-2)} \sum_{x=1}^{n-2} x^2 \\
&= \frac{2}{n-2} \cdot \frac{(n-2)(n-1)}{2} - \frac{2}{(n-1)(n-2)} \cdot \frac{(n-2)(n-1)(2(n-2)+1)}{6} \\
&= n-1 - \frac{2n-3}{3} \\
&= \frac{n}{3}.
\end{aligned}$$

Section 4.6:

- 4.6.1 First of all, since Z given $Y = y$ follows a $Poi(y^2 + ab)$ distribution, we know that $E(Z|Y = y) = y^2 + ab$. Therefore, by the law of total expectation, we obtain

$$\begin{aligned}
 E(Z) &= E(E(Z|Y)) \\
 &= E(Y^2 + ab) \\
 &= E(Y^2) + ab \\
 &= Var(Y) + (E(Y))^2 + ab \\
 &= Var(X + b) + (E(X + b))^2 + ab \\
 &= Var(X) + (E(X) + b)^2 + ab \\
 &= a + (a + b)^2 + ab \text{ since } X \sim Poi(a) \\
 &= a(b + 1) + (a + b)^2.
 \end{aligned}$$

- 4.6.2 Let X be the random variable which counts the number of traffic accidents in Waterloo tomorrow. Define a random variable Y such that

$$Y = \begin{cases} 1 & \text{if the weather is clear tomorrow,} \\ 2 & \text{if the weather is rainy tomorrow,} \\ 3 & \text{if the weather is snowy tomorrow.} \end{cases}$$

Since it is equally likely that the weather tomorrow will be clear, rainy, or snowy, the pmf of Y is given by

$$f_Y(y) = \frac{1}{3} \text{ for } y = 1, 2, 3.$$

Based on the given information, X given $Y = 1$ has a $Poi(2.5)$ distribution, X given $Y = 2$ has a $Bin(8, 0.5)$ distribution, and X given $Y = 3$ has a $Geo(\frac{1}{7})$ distribution. By the law of total expectation, we obtain

$$\begin{aligned}
 E(X) &= \sum_{y=1}^3 E(X|Y = y)f_Y(y) \\
 &= \frac{1}{3}E(X|Y = 1) + \frac{1}{3}E(X|Y = 2) + \frac{1}{3}E(X|Y = 3) \\
 &= \frac{1}{3} \cdot 2.5 + \frac{1}{3} \cdot 8(0.5) + \frac{1}{3} \cdot \frac{1 - \frac{1}{7}}{\frac{1}{7}} \\
 &= \frac{2.5 + 4 + 6}{3} \\
 &= \frac{25}{6} \approx 4.16667.
 \end{aligned}$$

and

$$\begin{aligned}
E(X^2) &= \sum_{y=1}^3 E(X^2|Y=y)f_Y(y) \\
&= \frac{1}{3}E(X^2|Y=1) + \frac{1}{3}E(X^2|Y=2) + \frac{1}{3}E(X^2|Y=3) \\
&= \frac{1}{3} \cdot (2.5 + (2.5)^2) + \frac{1}{3} \cdot (8(0.5)(0.5) + 4^2) + \frac{1}{3} \cdot \left(\frac{1 - \frac{1}{7}}{(\frac{1}{7})^2} + 6^2 \right) \\
&= \frac{8.75 + 18 + 78}{3} \\
&= \frac{419}{12}.
\end{aligned}$$

Therefore, $SD(X) = \sqrt{\frac{419}{12} - \frac{625}{36}} = \sqrt{\frac{158}{9}} = \frac{\sqrt{158}}{3} \approx 4.18994$.

4.6.3 Let T be the number of minutes travelled until the rat finds the cheese. Also, let Y be the random variable denoting which door the rat selects first, so that

$$f_Y(y) = P(\text{rat first selects door } y) = \frac{1}{4} \text{ for } y = 1, 2, 3, 4.$$

For $i = 1, 2, 3$, let $X_i \sim Poi(0.5i)$, representing the number of minutes to return to the start of the maze after having door i be the first selection. From the description of the problem, we deduce that T given $Y = y$ has the same probability distribution as $X_y + T_2$ for $y = 1, 2, 3$, where T_2 is the number of minutes travelled, starting from the second attempt, until the rat finds the cheese. On the other hand, T given $Y = 4$ is simply equal to $\frac{10}{60} = \frac{1}{6}$ with probability 1. Therefore, conditioning on the door that the rat initially selects, we have that

$$\begin{aligned}
E(T) &= \sum_{y=1}^4 E(T|Y=y)P(Y=y) \\
&= \frac{1}{4} \sum_{y=1}^3 E(T|Y=y) + \frac{1}{4}E(T|Y=4) \\
&= \frac{1}{4} \sum_{y=1}^3 E(X_y + T_2) + \frac{1}{4} \cdot \frac{1}{6} \\
&= \frac{1}{4} \sum_{y=1}^3 E(X_y) + \frac{3}{4}E(T_2) + \frac{1}{24} \\
&= \frac{0.5 + 1 + 1.5}{4} + \frac{3}{4}E(T) + \frac{1}{24} \text{ since } T_2 \text{ has the same distribution as } T \\
&= \frac{3}{4}E(T) + \frac{19}{24}.
\end{aligned}$$

Isolating the above equation for $E(T)$, we obtain $E(T) = 4 \left(\frac{19}{24} \right) = \frac{19}{6} \approx 3.16667$.

Turning our attention next to the variance, we begin by obtaining

$$\begin{aligned}
& E(T^2) \\
&= \sum_{y=1}^4 E(T^2|Y=y)P(Y=y) \\
&= \frac{1}{4} \sum_{y=1}^3 E(T^2|Y=y) + \frac{1}{4} E(T^2|Y=4) \\
&= \frac{1}{4} \sum_{y=1}^3 E((X_y + T_2)^2) + \frac{1}{4} \cdot \left(\frac{1}{6} \right)^2 \\
&= \frac{1}{4} \sum_{y=1}^3 E(X_y^2 + 2T_2X_y + T_2^2) + \frac{1}{144} \\
&= \frac{1}{4} \sum_{y=1}^3 E(X_y^2) + \frac{1}{2} E(T_2) \sum_{y=1}^3 E(X_y) + \frac{3}{4} E(T_2^2) + \frac{1}{144} \text{ since } X_y \text{ and } T_2 \text{ are independent} \\
&= \frac{0.75+2+3.75}{4} + \frac{1}{2} \cdot \frac{19}{6} (0.5+1+1.5) + \frac{3}{4} E(T^2) + \frac{1}{144} \text{ since } T_2 \text{ has the same distribution as } T \\
&= \frac{13}{8} + \frac{19}{4} + \frac{3}{4} E(T^2) + \frac{1}{144} \\
&= \frac{3}{4} E(T^2) + \frac{919}{144}.
\end{aligned}$$

Isolating the above equation for $E(T^2)$, we get $E(T^2) = 4 \left(\frac{919}{144} \right) = \frac{919}{36}$. Thus, the variance is given by

$$Var(T) = \frac{919}{36} - \left(\frac{19}{6} \right)^2 = \frac{919 - 361}{36} = \frac{558}{36} = \frac{31}{2} = 15.5.$$

4.6.4 From the description of the problem, if Y represents the number of card selections made until the card numbered 1 is selected, then $Y - 1 \sim Geo(\frac{1}{k})$. Thus, we obtain

$$E(Y) = E(Y - 1 + 1) = E(Y - 1) + 1 = \frac{1 - \frac{1}{k}}{\frac{1}{k}} + 1 = k - 1 + 1 = k.$$

Moreover, if we let X be the total number of coin flips made, then X given $Y = y$ has conditional pmf given by

$$\begin{aligned}
f_{X|Y}(x|y) &= P(x \text{ total flips made to get } y \text{ heads}) \\
&= \binom{x-1}{y-1} p^y (1-p)^{x-y} \text{ for } x = y, y+1, y+2, \dots
\end{aligned}$$

Based on the above pmf, it follows that $X - y$ given $Y = y$ has a $NB(y, p)$ distribution, implying that

$$E(X|Y = y) = E(X - y + y|Y = y) = E(X - y|Y = y) + y = \frac{y(1-p)}{p} + y = \frac{y}{p}.$$

With these facts established, we now apply the law of total expectation to ultimately obtain

$$E(X) = E(E(X|Y)) = E\left(\frac{Y}{p}\right) = \frac{1}{p}E(Y) = \frac{k}{p}.$$

4.6.5 For each $i = 1, 2, \dots, m$, define

$$\begin{aligned} X_i &= 0 \text{ if the elevator does not stop at floor } i, \text{ and} \\ X_i &= 1 \text{ if the elevator does stop at floor } i. \end{aligned}$$

Let $Y \sim Poi(\mu)$ be the number of people entering the elevator on the ground floor. If N represents the total number of stops that the elevator makes, then we have that $N = \sum_{i=1}^m X_i$. To determine $E(N)$, we use the law of total expectation as follows:

$$\begin{aligned} E(N) &= \sum_{all\ y} E(N|Y = y)f_Y(y) \\ &= \sum_{y=0}^{\infty} E\left(\sum_{i=1}^m X_i \middle| Y = y\right) \frac{e^{-\mu}\mu^y}{y!} \\ &= \sum_{y=0}^{\infty} \sum_{i=1}^m E(X_i|Y = y) \frac{e^{-\mu}\mu^y}{y!} \\ &= \sum_{y=0}^{\infty} \sum_{i=1}^m P(X_i = 1|Y = y) \frac{e^{-\mu}\mu^y}{y!} \\ &= \sum_{y=0}^{\infty} \sum_{i=1}^m P(\text{at least 1 of } y \text{ people select floor } i \text{ to get off}) \frac{e^{-\mu}\mu^y}{y!} \\ &= \sum_{y=0}^{\infty} \sum_{i=1}^m \left\{1 - \left(\frac{m-1}{m}\right)^y\right\} \frac{e^{-\mu}\mu^y}{y!} \\ &= m \sum_{y=0}^{\infty} \frac{e^{-\mu}\mu^y}{y!} - m \sum_{y=0}^{\infty} \left(\frac{m-1}{m}\right)^y \frac{e^{-\mu}\mu^y}{y!} \\ &= m - me^{-\mu} \sum_{y=0}^{\infty} \frac{\{\mu(m-1)/m\}^y}{y!} \\ &= m(1 - e^{-\mu}e^{\mu(1-1/m)}) \\ &= m(1 - e^{-\mu/m}). \end{aligned}$$

Solutions to End of Section Chapter 5 Problems

Section 5.1:

- 5.1.1 (a) To determine the value of k , we make use of the property that $\int_{-\infty}^{\infty} f(x)dx = 1$ for a continuous random variable. This leads to

$$\begin{aligned}\int_{-\infty}^{\infty} f(x)dx &= 1 \\ \int_{-1}^3 k(1-x)^2 dx &= 1 \\ k \left(-\frac{(1-x)^3}{3} \right) \Big|_{-1}^3 &= 1 \\ k \left(\frac{8}{3} + \frac{8}{3} \right) &= 1 \\ \frac{16k}{3} &= 1 \\ k &= \frac{3}{16}.\end{aligned}$$

In addition, with this choice of k , note that $f(x) \geq 0$ for all $x \in \mathbb{R}$. Therefore, $k = \frac{3}{16}$.

- (b) First of all, for $x \leq -1$, note that

$$F(x) = \int_{-\infty}^x f(z)dz = \int_{-\infty}^x 0 dz = 0.$$

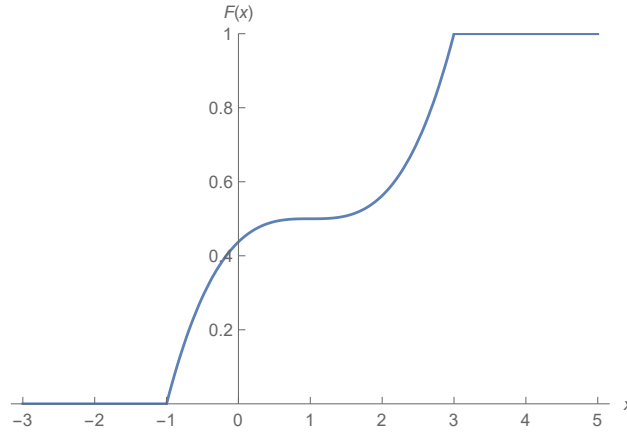
Next, for $-1 < x \leq 3$, we have

$$\begin{aligned}F(x) &= \int_{-\infty}^x f(z)dz \\ &= \int_{-\infty}^{-1} 0 dz + \int_{-1}^x \frac{3}{16}(1-z)^2 dz \\ &= 0 + \frac{3}{16} \left(-\frac{(1-z)^3}{3} \right) \Big|_{-1}^x \\ &= \frac{3}{16} \left(-\frac{(1-x)^3}{3} + \frac{8}{3} \right) \\ &= \frac{8 - (1-x)^3}{16}.\end{aligned}$$

Finally, we have that $F(x) = F(3) = 1$ for $x > 3$ since the pdf $f(x)$ equals 0 for all $x > 3$. Putting everything together, we ultimately obtain

$$F(x) = \begin{cases} 0 & \text{for } x \leq -1, \\ \frac{8-(1-x)^3}{16} & \text{for } -1 < x \leq 3, \\ 1 & \text{for } x > 3. \end{cases}$$

The plot of $F(x)$ is given below:



(c) We wish to calculate

$$\begin{aligned} P(-0.1 < X < 0.2) &= F(0.2) - F(-0.1) \\ &= \frac{8 - (0.8)^3}{16} - \frac{8 - (1.1)^3}{16} \\ &= 0.468 - 0.4168125 \\ &= 0.0511875. \end{aligned}$$

(d) We wish to find the value of $q(0.95)$ such that

$$\begin{aligned} F(q(0.95)) &= 0.95 \\ \frac{8 - (1 - q(0.95))^3}{16} &= 0.95 \\ \frac{1}{2} - \frac{(1 - q(0.95))^3}{16} &= 0.95 \\ (1 - q(0.95))^3 &= -7.2 \\ 1 - q(0.95) &= (-7.2)^{1/3} \\ q(0.95) &= 1 - (-7.2)^{1/3} \\ q(0.95) &\approx 2.93098. \end{aligned}$$

5.1.2 (a) Since $\alpha > 0$, the given cdf $F(x)$ is continuous for all $x \in \mathbb{R}$. Moreover, note that

$$F(x) = \frac{kx^\alpha}{1+x^\alpha} = \frac{k}{1+x^{-\alpha}} = \frac{k}{1+\left(\frac{1}{x}\right)^\alpha},$$

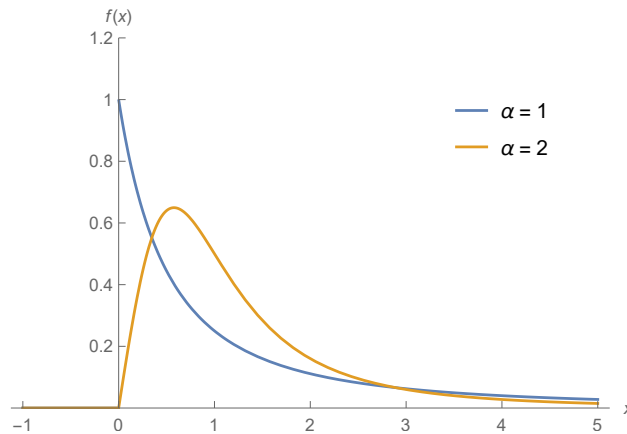
which clearly shows that $F(x)$ is an increasing function when $x > 0$. Therefore, to determine the value of k , we make use of the property that $\lim_{x \rightarrow \infty} F(x) = 1$. This leads to

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{k}{1+\left(\frac{1}{x}\right)^\alpha} &= 1 \\ \frac{k}{1+(0)^\alpha} &= 1 \\ k &= 1. \end{aligned}$$

(b) For $x > 0$, we have that the pdf of X is given by

$$\begin{aligned} f(x) &= F'(x) \\ &= \frac{d}{dx} \left(\frac{x^\alpha}{1+x^\alpha} \right) \\ &= \frac{(1+x^\alpha)(\alpha x^{\alpha-1}) - x^\alpha(\alpha x^{\alpha-1})}{(1+x^\alpha)^2} \\ &= \frac{\alpha x^{\alpha-1} + \alpha x^{2\alpha-1} - \alpha x^{2\alpha-1}}{(1+x^\alpha)^2} \\ &= \frac{\alpha x^{\alpha-1}}{(1+x^\alpha)^2}. \end{aligned}$$

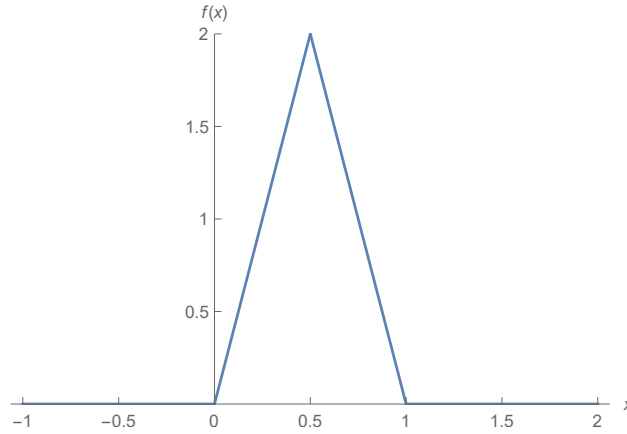
The plot of $f(x)$ is given below for $\alpha = 1$ and $\alpha = 2$:



(c) We wish to find the value of $m = q(0.5)$ such that

$$\begin{aligned}
 F(m) &= 0.5 \\
 \frac{m^\alpha}{1+m^\alpha} &= 0.5 \\
 \frac{1+m^\alpha}{m^\alpha} &= 2 \\
 1+m^\alpha &= 2m^\alpha \\
 m^\alpha &= 1 \\
 m &= 1^{1/\alpha} \\
 m &= 1.
 \end{aligned}$$

5.1.3 (a) The plot of the pdf $f(x)$ is given below:



We know that $\int_{-\infty}^{\infty} f(x)dx$ is equal to the area under the curve of $f(x)$. In this case, it is simply the area of the triangle, which we can easily calculate as

$$\frac{1}{2}(0.5 - 0)(2) + \frac{1}{2}(1 - 0.5)(2) = \frac{1}{2} + \frac{1}{2} = 1.$$

(b) First of all, for $x \leq 0$, note that

$$F(x) = \int_{-\infty}^x f(z)dz = \int_{-\infty}^x 0 dz = 0.$$

For $0 < x \leq \frac{1}{2}$, we have

$$F(x) = \int_{-\infty}^x f(z)dz = \int_{-\infty}^0 0 dz + \int_0^x 4z dz = 0 + (2z^2)\Big|_0^x = 2x^2.$$

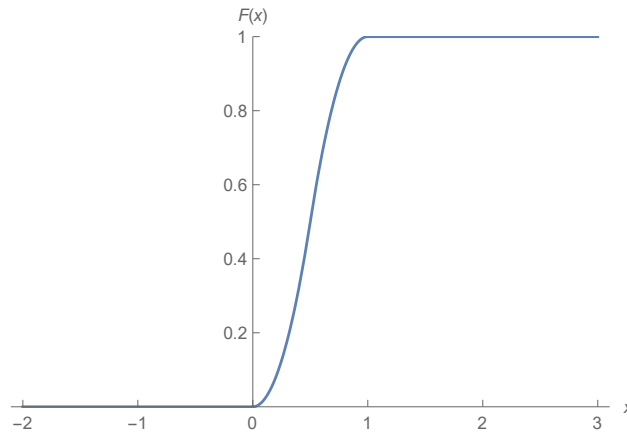
Next, for $\frac{1}{2} < x \leq 1$, we obtain

$$\begin{aligned}
 F(x) &= \int_{-\infty}^x f(z) dz \\
 &= \int_{-\infty}^0 0 dz + \int_0^{1/2} 4z dz + \int_{1/2}^x 4(1-z) dz \\
 &= 0 + (2z^2) \Big|_0^{1/2} + (-2(1-z)^2) \Big|_{1/2}^x \\
 &= \frac{1}{2} - 2(1-x)^2 + \frac{1}{2} \\
 &= 1 - 2(1-x)^2.
 \end{aligned}$$

Finally, we have that $F(x) = F(1) = 1$ for $x > 1$ since the pdf $f(x)$ equals 0 for all $x > 1$. Putting everything together, we ultimately obtain

$$F(x) = \begin{cases} 0 & \text{for } x \leq 0, \\ 2x^2 & \text{for } 0 < x \leq \frac{1}{2}, \\ 1 - 2(1-x)^2 & \text{for } \frac{1}{2} < x \leq 1, \\ 1 & \text{for } x > 1. \end{cases}$$

The plot of $F(x)$ is given below:



(c) We wish to calculate

$$\begin{aligned}
 P(0.25 < X < 0.8) &= F(0.8) - F(0.25) \\
 &= 1 - 2(0.2)^2 - 2(0.25)^2 \\
 &= 0.92 - 0.125 \\
 &= 0.795.
 \end{aligned}$$

- (d) To find the 10th percentile of the distribution of X , denoted by $q(0.1)$, we consider the equation

$$\begin{aligned} F(q(0.1)) &= 0.1 \\ 2(q(0.1))^2 &= 0.1 \\ (q(0.1))^2 &= 0.05 \\ q(0.1) &= \sqrt{0.05} \\ q(0.1) &\approx 0.223607. \end{aligned}$$

In addition, note that $F(0.5) = 2(0.5)^2 = 0.5$. Therefore, by definition, the median is given by $m = 0.5$.

- 5.1.4 (a) For $x \in (0, 1)$, we require $\theta + 1 > 0$, or equivalently $\theta > -1$, so as to ensure that the pdf $f(x) \geq 0$ for all $x \in \mathbb{R}$. Assuming that $\theta > -1$, note that

$$\int_{-\infty}^{\infty} f(x) dx = \int_0^1 (\theta + 1)x^\theta dx = (\theta + 1) \left(\frac{x^{\theta+1}}{\theta + 1} \right) \Big|_0^1 = 1^{\theta+1} - 0^{\theta+1} = 1.$$

Therefore, this is a valid pdf provided that $\theta > -1$.

- (b) We wish to calculate

$$\begin{aligned} P(X > 0.5) &= 1 - P(X \leq 0.5) \\ &= 1 - \int_0^{0.5} (\theta + 1)x^\theta dx \\ &= 1 - (\theta + 1) \left(\frac{x^{\theta+1}}{\theta + 1} \right) \Big|_0^{0.5} \\ &= 1 - (0.5)^{\theta+1}. \end{aligned}$$

- (c) For $p \in (0, 1)$, we wish to find the value of $q(p)$ such that

$$\begin{aligned} P(X \leq q(p)) &= p \\ \int_0^{q(p)} (\theta + 1)x^\theta dx &= p \\ (\theta + 1) \left(\frac{x^{\theta+1}}{\theta + 1} \right) \Big|_0^{q(p)} &= p \\ (q(p))^{\theta+1} &= p \\ q(p) &= p^{(\theta+1)^{-1}}. \end{aligned}$$

5.1.5 (a) To determine the value of k , we make use of the property that $\int_{-\infty}^{\infty} f(x)dx = 1$ for a continuous random variable. Keeping in mind that $\theta > 0$, this leads to

$$\begin{aligned}
 \int_{-\infty}^{\infty} f(x)dx &= 1 \\
 \underbrace{\int_0^{\infty} kxe^{-x^2/\theta} dx}_{\text{Let } y = \frac{x^2}{\theta} \text{ so that } dy = \frac{2x}{\theta} dx} &= 1 \\
 k \int_0^{\infty} e^{-y} \frac{\theta}{2} dy &= 1 \\
 \frac{k\theta}{2} (-e^{-y}) \Big|_0^{\infty} &= 1 \\
 \frac{k\theta}{2} (1 - \underbrace{\lim_{y \rightarrow \infty} e^{-y}}_{=0}) &= 1 \\
 \frac{k\theta}{2} &= 1 \\
 k &= \frac{2}{\theta}.
 \end{aligned}$$

In addition, with this choice of k , note that $f(x) \geq 0$ for all $x \in \mathbb{R}$. Therefore, $k = \frac{2}{\theta}$.

(b) First of all, for $x \leq 0$, note that

$$F(x) = \int_{-\infty}^x f(z)dz = \int_{-\infty}^x 0 dz = 0.$$

Next, for $x > 0$, we have

$$\begin{aligned}
 F(x) &= \int_{-\infty}^x f(z)dz \\
 &= \int_{-\infty}^0 0 dz + \int_0^x \frac{2}{\theta} ze^{-z^2/\theta} dz \\
 &= 0 + \frac{2}{\theta} \underbrace{\int_0^x ze^{-z^2/\theta} dz}_{\text{Let } y = \frac{z^2}{\theta} \text{ so that } dy = \frac{2z}{\theta} dz} \\
 &= \frac{2}{\theta} \int_0^{x^2/\theta} e^{-y} \frac{\theta}{2} dy \\
 &= (-e^{-y}) \Big|_0^{x^2/\theta} \\
 &= 1 - e^{-x^2/\theta}.
 \end{aligned}$$

Putting both cases together gives us

$$F(x) = \begin{cases} 0 & \text{for } x \leq 0, \\ 1 - e^{-x^2/\theta} & \text{for } x > 0. \end{cases}$$

(c) We wish to calculate

$$\begin{aligned} P(\sqrt{\theta} \leq X < 2\sqrt{\theta}) &= F(2\sqrt{\theta}) - F(\sqrt{\theta}) \\ &= 1 - e^{-(2\sqrt{\theta})^2/\theta} - \left(1 - e^{-(\sqrt{\theta})^2/\theta}\right) \\ &= e^{-1} - e^{-4} \\ &\approx 0.349564. \end{aligned}$$

(d) We wish to find the value of $m = q(0.5)$ such that

$$\begin{aligned} F(m) &= 0.5 \\ 1 - e^{-m^2/\theta} &= 0.5 \\ e^{-m^2/\theta} &= 0.5 \\ -\frac{m^2}{\theta} &= \ln 0.5 \\ m^2 &= \theta \ln 2 \\ m &= (\theta \ln 2)^{1/2}. \end{aligned}$$

Section 5.2:

5.2.1 (a) From part (a) of Problem 5.1.1, the pdf of X is given by

$$f(x) = \begin{cases} \frac{3}{16}(1-x)^2 & \text{for } -1 < x < 3, \\ 0 & \text{otherwise.} \end{cases}$$

Since $Y = 2X + 3$ is of the form $Y = aX + b$ with $a = 2$ and $b = 3$, we apply Corollary 5.2.1 to obtain the following pdf of Y :

$$f_Y(y) = \frac{1}{|2|} f\left(\frac{y-3}{2}\right) = \frac{1}{2} \cdot \frac{3}{16} \left(1 - \frac{y-3}{2}\right)^2 = \frac{3}{32} \left(\frac{2-y+3}{2}\right)^2 = \frac{3}{128} (5-y)^2 \text{ for } 1 < y < 9.$$

(b) We wish to find the pdf of $Z = X^2$. As a starting point, we use the result of Example 5.2.2 to first get

$$f_Z(z) = \frac{1}{2\sqrt{z}} f(\sqrt{z}) + \frac{1}{2\sqrt{z}} f(-\sqrt{z}) \text{ for } z > 0. \quad (13)$$

When $0 < z < 1$, (13) gives rise to

$$\begin{aligned} f_Z(z) &= \frac{1}{2\sqrt{z}} \cdot \frac{3}{16} (1 - \sqrt{z})^2 + \frac{1}{2\sqrt{z}} \cdot \frac{3}{16} (1 + \sqrt{z})^2 \\ &= \frac{3}{32\sqrt{z}} (1 - 2\sqrt{z} + z + 1 + 2\sqrt{z} + z) \\ &= \frac{3}{32\sqrt{z}} (2 + 2z) \\ &= \frac{3}{16\sqrt{z}} (1 + z). \end{aligned}$$

In the case when $1 < z < 9$, note that (13) simplifies to become

$$f_Z(z) = \frac{1}{2\sqrt{z}} \cdot \frac{3}{16} (1 - \sqrt{z})^2 + \frac{1}{2\sqrt{z}} \underbrace{f(-\sqrt{z})}_{= 0 \text{ when } z > 1} = \frac{3}{32\sqrt{z}} (1 - \sqrt{z})^2.$$

Putting both cases together, we obtain

$$f_Z(z) = \begin{cases} \frac{3}{16\sqrt{z}} (1 + z) & \text{for } 0 < z < 1, \\ \frac{3}{32\sqrt{z}} (1 - \sqrt{z})^2 & \text{for } 1 < z < 9, \\ 0 & \text{otherwise.} \end{cases}$$

5.2.2 (a) Since $Y = 2(X - 1) = 2X - 2$ is of the form $Y = aX + b$ with $a = 2$ and $b = -2$, we apply Corollary 5.2.1 to first obtain

$$f_Y(y) = \frac{1}{|2|} f\left(\frac{y+2}{2}\right) = \frac{1}{2} f\left(\frac{y+2}{2}\right) \text{ for } -2 < y < 0. \quad (14)$$

When $-2 < y < -1$, (14) becomes

$$f_Y(y) = \frac{1}{2} \cdot 4 \left(\frac{y+2}{2} \right) = y+2.$$

On the other hand, when $-1 < y < 0$, (14) gives rise to

$$f_Y(y) = \frac{1}{2} \cdot 4 \left(1 - \frac{y+2}{2} \right) = 2 \left(-\frac{y}{2} \right) = -y.$$

Altogether, we ultimately obtain

$$f_Y(y) = \begin{cases} y+2 & \text{for } -2 < y < -1, \\ -y & \text{for } -1 < y < 0, \\ 0 & \text{otherwise.} \end{cases}$$

- (b) Note that $Z = X^3$ is a strictly increasing function of X . Moreover, if $z = g(x) = x^3$, then we clearly have $x = g^{-1}(z) = z^{\frac{1}{3}}$. Therefore, by Theorem 5.2.1, the pdf of Z is given by

$$f_Z(z) = f(g^{-1}(z)) \left| \frac{dg^{-1}(z)}{dz} \right| = \frac{1}{3} z^{-\frac{2}{3}} f(z^{\frac{1}{3}}) \text{ for } 0 < z < 1. \quad (15)$$

When $0 < z < \frac{1}{8}$, (15) gives rise to

$$f_Z(z) = \frac{1}{3} z^{-\frac{2}{3}} \cdot 4z^{\frac{1}{3}} = \frac{4}{3} z^{-\frac{1}{3}}.$$

In the case when $\frac{1}{8} < z < 1$, note that (15) simplifies to become

$$f_Z(z) = \frac{1}{3} z^{-\frac{2}{3}} \cdot 4 \left(1 - z^{\frac{1}{3}} \right) = \frac{4}{3} z^{-\frac{2}{3}} \left(1 - z^{\frac{1}{3}} \right).$$

Putting both cases together, we obtain

$$f_Z(z) = \begin{cases} \frac{4}{3} z^{-\frac{1}{3}} & \text{for } 0 < z < \frac{1}{8}, \\ \frac{4}{3} z^{-\frac{2}{3}} \left(1 - z^{\frac{1}{3}} \right) & \text{for } \frac{1}{8} < z < 1, \\ 0 & \text{otherwise.} \end{cases}$$

- 5.2.3 (a) Since $Y = 1 - X$ is of the form $Y = aX + b$ with $a = -1$ and $b = 1$, we apply Corollary 5.2.1 to obtain the following pdf of Y :

$$f_Y(y) = \frac{1}{|-1|} f\left(\frac{y-1}{-1}\right) = (\theta+1)(1-y)^\theta \text{ for } 0 < y < 1.$$

- (b) Note that $Z = \frac{1}{X}$ is a strictly decreasing function of X . Moreover, if $z = g(x) = \frac{1}{x}$, then we clearly have $x = g^{-1}(z) = z^{-1}$. Therefore, by Theorem 5.2.1, the pdf of Z is given by

$$f_Z(z) = f(g^{-1}(z)) \left| \frac{dg^{-1}(z)}{dz} \right| = z^{-2} f(z^{-1}) = z^{-2} (\theta + 1) z^{-\theta} = (\theta + 1) z^{-(\theta+2)} \text{ for } z > 1.$$

5.2.4 Let us first find the cdf of the random variable X . For $0 < x \leq 1$, note that

$$F_X(x) = P(X \leq x) = \int_0^x 4y^3 dy = 4 \left(\frac{y^4}{4} \right) \Big|_0^x = x^4.$$

Next, the cdf of $Y = -\ln X^4$ is given by

$$\begin{aligned} F_Y(y) &= P(Y \leq y) \\ &= P(-\ln X^4 \leq y) \\ &= P(-4 \ln X \leq y) \\ &= P\left(\ln X \geq -\frac{y}{4}\right) \\ &= P\left(X \geq e^{-\frac{y}{4}}\right) \\ &= 1 - F_X\left(e^{-\frac{y}{4}}\right) \\ &= 1 - \left(e^{-\frac{y}{4}}\right)^4 \\ &= 1 - e^{-y} \text{ for } y > 0. \end{aligned}$$

Therefore, the pdf of Y is given by

$$f_Y(y) = F'_Y(y) = e^{-y} \text{ for } y > 0.$$

Based on the above form, we see that $Y \sim \text{Exp}(1)$.

- 5.2.5 Note that $Y = X^3$ is a strictly increasing function of X . Moreover, if $y = g(x) = x^3$, then we clearly have $x = g^{-1}(y) = y^{\frac{1}{3}}$. Therefore, by Theorem 5.2.1, the pdf of Y is given by

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dg^{-1}(y)}{dy} \right| = \frac{1}{3} y^{-\frac{2}{3}} f_X(y^{\frac{1}{3}}) = \frac{1}{3} y^{-\frac{2}{3}} \frac{|y^{\frac{1}{3}}|}{4} = \frac{|y|^{-\frac{1}{3}}}{12} \text{ for } -8 < y < 8.$$

Next, we find the cdf of Y . When $-8 < y \leq 0$, we obtain

$$\begin{aligned}
 F_Y(y) &= \int_{-8}^y \frac{1}{12} |z|^{-\frac{1}{3}} dz \\
 &= -\frac{1}{12} \int_{-8}^y z^{-\frac{1}{3}} dz \\
 &= -\frac{1}{12} \left(\frac{z^{\frac{2}{3}}}{\frac{2}{3}} \right) \Big|_{-8}^y \\
 &= -\frac{1}{8} \left(y^{\frac{2}{3}} - (-8)^{\frac{2}{3}} \right) \\
 &= \frac{1}{8} \left(4 - y^{\frac{2}{3}} \right).
 \end{aligned}$$

When $0 < y \leq 8$, we obtain

$$\begin{aligned}
 F_Y(y) &= \int_{-8}^0 \frac{1}{12} |z|^{-\frac{1}{3}} dz + \int_0^y \frac{1}{12} |z|^{-\frac{1}{3}} dz \\
 &= \frac{1}{8} \left(4 - 0^{\frac{2}{3}} \right) + \frac{1}{12} \int_0^y z^{-\frac{1}{3}} dz \\
 &= \frac{1}{2} + \frac{1}{12} \left(\frac{z^{\frac{2}{3}}}{\frac{2}{3}} \right) \Big|_0^y \\
 &= \frac{1}{2} + \frac{1}{8} \left(y^{\frac{2}{3}} - (0)^{\frac{2}{3}} \right) \\
 &= \frac{1}{8} \left(4 + y^{\frac{2}{3}} \right).
 \end{aligned}$$

Putting it altogether leads to

$$F_Y(y) = \begin{cases} 0 & \text{for } y \leq -8, \\ \frac{1}{8} \left(4 - y^{\frac{2}{3}} \right) & \text{for } -8 < y \leq 0, \\ \frac{1}{8} \left(4 + y^{\frac{2}{3}} \right) & \text{for } 0 < y \leq 8, \\ 1 & \text{for } y > 8. \end{cases}$$

Section 5.3:

5.3.1 From part (a) of Problem 5.1.1, the pdf of X is given by

$$f(x) = \begin{cases} \frac{3}{16}(1-x)^2 & \text{for } -1 < x < 3, \\ 0 & \text{otherwise.} \end{cases}$$

Using this pdf, we wish to calculate

$$\begin{aligned} E(X) &= \int_{-1}^3 x \frac{3}{16} (1-x)^2 dx \\ &= \frac{3}{16} \int_{-1}^3 (x - 2x^2 + x^3) dx \\ &= \frac{3}{16} \left(\frac{x^2}{2} - \frac{2x^3}{3} + \frac{x^4}{4} \right) \Big|_{-1}^3 \\ &= \frac{3}{16} \left(\frac{9}{2} - 18 + \frac{81}{4} - \frac{1}{2} - \frac{2}{3} - \frac{1}{4} \right) \\ &= \frac{3}{16} \cdot \frac{64}{12} \\ &= 1, \end{aligned}$$

$$\begin{aligned} E(X^2) &= \int_{-1}^3 x^2 \frac{3}{16} (1-x)^2 dx \\ &= \frac{3}{16} \int_{-1}^3 (x^2 - 2x^3 + x^4) dx \\ &= \frac{3}{16} \left(\frac{x^3}{3} - \frac{x^4}{2} + \frac{x^5}{5} \right) \Big|_{-1}^3 \\ &= \frac{3}{16} \left(9 - \frac{81}{2} + \frac{243}{5} + \frac{1}{3} + \frac{1}{2} + \frac{1}{5} \right) \\ &= \frac{3}{16} \cdot \frac{544}{30} \\ &= \frac{17}{5}, \end{aligned}$$

and

$$Var(X) = E(X^2) - E(X)^2 = \frac{17}{5} - 1^2 = \frac{12}{5}.$$

5.3.2 Using the pdf from Problem 5.1.3, we wish to calculate

$$\begin{aligned}
 E(X) &= \int_{-\infty}^{\infty} xf(x)dx \\
 &= \int_0^{1/2} 4x^2 dx + \int_{1/2}^1 4x(1-x)dx \\
 &= \left(\frac{4x^3}{3} \right) \Big|_0^{1/2} + \left(2x^2 - \frac{4x^3}{3} \right) \Big|_{1/2}^1 \\
 &= \frac{1}{6} + \left(2 - \frac{4}{3} - \frac{1}{2} + \frac{1}{6} \right) \\
 &= \frac{1}{6} + \frac{12 - 8 - 3 + 1}{6} \\
 &= \frac{1}{6} + \frac{2}{6} \\
 &= \frac{1}{2},
 \end{aligned}$$

$$\begin{aligned}
 E(X^2) &= \int_{-\infty}^{\infty} x^2 f(x)dx \\
 &= \int_0^{1/2} 4x^3 dx + \int_{1/2}^1 4x^2(1-x)dx \\
 &= x^4 \Big|_0^{1/2} + \left(\frac{4x^3}{3} - x^4 \right) \Big|_{1/2}^1 \\
 &= \frac{1}{16} + \left(\frac{4}{3} - 1 - \frac{1}{6} + \frac{1}{16} \right) \\
 &= \frac{1}{16} + \frac{64 - 48 - 8 + 3}{48} \\
 &= \frac{1}{16} + \frac{11}{48} \\
 &= \frac{7}{24},
 \end{aligned}$$

and

$$SD(X) = \sqrt{\frac{7}{24} - \left(\frac{1}{2}\right)^2} = \sqrt{\frac{1}{24}} = \frac{1}{2\sqrt{6}} \approx 0.2041.$$

5.3.3 Using the pdf from Problem 5.1.4, the n^{th} moment of X is given by

$$\begin{aligned}
 E(X^n) &= \int_{-\infty}^{\infty} x^n f(x) dx \\
 &= \int_0^1 x^n (\theta + 1) x^\theta dx \\
 &= (\theta + 1) \int_0^1 x^{n+\theta} dx \\
 &= (\theta + 1) \left(\frac{x^{n+\theta+1}}{n + \theta + 1} \right) \Big|_0^1 \\
 &= \frac{\theta + 1}{n + \theta + 1} \text{ for } n = 1, 2, 3, \dots
 \end{aligned}$$

5.3.4 The area of the rectangle is given by the random variable $A = X(1 - X)$. We wish to calculate

$$\begin{aligned}
 E(A) &= \int_0^1 x(1 - x) 3x^2 dx \\
 &= 3 \int_0^1 (x^3 - x^4) dx \\
 &= 3 \left(\frac{x^4}{4} - \frac{x^5}{5} \right) \Big|_0^1 \\
 &= 3 \left(\frac{1}{4} - \frac{1}{5} \right) \\
 &= \frac{3}{20},
 \end{aligned}$$

$$\begin{aligned}
 E(A^2) &= \int_0^1 x^2(1 - x)^2 3x^2 dx \\
 &= 3 \int_0^1 (x^4 - 2x^5 + x^6) dx \\
 &= 3 \left(\frac{x^5}{5} - \frac{x^6}{3} + \frac{x^7}{7} \right) \Big|_0^1 \\
 &= 3 \left(\frac{1}{5} - \frac{1}{3} + \frac{1}{7} \right) \\
 &= \frac{3(21 - 35 + 15)}{105} \\
 &= \frac{1}{35},
 \end{aligned}$$

and

$$Var(A) = E(A^2) - (E(A))^2 = \frac{1}{35} - \left(\frac{3}{20}\right)^2 = \frac{17}{2800}.$$

5.3.5 First of all, we obtain the pdf of X as follows:

$$f_X(x) = F'_X(x) = \frac{d}{dx} \left(\frac{x}{2} - \frac{x^2}{16} \right) = \frac{1}{2} - \frac{x}{8} = \frac{4-x}{8} \text{ for } 0 < x < 4.$$

With the above pdf and $Y = 2(9 - X^2)$, we can now calculate

$$\begin{aligned} E(Y) &= \int_0^4 2(9 - x^2) \frac{4-x}{8} dx \\ &= \frac{1}{4} \int_0^4 (36 - 9x - 4x^2 + x^3) dx \\ &= \frac{1}{4} \left(36x - \frac{9x^2}{2} - \frac{4x^3}{3} + \frac{x^4}{4} \right) \Big|_0^4 \\ &= \frac{1}{4} \left(144 - 72 - \frac{256}{3} + 64 \right) \\ &= \frac{1}{4} \cdot \frac{152}{3} \\ &= \frac{38}{3} \end{aligned}$$

and

$$\begin{aligned} E(Y^2) &= \int_0^4 4(9 - x^2)^2 \frac{4-x}{8} dx \\ &= \frac{1}{2} \int_0^4 (81 - 18x^2 + x^4)(4 - x) dx \\ &= \frac{1}{2} \int_0^4 (324 - 81x - 72x^2 + 18x^3 + 4x^4 - x^5) dx \\ &= \frac{1}{2} \left(324x - \frac{81x^2}{2} - 24x^3 + \frac{9x^4}{2} + \frac{4x^5}{5} - \frac{x^6}{6} \right) \Big|_0^4 \\ &= \frac{1}{2} \left(1296 - 648 - 1536 + 1152 + \frac{4096}{5} - \frac{2048}{3} \right) \\ &= \frac{1}{2} \cdot \frac{6008}{15} \\ &= \frac{3004}{15}. \end{aligned}$$

Therefore, the variance of Y is given by

$$Var(Y) = E(Y^2) - (E(Y))^2 = \frac{3004}{15} - \left(\frac{38}{3}\right)^2 = \frac{9012 - 7220}{45} = \frac{1792}{45}.$$

5.3.6 (a) For a random variable X with pdf $f(x)$, we know that

$$\begin{aligned}\int_{-\infty}^{\infty} f(x)dx &= 1, \\ \int_{-\infty}^{\infty} xf(x)dx &= E(X) = 2, \text{ and} \\ \int_{-\infty}^{\infty} x^2f(x)dx &= E(X^2) = Var(X) + (E(X))^2 = 1 + 4 = 5.\end{aligned}$$

Applying these results to this particular distribution, we obtain:

$$\int_0^3 (a + bx + cx^2)dx = 1 \implies 3a + \frac{9}{2}b + 9c = 1, \quad (16)$$

$$\int_0^3 x(a + bx + cx^2)dx = 2 \implies \frac{9}{2}a + 9b + \frac{81}{4}c = 2, \quad (17)$$

$$\int_0^3 x^2(a + bx + cx^2)dx = 5 \implies 9a + \frac{81}{4}b + \frac{243}{5}c = 5. \quad (18)$$

Note that the above equations are linear and can be solved to find a , b , and c . We proceed as follows:

$$3 \times (16) - 2 \times (17) \implies -\frac{9}{2}b - \frac{27}{2}c = -1 \implies b = -3c + \frac{2}{9}, \quad (19)$$

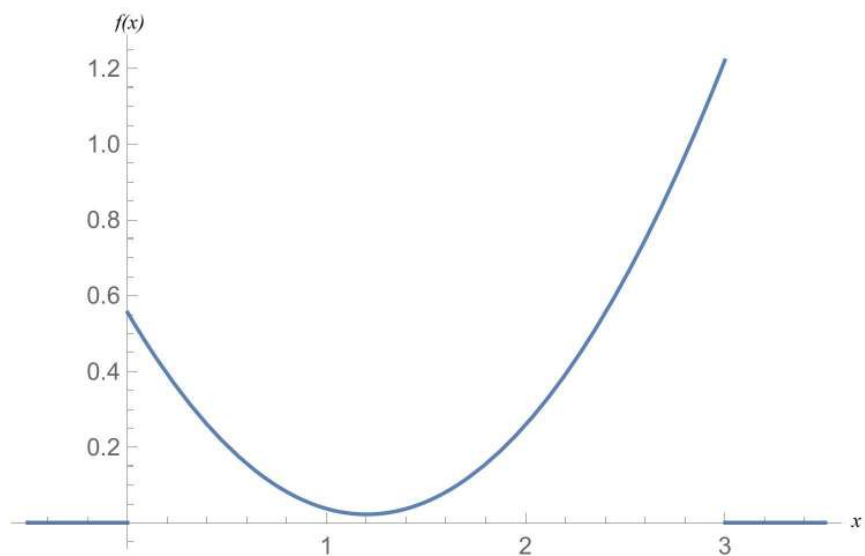
$$3 \times (16) - (18) \implies -\frac{27}{4}b - \frac{108}{5}c = -2. \quad (20)$$

Plugging (19) into (20), we get:

$$\frac{27}{4} \left(-3c + \frac{2}{9} \right) + \frac{108}{5}c = 2 \implies c = \frac{10}{27}.$$

Back substituting into the earlier equations, we then obtain $b = -\frac{8}{9}$ and $a = \frac{5}{9}$.

(b) With the computed values of a, b , and c from part (a), we plot the graph of $f(x)$:



5.3.7 Consider the mgf of X which is given by

$$\begin{aligned}
 M(t) &= \int_{-\infty}^{\infty} e^{tx} f(x) dx \\
 &= \int_{-\infty}^0 e^{tx} f(x) dx + \int_0^{\infty} e^{tx} f(x) dx \\
 &= \int_{\infty}^0 e^{-ty} f(-y) (-dy) + \int_0^{-\infty} e^{-ty} f(-y) (-dy) \quad \text{if we let } y = -x \text{ so that } dy = -dx \\
 &= \int_0^{\infty} e^{-ty} f(-y) dy + \int_{-\infty}^0 e^{-ty} f(-y) dy \\
 &= \int_0^{\infty} e^{-ty} f(y) dy + \int_{-\infty}^0 e^{-ty} f(y) dy \quad \text{since } f \text{ is symmetric about } 0 \\
 &= \int_{-\infty}^{\infty} e^{-ty} f(y) dy \\
 &= M(-t).
 \end{aligned}$$

5.3.8 (a) First of all, looking at the form of $f(x)$, we note that in order for $f(x) \geq 0$ for all $x > 0$, we must have that $c \geq 1$. To find the value of c , we consider the condition

$$\begin{aligned}\int_{-\infty}^{\infty} f(x) dx &= 1 \\ \int_0^{\infty} \left(\frac{3}{2} e^{-x} - \frac{3}{2} e^{-cx} \right) dx &= 1 \\ \frac{3}{2} \left(\frac{e^{-x}}{-1} \right) \Big|_0^{\infty} - \frac{3}{2} \left(\frac{e^{-cx}}{-c} \right) \Big|_0^{\infty} &= 1 \\ \frac{3}{2} - \frac{3}{2c} &= 1 \\ 1 - \frac{1}{c} &= \frac{2}{3} \\ c &= 3.\end{aligned}$$

(b) Using the value of c found in part (a), the mgf of X is given by

$$\begin{aligned}M(t) &= \int_0^{\infty} e^{tx} \left(\frac{3}{2} e^{-x} - \frac{3}{2} e^{-3x} \right) dx \\ &= \frac{3}{2} \int_0^{\infty} e^{(t-1)x} dx - \frac{3}{2} \int_0^{\infty} e^{(t-3)x} dx \\ &= \frac{3}{2} \left(\frac{e^{(t-1)x}}{t-1} \right) \Big|_0^{\infty} - \frac{3}{2} \left(\frac{e^{(t-3)x}}{t-3} \right) \Big|_0^{\infty} \\ &= \frac{3}{2(t-1)} (0-1) - \frac{3}{2(t-3)} (0-1) \quad \text{provided that } t-1 < 0 \text{ and } t-3 < 0 \\ &= \frac{3}{2} \left(\frac{1}{1-t} - \frac{1}{3-t} \right) \\ &= \frac{3}{(1-t)(3-t)} \quad \text{provided that } t < 1.\end{aligned}$$

(c) Taking the first two derivatives of $M(t)$ yields

$$M^{(1)}(t) = 3(1-t)^{-1}(3-t)^{-2} + 3(1-t)^{-2}(3-t)^{-1}$$

and

$$M^{(2)}(t) = 6(1-t)^{-1}(3-t)^{-3} + 6(1-t)^{-2}(3-t)^{-2} + 6(1-t)^{-3}(3-t)^{-1}.$$

Therefore, we readily obtain

$$E(X) = M^{(1)}(0) = 3(1-0)^{-1}(3-0)^{-2} + 3(1-0)^{-2}(3-0)^{-1} = \frac{3}{9} + 1 = \frac{4}{3},$$

$$\begin{aligned}
E(X^2) &= M^{(2)}(0) \\
&= 6(1-0)^{-1}(3-0)^{-3} + 6(1-0)^{-2}(3-0)^{-2} + 6(1-0)^{-3}(3-0)^{-1} \\
&= \frac{6}{27} + \frac{6}{9} + 2 \\
&= \frac{26}{9},
\end{aligned}$$

and

$$Var(X) = E(X^2) - (E(X))^2 = \frac{26}{9} - \frac{16}{9} = \frac{10}{9}.$$

Section 5.4:

- 5.4.1 First of all, if $c = 0$, then the random variable Y is simply equal to the constant d with probability 1. Let us next consider the case where $c > 0$. Since $Y = cX + d$ is a linear function of X , we apply Corollary 5.2.1 to obtain the following pdf of Y :

$$f_Y(y) = \frac{1}{|c|} f_X\left(\frac{y-d}{c}\right) = \frac{1}{c} \cdot \frac{1}{b-a} = \frac{1}{(b-a)c} = \frac{1}{(bc+d) - (ac+d)} \text{ for } ac+d < y < bc+d.$$

From the form of the above pdf, we see that $Y \sim U(ac+d, bc+d)$.

On the other hand, if $c < 0$, then applying Corollary 5.2.1 yields

$$f_Y(y) = \frac{1}{|c|} f_X\left(\frac{y-d}{c}\right) = \frac{1}{-c} \cdot \frac{1}{b-a} = \frac{1}{(a-b)c} = \frac{1}{(ac+d) - (bc+d)} \text{ for } bc+d < y < ac+d.$$

From the form of the above pdf, we now have that $Y \sim U(bc+d, ac+d)$.

- 5.4.2 If we let X represent the diameter of a spherical particle measured in centimetres, then $X \sim U(0.6, 1.0)$ with pdf

$$f_X(x) = \frac{1}{1.0 - 0.6} = \frac{5}{2} \text{ for } 0.6 < x < 1.0.$$

The volume of a spherical particle is given by the random variable

$$V = \frac{4}{3}\pi \left(\frac{X}{2}\right)^3 = \frac{\pi}{6}X^3.$$

Note that V is a strictly increasing function of X . Moreover, if $v = g(x) = \frac{\pi}{6}x^3$, then we clearly have

$$x = g^{-1}(v) = \left(\frac{6v}{\pi}\right)^{\frac{1}{3}}.$$

Therefore, by Theorem 5.2.1, the pdf of V is given by

$$\begin{aligned} f_V(v) &= f_X(g^{-1}(v)) \left| \frac{dg^{-1}(v)}{dv} \right| \\ &= f_X\left(\left(\frac{6v}{\pi}\right)^{\frac{1}{3}}\right) \cdot \frac{1}{3} \left(\frac{6v}{\pi}\right)^{-\frac{2}{3}} \left(\frac{6}{\pi}\right) \\ &= \frac{5}{2} \cdot \frac{2}{\pi} \left(\frac{6v}{\pi}\right)^{-\frac{2}{3}} \\ &= \frac{5}{\pi} \left(\frac{6v}{\pi}\right)^{-\frac{2}{3}} \text{ for } \frac{0.216\pi}{6} < v < \frac{\pi}{6}. \end{aligned}$$

5.4.3 First of all, we find the cdf of X . For $-1 < x \leq 1$, note that

$$F(x) = \int_{-1}^x \frac{3}{2}y^2 dy = \left(\frac{y^3}{2} \right) \Big|_{-1}^x = \frac{x^3 + 1}{2}.$$

If we define the random variable $Y = F(X) = \frac{X^3+1}{2}$, then we know that $Y \sim U(0, 1)$ from Theorem 5.4.1. Moreover, if $y = F(x) = \frac{x^3+1}{2}$, then $x = F^{-1}(y) = (2y - 1)^{1/3}$. As a result, if the $U(0, 1)$ random number generated is $y = 0.27125$, then we would take as the value from the distribution of X :

$$x = (2(0.27125) - 1)^{1/3} = (-0.4575)^{1/3} = -0.770543.$$

5.4.4 (a) Let X be the magnitude of a given earthquake. It is specified that $X \sim \text{Exp}(2.5)$. Moreover, the probability that an earthquake exceeds 5 on the Richter scale is given by

$$P(X > 5) = 1 - P(X \leq 5) = 1 - F_X(5) = 1 - (1 - e^{-5/2.5}) = e^{-2}.$$

For the 3 independent earthquakes taking place in a given month, let Y be the number of earthquakes which exceed 5 on the Richter scale. It follows that $Y \sim \text{Bin}(3, e^{-2})$. Therefore, we wish to calculate

$$P(Y = 0) = \binom{3}{0} (e^{-2})^0 (1 - e^{-2})^3 = (1 - e^{-2})^3 \approx 0.646462.$$

(b) Applying the memoryless property of the exponential distribution, we have that

$$P(X > 5 | X > 4) = P(X > 4+1 | X > 4) = P(X > 1) = 1 - (1 - e^{-1/2.5}) = e^{-0.4} \approx 0.67032.$$

5.4.5 (a) Let X represent the total number of thousands of kilometres that a used car can be driven before it needs to be scrapped. According to Jamie, $X \sim \text{Exp}(20)$. We wish to calculate

$$\begin{aligned} P(X > 10 + 20 | X > 10) &= P(X > 20) \text{ by the memoryless property of the exponential distribution} \\ &= 1 - F_X(20) \\ &= 1 - (1 - e^{-20/20}) \\ &= e^{-1} \\ &\approx 0.367879. \end{aligned}$$

(b) If we now suppose that $X \sim U(0, 40)$, then the probability in part (a) becomes

$$P(X > 10 + 20 | X > 10) = \frac{P(X > 30, X > 10)}{P(X > 10)} = \frac{P(X > 30)}{P(X > 10)} = \frac{0.25}{0.75} = \frac{1}{3}.$$

Note that $P(X > 20) = 0.5 \neq P(X > 10 + 20 | X > 10)$, thereby confirming that the $U(0, 40)$ distribution is not memoryless.

5.4.6 Let X represent the time the system runs between disruptions. We are told that $X \sim \text{Exp}(\theta)$, implying that $F_X(x) = 1 - e^{-x/\theta}$ for $x > 0$. We wish to find the value of θ which satisfies

$$\begin{aligned} P(X < 8) &= 0.25 \\ 1 - e^{-8/\theta} &= 0.25 \\ e^{-8/\theta} &= 0.75 \\ -\frac{8}{\theta} &= \ln 0.75 \\ \theta &= -\frac{8}{\ln 0.75} \\ \theta &\approx 27.8085. \end{aligned}$$

5.4.7 If $X \sim \text{Exp}(\theta)$, then the $100p^{\text{th}}$ percentile of X is given by the formula

$$q(p) = \theta \ln(1 - p)^{-1}.$$

We wish to determine

$$\begin{aligned} q(0.25) &= \theta \ln(1 - 0.25)^{-1} = \theta \ln \left(\frac{4}{3} \right), \\ q(0.5) &= \theta \ln(1 - 0.5)^{-1} = \theta \ln 2, \text{ and} \\ q(0.75) &= \theta \ln(1 - 0.75)^{-1} = \theta \ln 4. \end{aligned}$$

5.4.8 Suppose that $X \sim \text{Exp}(\theta)$. By definition, it follows that

$$E(X) = \int_0^\infty x \frac{1}{\theta} e^{-x/\theta} dx.$$

We will apply integration by parts (i.e., $\int u dv = uv - \int v du$) to the above integral. In particular, choose

$$u = x \implies \frac{du}{dx} = 1 \implies du = dx$$

and

$$dv = \frac{1}{\theta} e^{-x/\theta} dx \implies \int 1 \cdot dv = \int \frac{1}{\theta} e^{-x/\theta} dx \implies v = -e^{-x/\theta},$$

so that

$$\begin{aligned} E(X) &= \left(-xe^{-x/\theta} \right) \Big|_0^\infty - \int_0^\infty (-e^{-x/\theta}) dx \\ &= \underbrace{-\lim_{x \rightarrow \infty} xe^{-x/\theta}}_{=0} + \left(\frac{e^{-x/\theta}}{-1/\theta} \right) \Big|_0^\infty \\ &= \underbrace{-\theta \lim_{x \rightarrow \infty} e^{-x/\theta}}_{=0} + \theta \\ &= \theta. \end{aligned}$$

Next, we look to determine $E(X^2)$ by considering

$$E(X^2) = \int_0^\infty x^2 \frac{1}{\theta} e^{-x/\theta} dx.$$

We will again apply integration by parts to the above integral. This time, we choose

$$u = x^2 \implies \frac{du}{dx} = 2x \implies du = 2x dx$$

and

$$dv = \frac{1}{\theta} e^{-x/\theta} dx \implies \int 1 \cdot dv = \int \frac{1}{\theta} e^{-x/\theta} dx \implies v = -e^{-x/\theta}.$$

This leads to

$$\begin{aligned} E(X^2) &= \left(-x^2 e^{-x/\theta} \right) \Big|_0^\infty - \int_0^\infty (-e^{-x/\theta}) 2x dx \\ &= \underbrace{-\lim_{x \rightarrow \infty} x^2 e^{-x/\theta}}_{=0} + 2\theta \underbrace{\int_0^\infty x \frac{1}{\theta} e^{-x/\theta} dx}_{=E(X)=\theta} \\ &= 2\theta^2. \end{aligned}$$

5.4.9 From part (a) of Problem 5.1.5, the pdf of X is given by

$$f(x) = \begin{cases} \frac{2}{\theta} x e^{-x^2/\theta} & \text{for } x > 0, \\ 0 & \text{otherwise.} \end{cases}$$

Note that $Y = \frac{X^2}{\theta}$ is a strictly increasing function of X on the range of X (which is positive). Moreover, if $y = g(x) = \frac{x^2}{\theta}$, then $x = g^{-1}(y) = \sqrt{\theta y}$. Therefore, by Theorem 5.2.1, the pdf of Y is given by

$$f_Y(y) = f(g^{-1}(y)) \left| \frac{dg^{-1}(y)}{dy} \right| = f(\sqrt{\theta y}) \cdot \frac{\theta}{2\sqrt{\theta y}} = \frac{\theta}{2\sqrt{\theta y}} \cdot \frac{2}{\theta} \sqrt{\theta y} e^{-(\sqrt{\theta y})^2/\theta} = e^{-y} \text{ for } y > 0.$$

Recognizing the form of the above pdf, we conclude that $Y \sim \text{Exp}(1)$.

5.4.10 With $X \sim \text{Exp}(\theta)$, its pdf is given by

$$f(x) = \frac{1}{\theta} e^{-x/\theta} \text{ for } x > 0,$$

and its cdf is given by

$$F(x) = 1 - e^{-x/\theta} \text{ for } x > 0.$$

Noting that $|x - a| = a - x$ if $x \leq a$ and $|x - a| = x - a$ if $x \geq a$, we begin by obtaining the expression for

$$\begin{aligned}
E(|X - a|) &= \int_0^\infty |x - a|f(x)dx \\
&= \int_0^a (a - x)f(x)dx + \int_a^\infty (x - a)f(x)dx \\
&= a \int_0^a f(x)dx - \int_0^a xf(x)dx + \int_a^\infty xf(x)dx - a \int_a^\infty f(x)dx \\
&= a \left(\int_0^a f(x)dx - 1 + \int_0^a f(x)dx \right) - \int_0^a xf(x)dx + E(X) - \int_0^a xf(x)dx \\
&= 2aF(a) - a + E(X) - 2 \int_0^a xf(x)dx \\
&= 2a(1 - e^{-a/\theta}) - a + \theta - 2 \int_0^a x \frac{1}{\theta} e^{-x/\theta} dx \\
&= a - 2ae^{-a/\theta} + \theta - 2 \left((-xe^{-x/\theta}) \Big|_0^a + \int_0^a e^{-x/\theta} dx \right) \text{ by integration by parts} \\
&= a - 2ae^{-a/\theta} + \theta - 2(-ae^{-a/\theta} - \theta e^{-a/\theta} + \theta) \\
&= a - \theta + 2\theta e^{-a/\theta}.
\end{aligned}$$

If we now define $h(a) = a - \theta + 2\theta e^{-a/\theta}$, then we find the *critical points* by setting $h'(a) = 0$ which yields

$$1 + 2\theta e^{-a/\theta} \left(-\frac{1}{\theta} \right) = 0 \implies e^{-a/\theta} = 2^{-1} \implies a = -\theta \ln 2^{-1} = \theta \ln 2.$$

Since $h''(a) = -2e^{-a/\theta}(-\frac{1}{\theta}) = \frac{2}{\theta}e^{-a/\theta} > 0 \forall a \geq 0$, the *Second Derivative Test* confirms that the critical point $a^* = \theta \ln 2$ minimizes $E(|X - a|)$. As we observed in Problem 5.4.7, $\theta \ln 2$ is the median of the $Exp(\theta)$ distribution.

5.4.11 Suppose that $X \sim N(\mu, \sigma^2)$. Noting that $Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$, we wish to calculate:

$$\begin{aligned}
P(|X - \mu| < \sigma) &= P\left(\left|\frac{X - \mu}{\sigma}\right| < \frac{\sigma}{\sigma}\right) = P(|Z| < 1) = 2P(Z < 1) - 1 = 2(0.84134) - 1 = 0.68268, \\
P(|X - \mu| < 2\sigma) &= P\left(\left|\frac{X - \mu}{\sigma}\right| < \frac{2\sigma}{\sigma}\right) = P(|Z| < 2) = 2P(Z < 2) - 1 = 2(0.97725) - 1 = 0.9545, \\
P(|X - \mu| < 3\sigma) &= P\left(\left|\frac{X - \mu}{\sigma}\right| < \frac{3\sigma}{\sigma}\right) = P(|Z| < 3) = 2P(Z < 3) - 1 = 2(0.99865) - 1 = 0.9973.
\end{aligned}$$

Therefore, X lies within one standard deviation of the mean about 68.2% of the time, within two standard deviations of the mean about 95.5% of the time, and within three standard deviations of the mean about 99.7% of the time.

5.4.12 If $Z \sim N(0, 1)$, then the 60th and 80th percentiles of Z are 0.2533 and 0.8416, respectively. By the symmetry of the standard normal distribution, it follows that the 20th and 40th percentiles of Z are -0.8416 and -0.2533 , respectively. If $X \sim N(10, 16)$, then we have the following:

$$P(X < q(0.2)) = 0.2 \implies P\left(Z < \frac{q(0.2) - 10}{4}\right) = 0.2 \implies \frac{q(0.2) - 10}{4} = -0.8416,$$

$$P(X < q(0.4)) = 0.4 \implies P\left(Z < \frac{q(0.4) - 10}{4}\right) = 0.4 \implies \frac{q(0.4) - 10}{4} = -0.2533,$$

$$P(X < q(0.6)) = 0.6 \implies P\left(Z < \frac{q(0.6) - 10}{4}\right) = 0.6 \implies \frac{q(0.6) - 10}{4} = 0.2533,$$

$$P(X < q(0.8)) = 0.8 \implies P\left(Z < \frac{q(0.8) - 10}{4}\right) = 0.8 \implies \frac{q(0.8) - 10}{4} = 0.8416.$$

Solving for $q(0.2)$, $q(0.4)$, $q(0.6)$, and $q(0.8)$, we readily obtain:

$$\begin{aligned} q(0.2) &= 10 - 4(0.8416) = 6.6336, & q(0.4) &= 10 - 4(0.2533) = 8.9868, \\ q(0.6) &= 10 + 4(0.2533) = 11.0132, & q(0.8) &= 10 + 4(0.8416) = 13.3664. \end{aligned}$$

5.4.13 Let X represent the examination score of a student, which is known to have a $N(65, 10^2)$ distribution. We wish to calculate the following probabilities:

$$\begin{aligned} P(X > 80) &= P\left(Z > \frac{80-65}{10}\right) = P(Z > 1.5) = 1 - 0.93319 = 0.06681, \\ P(70 < X < 80) &= P\left(\frac{70-65}{10} < Z < \frac{80-65}{10}\right) = P(0.5 < Z < 1.5) = 0.93319 - 0.69146 = 0.24173, \\ P(60 < X < 70) &= P\left(\frac{60-65}{10} < Z < \frac{70-65}{10}\right) = P(-0.5 < Z < 0.5) = 2(0.69146) - 1 = 0.38292, \\ P(50 < X < 60) &= P\left(\frac{50-65}{10} < Z < \frac{60-65}{10}\right) = P(-1.5 < Z < -0.5) = \underbrace{P(0.5 < Z < 1.5)}_{= 0.24173}, \\ P(X < 50) &= P\left(Z < \frac{50-65}{10}\right) = P(Z < -1.5) = P(Z > 1.5) = 0.06681. \end{aligned}$$

Therefore, the letter grade percentages are:

$$A: 6.681\%, B: 24.173\%, C: 38.292\%, D: 24.173\%, F: 6.681\%.$$

5.4.14 (a) Assuming that $X \sim N(2, 0.01^2)$, we wish to calculate

$$P(X < 2) = P\left(Z < \frac{2-2}{0.01}\right) = P(Z < 0) = 0.5 \text{ since } Z \sim N(0, 1).$$

(b) Note that

$$0.9 = P(|X - \mu| \leq c) = P\left(\left|\frac{X - \mu}{0.01}\right| \leq \frac{c}{0.01}\right) = P(|Z| \leq 100c) = 2P(Z \leq 100c) - 1.$$

This implies that

$$P(Z \leq 100c) = 0.95.$$

Since $P(Z \leq 1.6449) = 0.95$, it follows that $100c = 1.6449$, or simply, $c = 0.016449$.

(c) We wish to determine μ such that

$$\begin{aligned} P(X < 2) &\leq 0.01 \\ P\left(\frac{X - \mu}{0.01} < \frac{2 - \mu}{0.01}\right) &\leq 0.01 \\ P\left(Z < \frac{2 - \mu}{0.01}\right) &\leq 0.01 \quad \text{where } Z \sim N(0, 1) \\ P\left(Z > -\frac{2 - \mu}{0.01}\right) &\leq 0.01 \quad \text{by symmetry of the } N(0, 1) \text{ distribution} \\ 1 - P\left(Z \leq -\frac{2 - \mu}{0.01}\right) &\leq 0.01 \\ P\left(Z \leq \frac{\mu - 2}{0.01}\right) &\geq 0.99. \end{aligned}$$

Since $P(Z < 2.3263) = 0.99$, it follows that

$$\frac{\mu - 2}{0.01} \geq 2.3263 \implies \mu \geq 2 + 2.3263(0.01) = 2.023263.$$

Therefore, μ should be set to at least 2.023263 to ensure that the probability a bottle has less than 2 litres is less than 0.01.

5.4.15 Let X represent the diameter of a bolt measured in inches. Since $X \sim N(1.20, 0.005^2)$, we wish to calculate

$$\begin{aligned} P(1.19 < X < 1.21) &= P\left(\frac{1.19 - 1.20}{0.005} < Z < \frac{1.21 - 1.20}{0.005}\right) \quad \text{where } Z \sim N(0, 1) \\ &= P(-2 < Z < 2) \\ &= 2P(Z < 2) - 1 \\ &= 2(0.97725) - 1 \\ &= 0.9545. \end{aligned}$$

Therefore, 4.55% of the bolts will not meet specifications.

5.4.16 Let X represent the diameter of an egg measured in millimetres. Since $X \sim N(40, 4)$, we first determine

$$P(X < 37) = P\left(Z < \frac{37 - 40}{2}\right) = P(Z < -1.5) = P(Z > 1.5) = 1 - 0.93319 = 0.06681,$$

$$P(X > 42) = P\left(Z < \frac{42 - 40}{2}\right) = P(Z > 1) = 1 - 0.84134 = 0.15866,$$

and

$$P(37 < X < 42) = 1 - 0.06681 - 0.15866 = 0.77453.$$

Let C represent the wholesale price per egg measured in cents. Based on the given information, we now calculate

$$E(C) = 5(0.06681) + 6(0.77453) + 7(0.15866) = 6.09185.$$

5.4.17 If $Z \sim N(0, 1)$, then its pdf is given by

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} \text{ for } -\infty < z < \infty.$$

We wish to find the pdf of $Y = Z^2$. Applying the result of Example 5.2.2, we obtain

$$\begin{aligned} f_Y(y) &= \frac{1}{2\sqrt{y}} f_Z(\sqrt{y}) + \frac{1}{2\sqrt{y}} f_Z(-\sqrt{y}) \\ &= \frac{1}{2\sqrt{y}} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(\sqrt{y})^2} + \frac{1}{2\sqrt{y}} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(-\sqrt{y})^2} \\ &= \frac{y^{-\frac{1}{2}}}{2\sqrt{2\pi}} e^{-\frac{y}{2}} + \frac{y^{-\frac{1}{2}}}{2\sqrt{2\pi}} e^{-\frac{y}{2}} \\ &= \frac{y^{-\frac{1}{2}} e^{-\frac{y}{2}}}{\sqrt{2\pi}} \text{ for } y > 0. \end{aligned}$$

Section 5.5:

- 5.5.1 (a) Let $X_1, X_2, \dots, X_{100}$ be independent random variables selected from the distribution with pdf

$$f(x) = \begin{cases} 4x & \text{for } 0 < x < \frac{1}{2}, \\ 4(1-x) & \text{for } \frac{1}{2} < x < 1, \\ 0 & \text{otherwise.} \end{cases}$$

From Problem 5.3.2, the mean and variance of the above distribution was found to be $\mu = \frac{1}{2}$ and $\sigma^2 = \frac{1}{24}$, respectively. Using the CLT, we have that

$$\begin{aligned} P\left(49.0 < \sum_{i=1}^{100} X_i < 50.5\right) &= P\left(\frac{49.0 - 100(1/2)}{\sqrt{100/24}} < \frac{\sum_{i=1}^{100} X_i - 100\mu}{\sqrt{100\sigma^2}} < \frac{50.5 - 100(1/2)}{\sqrt{100/24}}\right) \\ &\approx P(-0.49 < Z < 0.24) \text{ where } Z \sim N(0, 1) \\ &= P(Z < 0.24) + P(Z < 0.49) - 1 \\ &= 0.59484 + 0.68793 - 1 \\ &= 0.28277. \end{aligned}$$

- (b) Instead, suppose that $X_1, X_2, \dots, X_{100}$ are independent random variables selected from the $U(0, 1)$ distribution, having mean $\mu = \frac{1}{2}$ and variance $\sigma^2 = \frac{1}{12}$. Using the CLT, we now obtain

$$\begin{aligned} P\left(49.0 < \sum_{i=1}^{100} X_i < 50.5\right) &= P\left(\frac{49.0 - 100(1/2)}{\sqrt{100/12}} < \frac{\sum_{i=1}^{100} X_i - 100\mu}{\sqrt{100\sigma^2}} < \frac{50.5 - 100(1/2)}{\sqrt{100/12}}\right) \\ &\approx P(-0.35 < Z < 0.17) \text{ where } Z \sim N(0, 1) \\ &= P(Z < 0.17) + P(Z < 0.35) - 1 \\ &= 0.56750 + 0.63683 - 1 \\ &= 0.20433. \end{aligned}$$

- 5.5.2 Let X represent the number of tomato seeds which germinate. Since seeds germinate independently of each other, it follows that $X \sim \text{Bin}(100, 0.8)$. We wish to determine $P(X \geq 75)$. Since $n = 100$, we can use the CLT and apply the continuity correction to get

$$\begin{aligned} P(X \geq 75) &= P(X \geq 74.5) \\ &\approx P\left(Z \geq \frac{74.5 - 100(0.8)}{\sqrt{100(0.8)(0.2)}}\right) \text{ where } Z \sim N(0, 1) \\ &= P(Z \geq -1.38) \\ &= P(Z \leq 1.38) \text{ by symmetry of the } N(0, 1) \text{ distribution} \\ &= 0.91621. \end{aligned}$$

5.5.3 For $i = 1, 2, \dots, 80$, let X_i be the cost associated with the i^{th} metal part. We assume that $X_1, X_2, \dots, X_{80}$ are independent random variables with common mean

$$\mu = 0(0.6) + 10(0.3) + 100(0.1) = 13$$

and common variance

$$\sigma^2 = 0^2(0.6) + 10^2(0.3) + 100^2(0.1) - 13^2 = 1030 - 169 = 861.$$

Let $C = \sum_{i=1}^{80} X_i$ represent the total cost, which is a discrete random variable. Note that the range of the random variable C is given by the set $\{0, 10, 20, \dots, 8000\}$. Since C increases in \$10 increments, we use the CLT (along with continuity correction) to get

$$\begin{aligned} P(C > 1200) &= P(C \geq 1205) \\ &\approx P\left(Z \geq \frac{1205 - 80(13)}{\sqrt{80(861)}}\right) \text{ where } Z \sim N(0, 1) \\ &= P(Z \geq 0.63) \\ &= 1 - P(Z < 0.63) \\ &= 1 - 0.73565 \\ &= 0.26435. \end{aligned}$$

5.5.4 For $i = 1, 2, \dots, n$, let X_i be the examination score of the i^{th} student. We assume that $X_1, X_2, \dots, X_n$ are independent random variables such that each $X_i \sim N(65, 100)$. As a result, we know that

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \sim N\left(65, \frac{100}{n}\right).$$

Therefore, when $n = 10$, we obtain

$$P(\bar{X} > 70) = P\left(Z > \frac{70 - 65}{\sqrt{\frac{100}{10}}}\right) = P(Z > 1.58) = 1 - 0.94295 = 0.05705.$$

On the other hand, when $n = 25$, we get

$$P(\bar{X} > 70) = P\left(Z > \frac{70 - 65}{\sqrt{\frac{100}{25}}}\right) = P(Z > 2.5) = 1 - 0.99379 = 0.00621.$$

5.5.5 For $i = 1, 2, 3, 4$, let X_i be the length of the i^{th} section. We are told that X_1, X_2, X_3 , and X_4 are independent normally distributed random variables. Let $Y = X_1 + X_2 + X_3 + X_4$ be the length of the turbine shaft, which we know has a normal distribution with mean

$$\mu_Y = 8.10 + 7.25 + 9.75 + 3.10 = 28.2$$

and variance

$$\sigma_Y^2 = 0.22^2 + 0.20^2 + 0.24^2 + 0.20^2 = 0.186.$$

Therefore, the probability that an assembled turbine shaft meets the desired specifications is given by

$$\begin{aligned} P(28 - 0.26 \leq Y \leq 28 + 0.26) &= P\left(\frac{27.74 - 28.2}{\sqrt{0.186}} < Z < \frac{28.26 - 28.2}{\sqrt{0.186}}\right) \text{ where } Z \sim N(0, 1) \\ &= P(-1.07 < Z < 0.14) \\ &= P(Z < 0.14) + P(Z < 1.07) - 1 \\ &= 0.55567 + 0.85769 - 1 \\ &= 0.41336. \end{aligned}$$

5.5.6 In a survey of n voters, let X be the number of voters who say they would vote Conservative. Although the exact distribution of X is hypergeometric, we can view X as approximately binomial, since it is reasonable to think that the total population in this given riding from which we are sampling is large. As a result, we simply assume that $X \sim \text{Bin}(n, 0.16)$. If we now apply the normal approximation to the binomial distribution, then X is approximately normal with mean $0.16n$ and variance $n(0.16)(0.84) = 0.1344n$. We wish to find the value of n such that

$$P\left(\left|\frac{X}{n} - 0.16\right| \leq 0.03\right) = 0.95,$$

or equivalently,

$$\begin{aligned} P(|X - 0.16n| \leq 0.03n) &= 0.95 \\ P\left(\frac{|X - 0.16n|}{\sqrt{0.1344n}} \leq \frac{0.03n}{\sqrt{0.1344n}}\right) &= 0.95 \\ P\left(|Z| \leq \frac{0.03\sqrt{n}}{\sqrt{0.1344}}\right) &= 0.95 \text{ where } Z \sim N(0, 1) \\ 2P\left(Z \leq \frac{0.03\sqrt{n}}{\sqrt{0.1344}}\right) - 1 &= 0.95 \\ P\left(Z \leq \frac{0.03\sqrt{n}}{\sqrt{0.1344}}\right) &= 0.975. \end{aligned}$$

However, we have that $P(Z \leq 1.96) = 0.975$, which immediately implies that

$$\frac{0.03\sqrt{n}}{\sqrt{0.1344}} = 1.96 \implies n = \left(\frac{1.96\sqrt{0.1344}}{0.03}\right)^2 = 573.679.$$

Therefore, at least 574 voters should be surveyed.

- 5.5.7 (a) In a random sample of size 10,000, let X count the number of unemployed persons. If the unemployment rate in Canada is 7%, then it is reasonable to assume that $X \sim \text{Bin}(10000, 0.07)$ (since the Canadian labour force population from which we are sampling can be regarded as being large). If we apply the normal approximation to the binomial distribution (and do not include a continuity correction), we have that

$$\begin{aligned}
 P(675 \leq X \leq 725) &\approx P\left(\frac{675 - 10000(0.07)}{\sqrt{10000(0.07)(0.93)}} \leq Z \leq \frac{725 - 10000(0.07)}{\sqrt{10000(0.07)(0.93)}}\right) \text{ where } Z \sim N(0, 1) \\
 &= P(-0.98 < Z < 0.98) \\
 &= 2P(Z < 0.98) - 1 \\
 &= 2(0.83646) - 1 \\
 &= 0.67292.
 \end{aligned}$$

- (b) With $X \sim \text{Bin}(n, 0.07)$, we wish to find the value of n such that

$$\begin{aligned}
 P\left(0.069 < \frac{X}{n} < 0.071\right) &= 0.95 \\
 P(0.069n < X < 0.071n) &= 0.95 \\
 P\left(\frac{0.069n - 0.07n}{\sqrt{n(0.07)(0.93)}} < Z < \frac{0.071n - 0.07n}{\sqrt{n(0.07)(0.93)}}\right) &= 0.95 \text{ where } Z \sim N(0, 1) \\
 P\left(-\frac{0.001\sqrt{n}}{\sqrt{0.0651}} < Z < \frac{0.001\sqrt{n}}{\sqrt{0.0651}}\right) &= 0.95 \\
 2P\left(Z < \frac{0.001\sqrt{n}}{\sqrt{0.0651}}\right) - 1 &= 0.95 \\
 P\left(Z < \frac{0.001\sqrt{n}}{\sqrt{0.0651}}\right) &= 0.975.
 \end{aligned}$$

However, we have that $P(Z \leq 1.96) = 0.975$, which immediately implies that

$$\frac{0.001\sqrt{n}}{\sqrt{0.0651}} = 1.96 \implies n = \left(\frac{1.96\sqrt{0.0651}}{0.001}\right)^2 = 250088.16.$$

Therefore, a random sample of size at least 250,089 would be necessary.

- 5.5.8 For $i = 1, 2, \dots, n$, let W_i be the profit from the i^{th} independent play of the game. As a result, $X = \sum_{i=1}^n W_i$. On a single play of the game involving 3 dice being rolled, let Y be the number of hearts appearing. It immediately follows that $Y \sim \text{Bin}(3, \frac{1}{6})$. With this definition of Y , we note that

$$\mu = E(W_i) = E(Y - 1) = E(Y) - 1 = 3\left(\frac{1}{6}\right) - 1 = -0.5$$

and

$$\sigma^2 = \text{Var}(W_i) = \text{Var}(Y - 1) = \text{Var}(Y) = 3 \left(\frac{1}{6} \right) \left(\frac{5}{6} \right) = \frac{5}{12}.$$

Therefore, applying the CLT (along with continuity correction), we obtain

$$\begin{aligned} P(X > 0) &= P(X \geq 0.5) \\ &= P\left(\frac{X - n\mu}{\sqrt{n\sigma^2}} \geq \frac{0.5 - n(-0.5)}{\sqrt{n(5/12)}}\right) \\ &\approx P\left(Z \geq \frac{0.5(n+1)}{\sqrt{n(5/12)}}\right) \quad \text{where } Z \sim N(0, 1). \end{aligned} \tag{21}$$

When $n = 10$, (21) becomes

$$P(X > 0) \approx P(Z \geq 2.69) = 1 - 0.99643 = 0.00357.$$

When $n = 25$, (21) becomes

$$P(X > 0) \approx P(Z \geq 4.03) \approx 1 - 1 = 0.$$

When $n = 50$, (21) becomes

$$P(X > 0) \approx P(Z \geq 5.59) \approx 1 - 1 = 0.$$